



High-Level Solution Design

HLSD

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Document Information

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Version	Date	Author	Summary of Changes
1.0	12/05/2026	Joey Shao	Initial draft
1.1	20/05/2026	Joey Shao	Revised based on solution owner feedback

HLSD Review Checklist

All mandatory items in red must be confirmed before submission to the Steering Group. Tick marks indicate which stakeholders should review each item.

✓	Review Item	Sec.	OT	SO	AC	CAIT	SGG	SH	CAIC	ARB
Strategy & Business										
☐	Business problem articulated with measurable impact	<u>2.1</u>	✓	✓	✓	✓				
☐	Strategic alignment to NHCI, MOMAH SSO, and AI strategy	<u>2.2</u>	✓	✓	✓	✓				
☐	Business case metrics defined with clear assumptions	<u>2.5</u>	✓	✓	✓	✓				
☐	Existing solution landscape documented	<u>2.6</u>	✓	✓	✓	✓				
☐	Key requirements identified	<u>4</u>	✓	✓	✓	✓				
☐	Success metrics (i.e., KPIs) with baseline and targets	<u>11</u>	✓	✓	✓	✓				
☐	Executive summary self-contained	<u>Exec.</u>	✓			✓	✓	✓	✓	
Solution Design										
☐	Solution description accessible to non-technical audience	<u>3.1</u>	✓	✓	✓	✓		✓		
☐	Personas, capabilities, and user journeys defined and linked	<u>3.2–3.3–3.4–3.5</u>	✓	✓		✓				
☐	Productization potential assessed (if applicable)	<u>3.6</u>	✓	✓		✓				
☐	Architecture diagram and key components described	<u>5.1–5.2</u>	✓			✓				✓
☐	External and internal integrations identified	<u>5.3</u>	✓	✓		✓				✓
☐	Security covers authentication, encryption, audit logging	<u>5.5</u>	✓			✓				✓
AI & Responsible AI										
☐	Model type justified with alternatives considered	<u>6.1</u>	✓			✓				
☐	Intended and out-of-scope use cases stated	<u>6.2</u>	✓	✓		✓				
☐	Explainability and human-in-the-loop design defined	<u>6.3–6.4</u>	✓	✓		✓				
☐	Model monitoring: drift detection, alerting, retraining	<u>6.6</u>	✓			✓				
☐	Agentic AI design documented	<u>6.7</u>	✓			✓				
☐	AI evaluation pipeline defined	<u>6.8</u>	✓			✓				
☐	AI risk classification assigned with rationale	<u>8.1</u>	✓			✓				
☐	Compliance & Ethical AI checklist completed with evidence	<u>8.3</u>	✓			✓				
Data										
☐	Data sources identified with type, owner, sensitivity	<u>7.1</u>	✓	✓	✓	✓				
☐	Data quality assessed per source (5 dimensions)	<u>7.2</u>		✓						
Infrastructure & Readiness										
☐	Infrastructure sizing estimated with cost projection	<u>9.1</u>	✓	✓	✓					✓
☐	Testing strategy covers all test types with entry/exit criteria	<u>9.4</u>	✓	✓						✓
☐	Business continuity and DR targets (RTO/RPO) defined	<u>9.5–9.6</u>	✓							
☐	Prerequisites identified with owners and target dates	<u>10.3</u>	✓	✓	✓	✓				
☐	Roadmap covers all phases; MVP scope defined	<u>12.1–12.2</u>	✓	✓	✓	✓				
☐	Execution risks identified with mitigation strategies	<u>12.3</u>	✓		✓	✓				
Governance										
☐	Stakeholders identified with responsibilities and contacts	<u>2.3</u>	✓	✓	✓	✓				
☐	Approval sign-off table complete	<u>App. C</u>	✓			✓				

OT = Oversight Team | SO = Solution Owner | AC = AI Champion | CAIT = Central AI Team | SGG = AI Steering Group | SH = Sector Head | CAIC = AI Central Committee | ARB = NHCI ARB

Note: Other bodies might be involved during the development of the HLSD (e.g. DT, MOMAH EA). Section 1 (Introduction) must be completed prior to Stage-Gate 1 but is not subject to formal review at the gate. All other sections not listed above are not mandatory for Stage-Gate 1 approval. Both will be reviewed by the OT for completeness and alignment.

In red are the mandatory section to be completed and approved by stage-gate 1

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Executive Summary

The Policy AI Lab is a high-performance decision-support platform designed to transform MOMAH's policy lifecycle from a fragmented, document-centric, and reactive process into a unified, AI-augmented, predictive, and evidence-based decision-intelligence capability. By integrating advanced simulation, generative AI, conflict detection, and a canonical policy registry, the platform ensures that policies requiring simulation under approved trigger rules are stress-tested against Vision 2030 KPIs and fiscal responsibility criteria, while all policies complete required conflict, legal, and governance checks before enactment.

Solution Purpose

The platform provides MOMAH Policy Analysts, Housing Economists, Legal Officers, Senior Leaders, and the Minister with role-based AI tools to draft, simulate, benchmark, and approve policies within a single governed environment. It directly addresses the root cause of MOMAH's current policy challenges: a fully manual, document-centric process with no simulation capability — where the only means to test a policy is to launch it in a limited area of KSA.

Expected Impact and Outcomes

- Reduce average policy development cycle from 6–12 months to under 6 months (target: 50% reduction by Year 2)
- Embed predictive simulation so policies requiring simulation under SDD-confirmed trigger rules are tested for fiscal, social, and market impact across multiple scenarios before enactment
- Eliminate policy silos through cross-sector conflict and ripple-effect detection across housing, municipal, fiscal, environmental, and smart-city domains
- Build institutional memory through a version-controlled Policy Registry with full assumption and decision traceability
- Systematically score policies against Vision 2030 KPIs, particularly the 70% homeownership target and fiscal sustainability goals

Success Metrics (Baseline → 6M / Y1 / Y2)

- Active platform users: 0 → 10 / 20 / 50
- Policies simulated through the platform: 0 → 3 / 5 / 10
- Average policy development time reduction: 0 → 10% / 30% / 50%
- Simulation accuracy (HITL approvals without edits): baseline 0; target 80% by Year 2 (per BRD Strategic and Operational Metrics)
- Conflict detection recall prior to ministerial review: baseline 0; target $\geq 85\%$ by Year 2 (per BRD Strategic and Operational Metrics). Ground truth established through Ma'rifa/Sanad cross-referencing, MOMAH legal expert validation, and a labelled benchmark dataset
- Assumption documentation completeness: baseline 0; target 90% by Year 2 (per BRD Strategic and Operational Metrics). Metadata schema and required fields to be finalized in SDD.

- Automated risk threshold alerts: baseline 0; target >90% by Year 2 (per BRD Strategic and Operational Metrics). Fiscal/legal risk taxonomy and trigger rules to be finalized in SDD
- Policy simulation and conflict check coverage: baseline 0; target threshold to be confirmed in SDD after lifecycle trigger rules, simulation applicability, and conflict-check coverage requirements are finalized

Business Case

There are currently no automated processes or systems for policy launch at MOMAH. The Policy AI Lab replaces a fully manual process, targeting measurable efficiency gains: a 50% reduction in policy development cycle time by Year 2, reduction in post-approval policy reversals, and improved fiscal accountability through systematic impact simulation. A formal NPV/IRR financial model will be developed once simulation output formulas and fiscal assumptions are finalized with the client. Formula methodology confirmed in BO QA Reply: (1) NPV uses standard Discounted Cash Flow — $NPV = \sum (net_cash_flow_t / (1+r)^t)$, using MoF-published discount-rate inputs; the baseline analysis horizon shall follow the BRD fiscal-impact horizon, with any extended horizon to be confirmed with MOMAH/MoF in the SDD; (2) GDP contribution uses direct + indirect + induced economic impact methodology aligned with GASTAT national accounts and MoF input-output tables; (3) Social Equity Impact uses distributional analysis across GASTAT income deciles, MOMAH beneficiary data, and SAMA mortgage accessibility data. Financial input values (CAPEX, OPEX, IRR, NPV) to be provided by MOMAH once the simulation engine is calibrated.

AI Risk Classification: High Risk

The system directly informs policy decisions affecting national housing strategy, multi-sector regulatory alignment, and Vision 2030 national KPIs. All AI outputs are advisory only; human-in-the-loop validation is mandatory at every approval stage (Policy Analyst → Director → Minister). No policy recommendation reaches enactment without expert review and formal sign-off.

Key Assumptions Underpinning the Design

- MOMAH internal system data (housing permits, Sakani, Balady) will be made accessible to the platform
- Integration with Ma'rifa (legal references) and Sanad (legal advisory) systems is technically and contractually feasible within project scope
- MOMAH SSO integration is available and will be provided for authentication and role-based access control
- All AI inference and data storage will remain within KSA, in full compliance with PDPL and NCA Essential Cybersecurity Controls
- Human-in-the-loop governance framework will be formally adopted by MOMAH stakeholders as a mandatory part of the policy lifecycle

1. Introduction

1.1 Purpose

This High-Level Solution Design (HLSD) document provides a high-level blueprint for the Policy AI Lab solution at the Ministry of Municipalities and Housing (MOMAH), translating the business requirements documented in the BRD v2.0 into an initial solution design. It defines the solution purpose, scope, key capabilities, user personas, data landscape, AI approach, governance model, and delivery roadmap. This document is intended to be reviewed and endorsed by the AI Solutions Steering Group before detailed design and development commences in Phase 2. It serves as the primary reference for all subsequent design, architecture, and delivery activities.

1.2 Scope

In Scope

The Policy AI Lab encompasses four integrated capability groups, all to be delivered within the current engagement:

- **Policy Registry and AI-Assisted Drafting:** Canonical, version-controlled policy registry with full metadata; AI-assisted policy drafting with legal basis extraction, Vision 2030 alignment validation, and assumption documentation; amendment workflows with structured justification and traceability; Institutional Memory Vault storing all policy iterations with full decision explainability.
- **Benchmarking and Conflict Detection:** Automated detection of overlapping, redundant, or contradictory policies across sectors; cross-sector ripple-effect mapping (housing, municipal, fiscal, environmental, smart-city); policy benchmarking against peer countries (e.g., Singapore, Norway) and global standards; integration with Ma'rifa and Sanad for legal conflict validation.
- **Policy Impact Simulation:** Multi-variable simulation engine with "what-if" scenario testing; configurable sensitivity analysis; Black Swan stress testing; and risk scoring across five dimensions (Legal, Fiscal, Operational, Strategic, International).
- **Consultation, Governance and Workflow:** Consultation comment synthesis with theme clustering and structured amendment proposals; tiered human-in-the-loop approval framework (Policy Analyst → Director → Minister); role-based access control; kill-switch mechanism; full audit trail; SSO integration; DMS integration; approval workflow system integration.

Out of Scope

- **Autonomous Policy Approval:** The system will not have authority to approve, enact, or publish policies. All final decisions remain human-led.
- **Binding Legal Interpretation:** The system does not provide legally binding interpretations or replace formal legal counsel. All outputs are advisory.

- **Direct Legislative Draft Submission:** No automatic submission to legislative bodies or gazette systems without human validation and formal approval workflows.
- **National Econometric Forecasting Infrastructure:** The system will not build or replace national econometric forecasting platforms. It relies on predefined models and integrated external data inputs.
- **Full Regulatory Enforcement Monitoring:** The system will not monitor real-time compliance or execute regulatory enforcement actions.
- **Custom Foundation Model Training:** The system will not develop foundational LLMs or large econometric models from scratch. It will leverage existing AI models and configurable frameworks.
- **Political Decision-Making:** The system does not determine political priorities or make value-based policy choices. Strategic direction remains with MOMAH policymakers.
- **Operational Municipal Service Delivery:** The platform does not manage day-to-day municipal operations (e.g., waste collection, utilities).
- **Citizen-Facing Applications:** The platform is an internal decision-support tool only, not a public-facing citizen service portal.

1.3 Intended Audience

Audience	Purpose of Review
Central AI Committee	<i>[For moonshot solutions only] Strategic review of an executive version of the HLSD and final stage-gate approval</i>
AI Solutions Steering Group	<i>Strategic and technical review of an executive version of the HLSD, stage-gate endorsement for moonshot solutions and stage-gate approval for standalone solutions</i>
Sector Head	<i>Review for alignment with sector needs and priorities, provide final approval before submission to Steering Group / Central AI Committee</i>
AI Champion	<i>Review for sector alignment</i>
NHCI ARB	<i>Review for technical architecture (no approval at this stage)</i>
NHCI Technical Lead(s)	<i>Day-to-day delivery management and review</i>

Oversight team	<i>Governance review, standards compliance, and delivery assurance across the program</i>
AI Solution Owner	<i>Solution-level owner; provide input, review key requirements, business outcomes, and user journeys</i>
Solution Vendor	<i>Document owner - produced and maintained by vendor. Responsible for completeness and accuracy of all sections.</i>

Note: For governing bodies' review (i.e., Central AI Committee and AI Solutions Steering Group) an executive summary presentation needs to be developed covering all elements listed in the Executive Summary (at the beginning of this document)

1.4 Definitions & Acronyms

Term / Acronym	Definition
HLSD	High-Level Solution Design
NHCI	National Housing Company — Innovation: the innovation programme within the National Housing Company (NHC, الشركة الوطنية للإسكان), responsible for governance, oversight, and architectural review of AI solutions delivered to MOMAH and related stakeholders. References in this document include the NHCI AI Programme Governance Framework, the NHCI ARB (Architecture Review Board), and NHCI Oversight / Technical Lead functions.
MOMAH	Ministry of Municipalities and Housing (المملكة العربية السعودية)
ARB	Architecture Review Board
AI	Artificial Intelligence
SDD	Solution Design Document (Phase 2 deliverable, follows HLSD approval)
BRD	Business Requirements Document
HITL	Human-in-the-Loop — governance model requiring mandatory human review of AI outputs before action is taken

ABM	Agent-Based Model — computational simulation model used to simulate the impact of housing and municipal policies across KSA's 13 regions. Agent categories are confirmed in the BO QA Reply; exact agent population scale, instance counts, and regional granularity will be confirmed in Phase 2 SDD after data assessment and SDAIA platform benchmarking.
PDPL	Saudi Personal Data Protection Law — national data privacy regulation governing processing and storage of personal data within KSA
NCA	National Cybersecurity Authority — Saudi authority issuing Essential Cybersecurity Controls applicable to this solution
SDAIA	Saudi Data and AI Authority — national body providing AI ethics guidelines and governance framework
SIBOR	Saudi Interbank Offered Rate — key interest rate benchmark used as a simulation input variable
LTV	Loan-to-Value Ratio — mortgage lending parameter used as a policy lever in housing simulations
GASTAT	General Authority for Statistics — Saudi national statistics body providing demographic and economic data
SAMA	Saudi Arabian Monetary Authority — central bank providing financial and mortgage lending data
SSO	Single Sign-On — authentication mechanism enabling users to access the platform with their existing MOMAH credentials
DMS	Document Management System — MOMAH's existing system for storing, versioning, and archiving policy documents
XAI	Explainable AI — AI design principle ensuring model outputs include interpretable explanations accessible to non-technical users
KPI	Key Performance Indicator
Vision 2030	Saudi Arabia's national transformation plan, including a target of 70% homeownership — a primary alignment benchmark for the Policy AI Lab

HDOC	Housing Data and Operations Catalogue — MOMAH internal operational database confirmed as a required integration source for the Policy AI Lab (confirmed by BO QA Reply). Contains MOMAH-specific operational records and housing programme data not available from external sources. Detailed database schema, integration interface specifications, and data quality assessment to be conducted in Phase 2 following a dedicated scoping session with MOMAH data owners.
EWG	Expert Working Group — MOMAH-led advisory body formed to provide and validate domain knowledge inputs for the Policy Simulation engine. Responsibilities include defining baseline policy assumptions, validating ABM parameters, and signing off on Low Confidence outputs. See Section 3.7.2.
Policy Instrument	The specific mechanism — such as a tax, subsidy, regulation, land grant, fee, or eligibility threshold — used by a policy to achieve a stated objective. Referenced in Section 3.3.6 (Simulation Input — Policy Variables) and Section 3.3.5 (Benchmarking — Mechanism Type). Aligned with BRD Terms & Definitions.
Fiscal Neutrality	A state where the implementation costs of a policy are fully offset by the economic value, revenue uplift, or savings the policy generates. Used as a design constraint in CAP-02 Gap Analysis intervention recommendations (Section 3.3.2) and in CAP-05 Benchmarking Fiscal Commitment scoring (Section 3.3.5). Aligned with BRD Terms & Definitions.
Policy Knowledge Graph	A structured, machine-readable digital map of the relationships between policies, regulations, entities, sectors, Vision 2030 KPIs, and outcomes. Forms the backbone of the Policy Registry & Institutional Memory Layer (Section 5.2 Table 26 — Policy Registry & Knowledge Graph) and supports cross-sector ripple-effect detection by CAP-04 Conflict Detection Engine (Section 3.3.4). Aligned with BRD Terms & Definitions.

2. Business Context

2.1 Business Problem

MOMAH operates at the center of Saudi Arabia's national urban development, municipal governance, and housing strategy. Its mandate directly influences economic diversification, infrastructure efficiency, private sector participation, and quality of life — all foundational pillars

of Vision 2030. As the Kingdom accelerates reform, the complexity, interdependency, and scale of policy decisions have significantly increased, yet MOMAH's current policy lifecycle is structurally unprepared for this level of systemic complexity.

Current State

Today, MOMAH's policy lifecycle is document-centric rather than system-centric, fragmented across data sources and stakeholders, dependent on manual analysis and siloed expertise, limited in predictive modeling capability, and reactive rather than simulation-driven. There are no automated processes for policy launch, and no system exists to support it. The only way MOMAH currently tests a policy is by launching it in a small area of KSA — a slow, costly, and high-risk approach to policy validation.

Five Core Pain Points

- **Limited System-Wide Visibility:** Policies are developed and reviewed without a unified, canonical registry. There is no single source of truth for active policies, their dependencies, or their status across ministries.
- **Insufficient Predictive and Simulation Capability:** Impact assessments are static reports. Scenario analysis is weak, particularly regarding risk identification and unintended consequences. Fiscal and social ripple effects are mostly either missed or not comprehensively analyzed.
- **Weak Institutional Memory and Feedback Loops:** Lessons from past policies are not systematically captured. There is no structured mechanism to trace assumptions and outcomes or compare policy performance over time.
- **Limited Global Benchmarking Integration:** International standards and peer-country benchmarks (e.g., Singapore, Norway) are not embedded into policy design in a structured way, limiting alignment with global best practices.
- **Extended Policy Development Timelines:** The current process requires many iterations and significant stakeholder bandwidth, resulting in policy development cycles of 6–12 months.

Policies now affect multiple sectors simultaneously — real estate, financial, industrial/construction, environmental, social, and smart-city sectors — carrying long-term fiscal and macroeconomic implications. Without a system capable of modeling these interdependencies, MOMAH risks regulatory conflicts, fiscal overruns, and failure to achieve Vision 2030 targets, including the 70% homeownership goal.

2.2 Strategic Alignment

Vision 2030 Objective	Enhance government effectiveness through data-driven policymaking, improved regulatory coherence, and accelerated reform delivery — directly supporting the 70% homeownership target, Quality of Life improvement, and fiscal sustainability under Vision 2030.
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NHCI Strategic Objective	Advance AI-powered decision-intelligence capabilities within MOMAH, operationalizing AI for governance, institutional learning, and predictive policy analysis in support of NHCI's AI programme goals.
MOMAH Sector Strategic Objectives (SSOs)	Strengthen evidence-based policy development, improve regulatory quality, reduce policy contradictions and sectoral fragmentation, accelerate policy development cycles, enhance housing affordability and supply planning through advanced simulation, and build institutional memory for long-term governance excellence.
MOMAH AI Strategy Alignment	Deploy AI-driven decision-support tools that enhance policy quality, predictive capability, and institutional efficiency. Operationalize AI for governance, insight generation, and institutional learning through automated drafting, conflict detection, multi-variable simulation, risk scoring, and continuous model improvement — all under a human-in-the-loop governance framework compliant with SDAIA AI Ethics Guidelines.

2.3 Key Stakeholders

Stakeholder	Role	Interests / Concerns	Responsibility	Name	Contact
AI Solution Owner	Solution-level owner	Requirements definition; ensure progress in line with business needs	Participate in vendor check-ins, Sprint Reviews, Steering Group, and design discussions	Dr. Ahmed Aldurwiesh	a.aldarwish@momah.gov.sa
AI Champion	Sector-level coordinator	Cross-solution coordination; surface risks, dependencies, and adoption needs within sector	Participate in Sprint Reviews and Bi-Weekly Sector meetings	Hamad Alsharida	halsharida@momah.gov.sa

Sector Head	Sector-level sponsor	Solution direction, progress alignment, escalation support	Participate in Bi-Weekly Sector meetings and Central AI Committee sessions	Majid bin Abdullah Al-Hogail	
Policy KPI Owner / Business Owner	MOMAH-side owner of KPI targets and policy outcomes; primary point of contact for business requirements and metric definition	Simulation accuracy, conflict detection quality, assumption completeness, risk alert coverage, and overall policy outcome measurement	Own and validate all KPI targets; approve simulation methodology and metric definitions; participate in Sprint Reviews and milestone sign-offs	Dr. Ahmed Aldurwiesh	a.alдарwish@momah.gov.sa

2.4 Expected Business Outcomes

- **Policy Quality Improvement:** Fewer contradictions and higher evidence standards across all MOMAH policies; systematic Vision 2030 alignment scoring for every policy draft.
- **Risk Reduction:** Early identification of unintended consequences, fewer regulatory escalations post-launch, and automated fiscal and legal risk alerts for policies exceeding defined thresholds.
- **Accelerated Reform Cycles:** Reduction in average policy development time from 6–12 months to under 6 months, enabling MOMAH to respond faster to market volatility, regulatory demands, and Vision 2030 milestones.

2.5 Expected Business Case Metrics

MOMAH business case

Metric Type	Metric	Value
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P&L impact	Revenue or cost savings in steady state	TBD
P&L impact	Total CAPEX	TBD
P&L impact	OPEX in steady state	TBD
P&L impact	Incremental Net Profit steady state	TBD
Profitability	IRR	TBD
Profitability	NPV	TBD

Commercialization business case (beyond MOMAH, if applicable)

Metric Type	Metric	Value
P&L impact	Revenue or cost savings in steady state	
P&L impact	Total CAPEX	
P&L impact	OPEX in steady state	
P&L impact	Incremental Net Profit steady state	
Profitability	IRR	
Profitability	NPV	

2.6 Current Solution Landscape

Solution Name	Owner / Provider	Dedicated Budget	Timeline / Status	Core features	Coverage of Problem (2.1)	Gaps / Limitations	Demo / Documentation
None (No existing system)	N/A	N/A	N/A — no system currently in use	N/A — there are no automated tools, AI systems, or digital	None — the business problem is entirely	All policy development activities are conducted manually using documents and spreadsheets. Simulation is performed by	N/A

				platforms currently in use for policy drafting, simulation, conflict detection, or benchmarking.	unaddressed by any existing digital system.	launching a policy in a small geographic area of KSA — a slow, costly, and irreversible validation method. There is no registry, no conflict detection, no institutional memory, and no AI assistance of any kind.	
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Initiative	Timeline	Owner / Provider	Approach taken	Outcome	Key Reason for Failure	Lessons Learned for Current Design	Demo / Documentation
N/A — no previous digital or AI initiative has been attempted for policy management at MOMAH.	N/A	N/A	N/A	N/A	N/A	N/A	N/A

3. Solution Overview

3.1 Solution Description

The Policy AI Lab is a centralized, AI-powered decision-intelligence platform that transforms MOMAH's policy lifecycle from fragmented and reactive to unified, predictive, and evidence-based. It provides government users — from Policy Analysts to the Minister — with role-specific tools to draft, simulate, conflict-check, benchmark, and approve policies within a single governed environment.

The platform operates across three integrated capability layers: (1) a canonical Policy Registry and AI-assisted drafting environment, enabling structured policy creation with legal basis extraction, Vision 2030 alignment validation, and assumption documentation; (2) a Benchmarking and Conflict Detection engine, automatically identifying overlapping or contradictory policies across sectors and benchmarking against peer countries such as Singapore and Norway; and (3) a Multi-Scenario Simulation engine, enabling Housing Economists to stress-test policies across economic, demographic, supply, and market variables — including interest rate shocks, land price changes, and developer behavior — before any policy reaches ministerial review.

All AI outputs are advisory. A tiered human-in-the-loop governance framework (Policy Analyst → Director → Minister) ensures no AI recommendation reaches enactment without expert validation. The platform is designed for full compliance with Saudi PDPL, NCA Essential Cybersecurity Controls, and SDAIA AI Ethics Guidelines, with all data and AI inference remaining within KSA.

3.2 User Personas

Persona	Description	Primary Need	Interaction with the Solution	Frequency of Use
Minister (H.E.)	CEO of MOMAH; ultimate decision-maker for all ministry policies.	High-level dashboard of policy health, Vision 2030 alignment scores, risk alerts, and Tier-3 escalations requiring ministerial decision.	Executive Intelligence Cockpit: scenario comparison, final approval gateway, override capability.	Weekly / On escalation
Senior Leader / Executive	High-level decision-makers within MOMAH's Executive Office; responsible for strategic direction and cross-ministerial alignment.	Portfolio view of all active policies, cross-sector ripple effects, regional impact visualization, and Tier-2 strategic briefings.	Policy portfolio dashboard, Vision 2030 Alignment Auditor, Regional Impact Heatmaps.	Weekly
Policy Analyst	Primary strategist responsible for drafting and	AI-assisted policy drafting with legal basis extraction, gap analysis, consultation comment	Policy drafting sandbox, version control, assumption	Daily

	refining policy frameworks.	synthesis, and access to institutional memory.	builder, knowledge registry.	
Housing Economist	Expert focused on the financial and macroeconomic impact of policy levers.	Multi-variable simulation lab, Black Swan stress testing, sensitivity analysis, and ability to manipulate macro, demographic, supply, and market variables.	Agent-Based Simulation engine, scenario configurator, risk escalation monitoring.	Daily / Per policy
Data Scientist / AI Unit	Technical team responsible for the platform's underlying models, data pipelines, and AI governance.	Model retraining, explainability audits, drift detection, raw log access, threshold configuration, and model versioning.	Model governance dashboard, retraining pipeline, explainability audit tools.	Weekly / On drift alert
Legal & Compliance Officer	Gatekeeper ensuring all policy drafts adhere to Saudi Law, Royal Decrees, and PDPL/NCA requirements.	Regulatory conflict detection, legal basis extraction, clause-level traceability, and compliance gateway.	Regulatory Conflict Engine, Ma'rifa/Sanad integration interface, override permissions.	Per policy / On conflict alert
Business Owner / Product Manager	Responsible for strategic direction and delivery of the Policy AI Lab solution.	Policy lifecycle orchestration, workflow management, system usage analytics, KPI and threshold configuration, and role-based access control.	System health dashboard, workflow configurator, usage analytics.	Daily

Stakeholder (Internal/External)	Representatives from other ministries or the public providing consultation feedback.	Consultation portal to submit feedback, view summarized responses, and track how input influenced policy drafts.	Feedback submission interface, consultation synthesis viewer.	Per consultation cycle
Simulation Expert	A technically qualified expert invoked for complex or high-stakes policy simulations that exceed the standard Housing Economist workflow.	Deep access to ABM model parameters, Monte Carlo configuration, calibration datasets, and scenario sensitivity controls; ability to document and justify parameter adjustments.	Advanced Simulation Lab access; ability to override default parameter bounds with written justification; responsible for confirming technical validity of simulation outputs before they are presented to the Policy Analyst for HITL review.	On demand — engaged only when simulation complexity, policy risk level, or Monte Carlo instability exceeds standard Housing Economist scope.

3.3 Key Capabilities

ID	Capability	Description	Linked Persona(s)
CAP-01	Policy Intelligence & Institutional Memory Layer	Canonical, version-controlled Policy Registry with full policy metadata, legal basis, Vision 2030 KPI linkages, sector relationships, decisions, assumptions, final approved texts, supporting records, and outcome history. Supports institutional memory and traceability required by the BRD and MOMAH methodology Step 8.	All Personas

CAP-02	Case Identification & Gap Opportunity Analysis	Allows policy owners to register and classify triggers for new or updated legislation, including regulatory gaps, new risks or incidents, strategic changes, organizational changes, and market or technology developments. The Gap & Opportunity Analysis engine compares Policy Registry coverage against Vision 2030 KPI targets, sector priorities, and peer-country benchmarks to produce gap reports, 0-100 opportunity scores, benchmark maps, recommended interventions, legal-basis linkage, confidence bands, and audit entries. Aligned with BRD UC-16 and MOMAH methodology Step 1.	Policy Analyst, Senior Leader, Minister
CAP-03	Policy Concept Builder & AI Drafting Sandbox	Structured concept builder and AI drafting workspace that captures the initial rationale, target sector, legal basis, affected stakeholders, Vision 2030 pillar, expected impacts, and constraints. Generates concept notes and policy memoranda including legal basis, objectives, key provisions, financial/social/economic impact summaries, risk considerations, and source-grounded explanation. Aligned with BRD concept-builder requirements and MOMAH methodology Step 2.	Policy Analyst, Legal & Compliance Officer, Senior Leader
CAP-04	Checklist Compliance & Legal Conflict Detection	Digitizes MOMAH's unified 39-item legislative checklist and validates completeness, flags missing items, generates	Policy Analyst, Legal & Compliance

		checklist status, and produces reviewer reports for GRC, Legal Affairs, BI Center, Finance, and the Policy & Regulation Committee. Conflict detection scans the draft against Policy Registry and KSA legal corpus through Ma'rifa and Sanad. Aligned with MOMAH Legislative Development Methodology (March 2026), including Step 3 and the unified checklist.	Officer, GRC, Finance, BI Center
CAP-05	Benchmarking & Comparative Policy Analysis	Benchmarks draft and active policies against peer countries and global standards, including Singapore, Norway, UAE, OECD, IMF, and World Bank sources where available. Applies data freshness checks, jurisdiction metadata, normalization assumptions to be finalized in SDD, and produces benchmark tables, risk flags, gap maps, and best-practice recommendations. Aligned with BRD benchmarking requirements.	Policy Analyst, Senior Leader, Housing Economist
CAP-06	Policy Simulation & Scenario Modeling	Multi-variable simulation engine enabling what-if scenario testing across economic, demographic, supply, and market variables. Includes Black Swan stress testing, sensitivity analysis, ABM and Monte Carlo execution, confidence bands, predictive success score, affordability measures, fiscal NPV, and regional heatmaps before policy enactment.	Housing Economist, Policy Analyst, Senior Leader, Minister

CAP-07	Consultation Synthesis & Amendment Recommendation	Arabic-English NLP pipeline for public and inter-ministerial feedback ingestion, PDPL-aware redaction, topic clustering, theme weighting, sentiment scoring, representative comment extraction, clause-level amendment recommendation, and explicit analyst approval or rejection. Every recommended amendment must be traceable to supporting feedback and stored in the audit trail. Aligned with BRD consultation requirements and MOMAH methodology Step 6.	Policy Analyst, Stakeholder, Legal & Compliance Officer
CAP-08	Estitlaa Consultation Publishing Support	Prepares consultation publishing packages for the Estitlaa public consultation platform, including formatting, metadata generation, document package validation, consultation window planning, deadlines, and handoff/status tracking. The platform does not directly submit, publish, or legally enact consultation materials; actual submission or publication remains outside automated decisioning and follows MOMAH's official human-led workflow. Aligned with MOMAH methodology Step 5.	Policy Analyst, Legal & Compliance Officer, Stakeholder
CAP-09	HITL Governance, Strategic Waiver & Approval Workflow	Multi-stakeholder lifecycle orchestration with mandatory human-in-the-loop stage gates, role-based approvals, digital signature, escalation, strategic waiver workflow for fiscal or strategic exceptions, MoF alignment where required, and immutable audit trail. Aligned	All Personas, Minister, MoF Reviewer

		with BRD waiver and approval controls.	
CAP-10	Official Publication Package & Formal Archiving	Supports preparation, validation, and handoff tracking of final approved publication packages for Official Gazette and MOMAH website workflows, including final text, approval evidence, metadata, attachments, checklist records, package status, and timestamps. The platform does not directly submit, publish, or enact legislation; formal publication remains human-led and outside automated decisioning. The Institutional Memory Vault and DMS archive the final legislation and supporting material. Aligned with MOMAH methodology Steps 7-8.	Legal & Compliance Officer, GRC, Minister
CAP-11	Post-Implementation Quality Measurement & KPI Tracking	Tracks policy implementation quality, KPI/outcome performance, actual versus simulated outcomes, data-backed variance indicators, and periodic reports for GRC, Senior Leaders, and the Minister. It does not perform regulatory enforcement monitoring, execute compliance actions, or determine legal penalties. Detailed KPI formulas and thresholds will be finalized in the SDD and KPI methodology workshop, with source systems including GASTAT, SAMA, MoF, and Vision 2030 PMO feeds where applicable. Aligned with MOMAH methodology Step 9 and BRD UC-17.	Senior Leader, Minister, GRC, KPI Owner

CAP-12	Executive Decision Intelligence	Minister and Senior Leader dashboards showing portfolio health, Vision 2030 alignment, strategic risks, approved alert notifications, regional impact heatmaps, and internal policy-risk intelligence based on configured MOMAH data feeds. The platform does not provide real-time macroeconomic forecasting infrastructure. Aligned with BRD executive dashboard and real-time alerting use cases.	Minister, Senior Leader
CAP-13	Responsible AI, Auditability & Model Governance	High-risk AI governance controls covering explainability, citation verification, HITL approval, model monitoring, drift detection, kill-switch, access control, audit logging, annual independent audit of policy quality, benchmarking accuracy, model performance, and governance compliance. Aligned with BRD audit requirements and Section 8 governance controls.	Data Scientist / AI Unit, Business Owner, GRC

Capability Design Scope at HLSD Stage

The 13 capabilities in Table 12 are organised into two layers. CAP-01 to CAP-07 form the Core AI Capability Layer — the canonical policy intelligence foundation (CAP-01) and the six AI-augmented capabilities along the policy lifecycle main line: gap and opportunity analysis (CAP-02), AI-assisted drafting (CAP-03), regulatory conflict detection (CAP-04), peer-country benchmarking (CAP-05), multi-scenario simulation (CAP-06), and consultation synthesis (CAP-07). These seven capabilities embody the novel agentic AI patterns, high-risk AI controls, explainability requirements, and integration dependencies that demand HLSD-level definition to support Stage-Gate 1 review and subsequent SDD development. They are detailed in Sections 3.3.1 – 3.3.7 below.

CAP-08 to CAP-13 form the Integration, Governance, and Cross-Cutting Layer — capabilities that orchestrate, integrate, govern, and audit the work produced by the Core AI Capability Layer. They are scoped at HLSD level through Table 12 above and the cross-cutting frameworks already defined elsewhere in this document:

- **CAP-08 (Estitlaa Consultation Publishing Support)** — see Section 3.4.1 Step 5 and the Estitlaa integration entry in Section 5.3 (Table 27).
- **CAP-09 (HITL Governance, Strategic Waiver & Approval Workflow)** — see Section 6.4 (Human-in-the-Loop Design), Section 6.7 (Agentic AI Orchestration), and Section 8.4 (Strategic Waiver Workflow).
- **CAP-10 (Official Publication Package & Formal Archiving)** — see Section 3.4.1 Steps 7–8 and the Gazette / DMS integration entries in Section 5.3 (Table 27).
- **CAP-11 (Post-Implementation Quality Measurement & KPI Tracking)** — see Section 3.8.2 (KPI Scoring & Vision 2030 Alignment Framework) and Section 11.3 (Technical Metrics).
- **CAP-12 (Executive Decision Intelligence)** — see Section 3.8.3 (Notification & Alert Framework) and the Executive Intelligence Cockpit references in Section 3.2 and Section 5.1.
- **CAP-13 (Responsible AI, Auditability & Model Governance)** — see Section 6.3 (Explainability Approach), Section 6.6 (Model Monitoring), Section 6.8 (AI Evaluation Pipeline), and Section 8.6 (Annual Independent Audit).

3.3.1 Capability Design Detail: CAP-01 — Policy Registry

Policy Ingestion Pathways

The Policy Registry supports three ingestion pathways to accommodate all policy sources, from new drafts to legacy and archived policies:

- **Pathway A -- Manual Input (Primary):** Policy Creators initiate a new policy directly in the platform using structured forms. Minimum required fields before the policy can proceed to workflow: policy title and summary, policy category (housing / planning / finance / etc.), objectives and expected outcomes, initial assumptions, relevant attachments (legal references, background studies, data inputs), and assigned owner and department. The system validates completeness before routing to the next stage. Suitable for new proposals and early-stage drafts.
- **Pathway B -- AI-Assisted Document Parsing (Recommended for Legacy Policies):** The system ingests existing policy documents (PDF, Word) and uses the LLM Policy Drafting Agent to extract structured content: titles, articles and clauses, definitions, policy objectives, and key parameters. AI extraction targets ≥80% field accuracy (BRD acceptance criterion). Users review and validate extracted content before formal registration. Reduces manual effort for onboarding legacy and archived policies.
- **Pathway C -- System Integration (For Existing Repositories):** Direct integration with MOMAH's Document Management System (DMS) and authorized policy repositories enables automatic ingestion of approved or archived policies, ensuring the registry remains synchronized with the official DMS source of truth. Automatic linkage to simulation, legal checks, and approvals is established upon ingestion.

Policy Object Data Model

Each policy in the registry is stored as a structured, machine-readable Policy Object with four layers:

- **Core Metadata:** Policy ID, title, category (enum from MOMAH sector taxonomy: Housing Supply / Housing Finance / Urban Planning / Land Regulation / etc.), department owner, version number, status (draft / under review / approved / archived), and Vision 2030 pillar linkage.
- **Policy Content Structure:** Objectives; articles and clauses (structured text); definitions; policy levers (variables that directly influence simulation inputs, e.g., subsidy amount, eligibility threshold, fee rate); target groups (households / developers / municipalities / etc.); geographic scope (region codes or GeoJSON coverage); and financial parameters (subsidies, fees, incentives, eligibility rules).
- **Governance & Workflow Fields:** Creator, reviewer(s), legal reviewer, approver, required simulation level (Level 1 / 2 / 3), conflict detection results and resolution status, legal validation results (Ma'rifa and Sanad), override logs with written justification, and complete immutable audit trail with timestamps and user IDs.
- **Attachments & References:** Supporting documents (background studies, data inputs), legal references retrieved from Ma'rifa, historical precedents from Sanad, and DMS version history links.

Hybrid Metadata Generation

Not all metadata fields require manual entry. The system uses a hybrid generation approach: (1) **User-provided** — mandatory minimum fields only (title, objectives, mechanism, target group, key parameters); (2) **AI-assisted extraction** — the Policy Drafting Agent automatically extracts sector/sub-sector, mechanism type, constraints, and initial KPI linkages from policy text and uploaded documents; users review and adjust but do not start from scratch; (3) **System-derived** — geospatial mapping, economic triggers, and interdependency relationships are generated automatically by the platform (via conflict detection engine and knowledge graph); (4) **Integration-derived** — legal references from Ma'rifa, legal interpretations from Sanad, and version history from DMS are populated through system integrations. This approach ensures metadata completeness while minimising the manual data-entry burden on Policy Creators.

3.3.2 Capability Design Detail: CAP-02 — Gap & Opportunity Analysis Engine

Detection Logic -- 3-Stage Pipeline

The Gap Analysis Engine operates in three sequential stages:

Stage 1: Coverage Mapping

The engine queries the Knowledge Graph to retrieve all active policies within the user-selected scope (Target Sector, Vision 2030 Pillar, Sub-Sector, Time Horizon). For each policy, it maps declared objectives and policy mechanisms against the corresponding Vision 2030 KPI targets and trendlines retrieved from the PMO data feed.

Stage 2: Gap Scoring

For each Vision 2030 KPI within scope, the engine calculates a coverage gap as the delta between the KPI target trajectory and the coverage provided by active policies. Gaps are classified by severity:

- **Critical:** KPI trajectory is falling >10% below the Vision 2030 target -- triggers automatic Tier-2 Strategic Briefing notification.
- **Moderate:** KPI gap is 5-10% below target.
- **Watch:** KPI gap is <5% below target.

Each gap receives an **Opportunity Score (0-100)** computed from the KPI gap magnitude, sector priority weight, and peer-country benchmark distance. Scores are immutable per analysis run.

Stage 3: Intervention Recommendation

For each identified gap, the engine proposes at least one candidate policy intervention grounded in peer-country benchmarks (Singapore, Norway, UAE, OECD). Each intervention cites: the source benchmark, the applicable legal basis from the KSA legal corpus, and the estimated KPI contribution. Proposed interventions that conflict with existing active policies are automatically cross-linked to the Regulatory Conflict Engine (CAP-04) before presentation.

XAI Mandate

Every identified gap must be traceable to at least one Vision 2030 KPI and at least one peer-country benchmark precedent. Gaps lacking this traceability are blocked from the report. Critical gaps cannot be dismissed without a logged justification stored in the audit trail.

Data Quality Controls

- **Data Latency Guard:** If any benchmark dataset is older than 30 days, all affected gap outputs are watermarked "Low Confidence -- Benchmark Data Stale".
- **KPI Feed Guard:** If the Vision 2030 PMO trendline feed is unavailable, the analysis run is blocked and the Data Scientist is alerted.

Output Example

Gap Identified -- Affordable Housing Finance (Housing Sector / Quality of Life Pillar): No active MOMAH policy addresses developer financing constraints for projects below SAR 300,000 per unit. Current homeownership trajectory: 63% (2025) vs. target 70% (2030). Opportunity Score: 84/100 [Critical]. Benchmark: Singapore Enhanced CPF Housing Grant reduced equivalent financing gap by 18% within 36 months. Recommended intervention: targeted developer subsidy mechanism with fiscal neutrality constraint. Estimated KPI contribution: +1.2pp toward homeownership target.

Performance Requirement

The Gap Analysis Engine response-time target will be baselined during Phase 2 SDD and validated through performance testing; no BRD NFR-02 SLA is asserted for this non-simulation capability.

Scope Selection (UI)

Users define the analysis scope via: Target Sector (Housing / Municipal / Infrastructure / Environmental / Social / Smart Cities), Vision 2030 Pillar (multi-select), Sub-Sector (optional), Region (optional), and Time Horizon (12m / 24m / 60m). Peer-country benchmark selection is configurable per run.

3.3.3 Capability Design Detail: CAP-03 — Policy Concept Builder & AI Drafting Sandbox

Overview

The Policy Project Concept Builder is a pre-drafting validation component of the Drafting Engine. Before a Policy Analyst initiates the formal drafting workflow (UC-01), this module systematically validates the policy concept against sector strategies, national strategies, and the legal framework, then auto-generates an initial Policy Concept Memorandum. The component ensures that every policy entering the drafting pipeline possesses strategic traceability and legal compliance at the point of origin.

Input Matrix

The Analyst provides the following parameters through a structured input interface:

- **Policy Objective:** Free-text description of the desired policy outcome
- **Target Sector:** Housing / Municipal / Infrastructure / Environmental / Social / Smart Cities
- **Vision 2030 Pillar:** Housing / Quality of Life / Economic Opportunity / Fiscal Sustainability (multi-select)
- **Legal Basis:** Referenced Royal Decree, Ministerial Resolution, or existing regulation number
- **Target KPIs:** Expected Vision 2030 KPIs to be impacted (selected from the PMO data feed)
- **Benchmark Reference:** Optional — referenced peer-country policy precedent

Three-Layer Alignment Validation — Sequential Execution

Layer 1 — Sector Strategy Alignment

The system queries the Knowledge Graph to retrieve the target sector's current sector strategy and performs semantic matching between the Analyst's policy objective and the sector's strategic objectives. Output: Sector Alignment Score (0–100) and a deviation list. Each deviation is labeled: Covered strategic objective / Uncovered strategic objective / Contradictory wording against a strategic objective. If the Score falls below 60, the system blocks progression to Layer 2 and generates a remediation checklist requiring the Analyst to revise before resubmission.

Layer 2 — National Strategy (Vision 2030) Alignment

The system maps the policy concept against the Vision 2030 national target framework, verifying whether it supports a positive contribution to at least one Vision 2030 KPI ($\geq 15\%$ improvement target, consistent with the 3.8.2 Policy Assessment Matrix). Scoring logic reuses the 3.8.2 Vision 2030 Alignment Scorer (KPI Retrieval → Clause-Level Scoring → Composite Score), employing the same PMO data source and pillar weights. Output: National Alignment Score (0–100) and per-pillar contribution pre-assessment. If no Vision 2030 Pillar is associated, the system blocks validation and prompts the Analyst to complete the Pillar Tag (consistent with the 3.8.2 Pillar Tag Requirement).

Layer 3 — Legal Basis Validation

The system queries the Knowledge Graph legal corpus (including Ma'rifa legal reference system and Sanad legal advisory system integration) for the referenced legal basis:

- Checks whether the referenced regulation is currently in force (Active / Repealed / Amended)
- Cross-checks whether the policy concept contradicts or conflicts with existing regulations
- Output: **Legal Basis Status: Valid / Partial / Invalid** + conflict details (if any)
- If a conflict is detected, the system auto-links to CAP-04 (Regulatory Conflict Engine) for deep conflict analysis
- If Status is "Invalid," the system blocks memorandum generation and requires the Analyst to correct the legal reference

Auto-Generated Policy Memorandum

Upon passing all three validation layers, the system auto-generates a **Policy Concept Memorandum** containing the following sections:

Memorandum Section	Content	Data Source
Policy Concept Summary	Policy objective summary + target sector + Vision 2030 Pillar mapping	Analyst input
Alignment Validation Results	Sector / National / Legal three-layer validation results and scores	Concept Builder validation engine
Suggested Mechanism Types	Recommended policy mechanism types (subsidy, tax incentive, land grant, loan guarantee, etc.)	CAP-02 Gap Report opportunity recommendations + CAP-05 Benchmarking best practices
Preliminary KPI Contribution	Initial KPI contribution estimates based on the 3.8.2 scoring matrix	Vision 2030 PMO feed + GASTAT / SAMA / MoF
Risk & Conflict Pre-Check	Potential conflict warnings and risk assessment	CAP-04 Conflict Detection
Recommended Drafting Pathway	Suggested drafting workflow and next-step actions	Linked to UC-01 Drafting Workspace

Integration with Existing Capabilities

- **CAP-02 (Gap Analysis):** Memorandum Suggested Mechanism Types reference opportunity recommendations and benchmark precedents from the Gap Report

- **CAP-04 (Conflict Detection):** Layer 3 legal basis validation automatically triggers conflict pre-screening
- **CAP-05 (Benchmarking):** Recommended mechanism types cite peer-country best practices, localized through the KSA Constraint Filter
- **3.8.2 (KPI Scoring):** Preliminary KPI Contribution uses the same scoring dimensions, data sources, and thresholds

Governance Rules

- Sector Alignment Score or National Alignment Score < 60 → system blocks entry to the drafting stage and generates a remediation checklist
- Legal Basis Status "Invalid" → blocks memorandum generation, requires correction of legal reference
- All validation results, scores, and memorandum versions are stored in the Policy Registry audit log and are immutable
- Memorandum must receive explicit Analyst confirmation before conversion to a formal policy draft (HITL requirement)
- Memorandum versions and subsequent formal draft versions maintain a complete provenance chain in the Policy Registry

Output Example

Policy Concept Memorandum — Residential Land Use Incentive

Target Sector: Housing | Vision 2030 Pillar: Housing, Fiscal Sustainability | Legal Basis: Royal Decree M/15 (Land Distribution) — Valid

Sector Alignment: 82/100 — 2 of 4 Housing sector strategic objectives directly addressed; 1 partially addressed (affordable rental supply); 1 not covered (green building compliance) → remediation hint generated

National Alignment: 76/100 | Pillar Breakdown — Housing: 88/100 | Fiscal Sustainability: 64/100

Suggested Mechanism: Land Value Capture + Zoning Density Bonus (referenced from Singapore Land Acquisition Act and Norway Husbanken Municipal Land Bank model, adapted via KSA Constraint Filter — Sharia-compliant structure confirmed; no SDAIA data residency flags)

Preliminary KPI Contribution: Homeownership Rate +0.8pp (36-month projection) | Affordability Index: PTI impact projected -0.3 (target PTI <5x) | Fiscal Impact: 15Y NPV cost SAR 1.4B (within approved envelope)

Conflict Pre-Check: 1 potential overlap with Municipal Regulation MR-2024-07 (zoning authority delegation) — linked to CAP-04 for detailed analysis

Recommended Pathway: Proceed to UC-01 Drafting Workspace with this Concept Memorandum as baseline document

3.3.4 Capability Design Detail: CAP-04 — Conflict Detection

Conflict Detection Inputs

The Conflict Detection module operates on four input layers simultaneously to ensure detection spans the full regulatory and policy landscape:

- **New or Modified Policy (Primary Input):** Full text of the policy, structured Policy Object (articles, clauses, definitions, parameters), and policy metadata (category, scope, target groups).
- **Full Policy Repository:** All active MOMAH policies, archived but still-relevant policies, and cross-sector policies that may create indirect conflicts. This ensures detection extends beyond obvious duplicates to subtler regulatory overlaps.
- **Legal Knowledge Bases:** Ma'rifa (regulatory hierarchy, legal references, existing Saudi laws and Royal Decrees) and Sanad (legal interpretations and precedents) to validate whether a detected conflict is legally meaningful, not just textually similar.
- **AI-Extracted Knowledge Graph:** A continuously maintained graph of policy relationships, dependencies, contradictions, overlaps, and term definitions. Enables deeper, context-aware conflict detection beyond direct text matching and supports cross-sector ripple-effect analysis.

Conflict Report Output Format

The system generates a structured, actionable Conflict Report within 30 seconds. Each report contains four components:

- **Conflict Summary:** Top-level status: No Conflict / Potential Conflict / Confirmed Conflict.
- **Detailed Conflict Entry (one per conflict detected):** (a) The exact clause or article in the new policy triggering the flag; (b) the conflicting clause in the existing policy or law; (c) conflict type — contradiction / duplication / overlap / hierarchy violation; (d) severity level — Low / Medium / High / Critical; (e) legal relevance assessment — whether the conflict is legally meaningful, validated via Ma'rifa and Sanad cross-reference.
- **Recommendations:** Suggested clause modifications, alternative wording, identification of other policies that may need alignment, and escalation recommendation (Legal Reviewer / Director) where required.
- **Traceability:** Links to source documents, specific legal article references, and a confidence score for each conflict flag. Required for XAI compliance (Section 6.3 Explainability Approach).

UI Presentation of Conflict Results

- **Highlighted Text in Policy Editor:** Conflicting clauses highlighted in red (Critical/High severity) or amber (Medium/Low); hover-over tooltips show conflict explanation; inline links navigate to the conflicting policy or legal article.
- **Structured Conflict List Panel:** All detected conflicts listed with severity, source policy or legal reference, and recommended actions; users click each item to navigate directly to the relevant clause.
- **Visual Knowledge Graph View (optional):** Nodes represent policies, articles, and legal references; edges show conflicts, overlaps, or dependencies; severity indicators applied

visually. Enables Legal Officers to understand the broader regulatory impact landscape before resolving flags.

- **Exportable Conflict Report:** PDF or structured export of all conflicts, their resolutions, and traceability references. Included in the approval package and retained in the immutable audit trail.

3.3.5 Capability Design Detail: CAP-05 — Benchmarking & Comparative Policy Analysis Engine

Comparison Methodology -- 3-Layer Architecture

The Benchmarking Engine processes every comparison run in three sequential layers: Data Ingestion, Normalization, and Comparison & Scoring. This layered approach ensures that raw international data is structurally adjusted for KSA context before any recommendation is surfaced to the analyst.

Layer 1: Data Ingestion & Source Repository

The engine maintains a curated benchmark repository drawing from the following authoritative sources:

- **OECD Housing and Urban Policy Database** -- housing affordability, subsidy structures, and land-use policies across member states.
- **IMF Government Finance Statistics** -- public expenditure on housing and fiscal commitment benchmarks.
- **World Bank Affordable Housing Indicators** -- homeownership rates, mortgage penetration, and supply metrics.
- **Peer-country policy databases:** Singapore (HDB, Enhanced CPF Housing Grant), Norway (Husbanken State Housing Bank), UAE (Affordable Housing Programme), GCC peer policies via bilateral data-sharing agreements where available.

Each dataset carries a freshness timestamp, jurisdiction metadata (legal system type, income classification, population density band), and a data confidence score used downstream in the normalization layer.

Layer 2: Normalization

Raw benchmark values are normalized across four structural dimensions before comparison, to account for the differences between KSA and peer economies:

- **Economic Scale (GDP per capita):** Raw policy expenditure and subsidy values are re-expressed as a percentage of per-capita GDP (KSA ~USD 32,000 vs. Singapore ~USD 65,000 vs. Norway ~USD 89,000), enabling like-for-like fiscal comparison independent of absolute currency values.
- **Housing Market Maturity:** Benchmark mechanisms are tagged by the homeownership baseline of the source country and adjusted for market stage. Policies designed for mature markets (homeownership >70%) are flagged with an applicability note when applied to KSA, which is targeting 70% by 2030.

- **Population Density:** Urban policy instruments (e.g., density bonuses, transit-oriented development incentives) are scaled to KSA Amana-level density profiles using Balady municipal data, to avoid recommending mechanisms suited to high-density cities in low-density Saudi contexts.
- **Fiscal Capacity:** Government housing expenditure benchmarks are normalized as a percentage of total government expenditure, to account for differences in fiscal structure between oil-revenue-dependent economies and tax-based economies.

Normalization weights and thresholds will be calibrated with MOMAH KPI owners and the Housing Economist in the SDD and KPI methodology workshop.

Layer 3: Comparison & Scoring

The engine compares MOMAH draft or active policy provisions against normalized peer benchmarks across four dimensions, producing a structured coverage score (0-100) per dimension:

- **Coverage Scope:** Which sectors and sub-sectors are addressed by the MOMAH policy versus peer-country equivalents, flagging unaddressed areas as coverage gaps.
- **Mechanism Type:** Policy mechanism type (subsidy, regulation, tax incentive, land grant, loan guarantee) is mapped against an international mechanism taxonomy to identify structural differences and adaptability.
- **Fiscal Commitment:** Policy budget commitment as a percentage of GDP versus peer-country norms for equivalent policy objectives.
- **KPI Contribution:** Expected contribution to homeownership, affordability, or housing supply targets, compared against realized outcomes from peer-country implementations.

KSA Contextualization -- Constraint Filter

Before any benchmark recommendation is surfaced, the engine applies a KSA Constraint Filter that automatically flags mechanisms requiring structural adaptation for the Saudi context. The four constraint categories are:

- **Sharia-Compliant Financing:** Mechanisms involving interest-bearing instruments are flagged and mapped to Sharia-compliant equivalents (Murabaha, Ijara, Musharaka) before being presented as recommendations.
- **Land Ownership Regulations:** Recommendations involving freehold land transfer or foreign ownership are cross-checked against KSA land ownership regulations and waqf land constraints.
- **SDAIA Sovereign Data Residency:** Any benchmark mechanism requiring data processing or model inference outside KSA is flagged as non-compliant with NFR-06 (Sovereign Compliance Layer) and excluded unless a KSA-sovereign equivalent exists.
- **Labor Market & Saudization:** Construction cost assumptions embedded in peer-country fiscal benchmarks are adjusted for KSA labor market conditions and Saudization requirements affecting material and labor cost structures.

Flagged adaptations are displayed inline alongside each recommendation so the Policy Analyst can review required modifications before initiating a draft.

Output Structure

Three output types are generated per benchmarking run:

- **Gap Table:** Structured table showing MOMAH policy provision versus best-in-class benchmark for each evaluated dimension, with coverage delta, normalized benchmark value, confidence score, and data source citation.
- **Risk Flags:** Policies that fall more than 25% below the normalized benchmark on any scored dimension are automatically flagged for analyst attention, with a severity rating (Critical / Moderate / Watch) consistent with the Gap Analysis Engine (CAP-02).
- **Best-Practice Recommendations:** Specific policy mechanisms from peer countries, adapted through the KSA Constraint Filter, each citing the source benchmark, applicable KSA legal basis, estimated KPI contribution, and implementation lead time.

Output Example

Benchmark Comparison -- Affordable Housing Delivery Mechanism (KSA Sakani vs. Singapore HDB): Singapore Enhanced CPF Housing Grant delivers SGD 80,000 direct buyer support (~8.5% of median unit price). Normalized to KSA GDP per capita and housing price index: equivalent support level is SAR 65,000-90,000. Current MOMAH Sakani scheme: SAR 100,000 for eligible citizens -- Fiscal Coverage Score 96/100 [No Gap]. Sub-gap identified: Singapore HDB reaches 78% of eligible applicants within 12 months; MOMAH Sakani reaches 43% of eligible citizens in Year 1 -- Delivery Mechanism Gap Score 44/100 [Critical]. KSA Constraint Filter: mechanism is Sharia-compliant (grant structure, no interest component). Recommendation: streamline Sakani application workflow to reduce approval cycle from 45 days to under 14 days, aligned with HDB digital-first application model. Estimated KPI contribution: +3pp eligible citizen coverage within 18 months.

Data Freshness Control

- **Data Latency Guard:** Benchmark datasets older than 30 days are automatically watermarked 'Low Confidence -- Benchmark Data Stale' on all outputs. Peer-country data is refreshed via scheduled pulls from OECD and IMF APIs, or manual upload where direct API access is unavailable.
- **Insufficient Data Handling:** Where a required peer-country dataset is unavailable or below the minimum confidence threshold, the affected benchmark dimension is excluded from scoring and flagged as 'Insufficient Benchmark Data' in the Gap Table.

Performance Requirement

Benchmarking response-time targets for a single policy against the peer-country dataset will be confirmed during Phase 2 SDD and validated through performance testing; no BRD NFR-02 SLA is asserted for this non-simulation capability.

3.3.6 Capability Design Detail: CAP-06 — Policy Simulation & Scenario Modeling

Simulation Input & Output Design

User-Provided Inputs

The simulation engine requires four categories of user-defined input:

- **Policy Variables (Core Inputs):** The specific levers or parameters defined in the policy that directly influence simulation outcomes: eligibility criteria, subsidy amounts, fees/taxes/incentives, geographic scope, target population, and implementation timelines.
- **Scenario Assumptions:** Users adjust assumptions to test different futures. The system provides default values based on GASTAT, SAMA, and MoF data; users can modify: economic assumptions (inflation, interest rates, income growth), demographic assumptions (household formation, migration), market assumptions (housing supply, construction costs), and behavioral assumptions (take-up rates, compliance rates).
- **KPIs and Outcome Metrics to Evaluate:** Users select which output metrics matter for this simulation: housing affordability (PTI), fiscal impact (15Y NPV), supply/demand balance, social equity impact, administrative cost, and time to impact. This ensures the simulation focuses on outcomes relevant to the specific policy under review.
- **Scenario Definitions:** Users configure at least one baseline scenario and optional alternative scenarios (Optimistic / Pessimistic / Black Swan). Multiple scenarios can be defined in parallel for side-by-side comparison.

System-Generated Outputs

- **Quantitative Metrics:** Fiscal impact (cost, savings, revenue), beneficiary counts, housing supply/demand changes, market price effects, administrative workload, and risk indicators.
- **Probabilistic Outcomes:** Probability of achieving each policy objective (Predictive Success Score; threshold for "Failed" verdict: <80% per BRD FR-11, see Section 8.4 Strategic Waiver Workflow), Monte Carlo confidence intervals, and sensitivity rankings of input variables.
- **Visualizations:** Line charts, bar charts, waterfall charts, multi-scenario comparison graphs, regional heatmaps, and KPI dashboards.
- **AI-Generated Narrative Summary:** Human-readable explanation covering key findings, risks and sensitivities, main drivers of impact, and improvement recommendations for policy design.
- **Exportable Simulation Report:** Full simulation results, assumptions used, scenario comparisons, and legal/conflict implications where relevant. Included in the approval package and logged for audit and traceability.

User Interaction Patterns

- **Parameter Control Panel:** Sliders for numeric values (subsidy amounts, thresholds), dropdowns for categorical choices (target group, region), toggles for enabling/disabling policy components.
- **Multi-Scenario Comparison:** Side-by-side view of baseline vs. alternative scenarios; save named scenario sets; batch simulation execution for sensitivity analysis.
- **Real-Time Updates:** Level 1 (Quick Estimate) simulations update immediately; Level 2/3 runs show progress indicators. Users can continue reviewing assumptions while a long-running simulation executes.

- **Drill-Down Analytics:** Click on any KPI to inspect the underlying driver variables, their weighting, and the data sources behind the result. Charts can be expanded to reveal the assumptions underlying each output figure.
- **Workflow Integration:** Simulation results are automatically forwarded to the Legal Reviewer workflow and HITL Approval stage on completion, with legal or conflict implications linked back to CAP-04 where relevant — no manual handoff required.

3.3.7 Capability Design Detail: CAP-07 — Consultation Synthesis & Amendment Recommendation

NLP Pipeline Architecture -- 4-Stage Processing

The Consultation Synthesis Module processes incoming feedback through four sequential stages, from raw ingestion to analyst-ready amendment proposals. All processing occurs within the KSA sovereign boundary per NFR-06.

Stage 1: Ingestion & Compliance Pre-processing

Feedback is ingested from three channels defined in BRD UC-05:

- Public comments submitted via the Absher public services portal during the open consultation window.
- Email submissions collected from citizens and industry bodies, routed through a dedicated consultation inbox.
- Inter-ministerial responses received via the Yesser Government Service Bus (GSB) from SAMA, MoF, MISA, and other regulatory counterparts.

On ingestion, two compliance controls are applied before any AI processing begins:

- **PDPL-Aware Redaction:** The PDPL Redaction Engine scans every submission for personally identifiable information (full name, national ID, phone number, address) and auto-redacts prior to analysis. If the redaction engine is unavailable or returns an error, the entire ingestion pipeline is paused, accumulated feedback is quarantined, and the Data Scientist is alerted (BRD A1 scenario).
- **Language Detection & Routing:** Each submission is classified as Arabic, English, or mixed-language. Arabic submissions are routed through the Arabic NLP path; English submissions through the English path; mixed submissions are split at the sentence level and processed through both paths before re-merging.

Stage 2: Theme Clustering

The AI Synthesis Engine applies semantic clustering to identify recurring themes across the full feedback corpus:

- **Embedding:** Each submission is encoded into a semantic embedding vector using the same Sovereign LLM deployed for policy drafting (CAP-03), ensuring consistent semantic representation across Arabic and English inputs.
- **Clustering:** Submissions are grouped into thematic clusters based on semantic proximity. The system identifies themes such as Affordability, Developer Incentives, Zoning,

Environmental Impact, and Infrastructure -- the exact taxonomy is derived dynamically from the corpus rather than from a fixed list, ensuring coverage of novel issues.

- **Theme Weighting:** Each theme is assigned a weight (percentage of total feedback volume), a list of representative verbatim excerpts (bilingual), and a confidence score reflecting cluster cohesion.

Material Threshold Rule (BRD UC-05 A2): If any single theme represents more than 25% of total feedback volume (to be defined in SDD), it is automatically flagged as "**Material**" and a Tier-2 notification is escalated to Senior Leaders. Material themes cannot be deferred -- the Analyst must record a disposition decision before the policy may advance.

Stage 3: Sentiment Scoring

Each submission and each thematic cluster receives independent sentiment scoring:

- **Submission-Level Scoring:** Every submission is classified as Positive, Neutral, or Negative, with an intensity score (0.0-1.0 float) reflecting the strength of the expressed sentiment. Arabic sentiment scoring uses an Arabic-specific lexicon and context model to handle dialectal variation.
- **Theme-Level Aggregation:** Each theme outputs an aggregated sentiment distribution showing the percentage breakdown across the three categories, enabling the Analyst to assess both the topic and the tone at a glance. Example: Affordability theme -- 72% Negative (avg intensity 0.81) / 20% Neutral / 8% Positive.
- **High-Intensity Flag:** Themes where Negative sentiment intensity exceeds 0.75 on average are flagged for priority review, independent of the Material Threshold rule.

Stage 4: Amendment Recommendation Generation

For each identified theme, the system generates a structured amendment proposal linked to the specific clause in the policy draft most relevant to the theme:

- **Clause-Level Linkage:** The system maps each theme to the draft clause(s) most semantically related to the feedback content, using the Knowledge Graph policy structure.
- **Amendment Drafting:** A candidate amendment is drafted by the LLM, grounded in the theme's representative excerpts and constrained to the legal basis and policy objectives declared in the draft. The amendment includes: proposed change text, a diff preview against the current clause, and citations to the supporting feedback excerpts.
- **No Auto-Apply (BRD Mandate):** No amendment may be auto-applied to the draft. Each proposal requires explicit Analyst decision (BRD Business Rule).

HITL Review Workflow

The Analyst reviews each amendment proposal through a structured decision interface aligned with BRD UC-05 Steps 4-5:

- **Accept:** The proposed amendment is applied to the draft and the resulting version is submitted for the next lifecycle stage.
- **Modify:** Opens a clause-level diff editor allowing the Analyst to adjust the wording before applying. The edited version is stored with a "Analyst-Modified" flag in the audit trail.

- **Reject:** The proposal is discarded. A rejection rationale field is mandatory for Material-threshold themes, and the rationale is stored in the audit trail.

Once all proposals have been actioned, the Analyst triggers Generate New Version, which produces a new versioned draft in the Policy Registry carrying a full provenance record linking every change to its source feedback submissions.

Compliance Controls

- **PDPL & NCA Compliance:** All feedback data is anonymized before analysis and stored only within the KSA sovereign boundary per PDPL and NFR-06. The PDPL Redaction Status badge is always displayed on the Analyst interface; ingestion is blocked if the status is not confirmed active.
- **Citation Mandate:** Every amendment proposal must cite at least one supporting verbatim excerpt from the ingested feedback. Proposals that cannot be grounded in source feedback are suppressed and flagged as ungrounded.
- **Audit Trail:** Every Accept, Modify, and Reject decision -- including rejection rationales for Material themes -- is written to an immutable audit log entry linked to the policy version and the consultation run.

Output Example

Theme: Housing Affordability | Feedback volume: 847 submissions (31% of total) [MATERIAL] | Sentiment: 72% Negative (avg intensity 0.81) / 20% Neutral / 8% Positive

Representative excerpt: "Monthly mortgage repayments exceed 40% of household income for median earners; recommend reducing the minimum down payment from 20% to 10% for first-time buyers below a defined income threshold."

Amendment Proposal: Insert new sub-clause in Section 4.2 (Mortgage Conditions): reduce the minimum down payment requirement for first-time buyers with annual household income below SAR 180,000 from 20% to 10%, subject to lender risk assessment and MoF fiscal envelope approval. | Linked clause: Section 4.2 | Diff preview available | Grounded in 312 supporting submissions | Status: Awaiting Analyst decision.

Business Rules

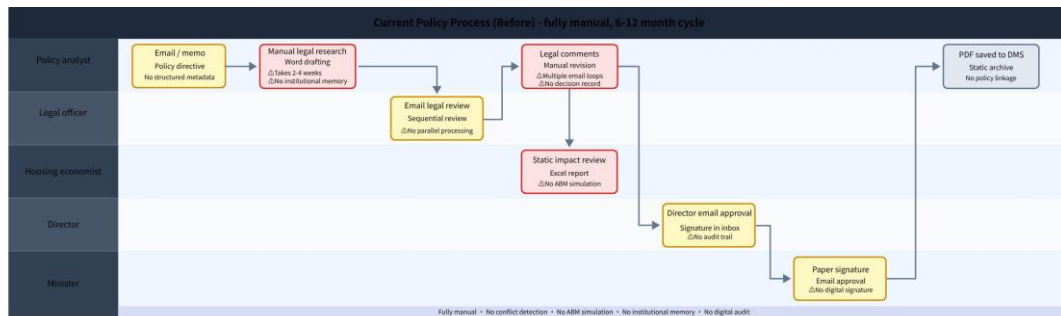
- All ingested feedback must pass PDPL and NCA anonymization checks before entering the analysis pipeline (BRD Business Rule 1).
- Every amendment proposal must cite supporting feedback excerpts; ungrounded proposals are suppressed (BRD Business Rule 2).
- No amendment may be auto-applied to any draft version; every proposal requires explicit Analyst approval (BRD Business Rule 3).
- Material themes (>25% of feedback volume) must receive an Analyst disposition decision before the policy may advance to the next lifecycle stage (BRD UC-05 A2).

Performance Requirement

The target time for ingestion, PDPL redaction, clustering, sentiment scoring, and amendment generation will be baselined during Phase 2 SDD based on consultation dataset size, redaction complexity, and NLP workload, then validated through performance testing.

3.4 Process / Workflow

Current Policy Development Process



Current State — Policy Development Without the Policy AI Lab (Before)

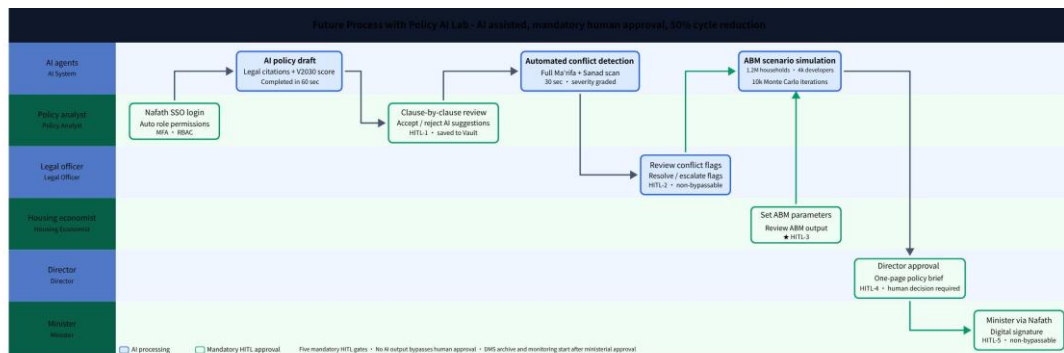
MOMAH's current policy lifecycle is entirely manual and document-centric. There are no automated systems, registries, or simulation tools in use. The process follows these steps:

- 1. Policy Initiation (Manual):** A Policy Analyst receives a directive from a Director or the Minister's Office, typically via email or physical memo. Policy context, relevant legislation, and prior precedents are located manually by the Analyst through personal knowledge, hard-copy archives, or informal queries to colleagues.
- 2. Research and Drafting (Manual):** The Analyst researches comparable policies and legal references by browsing government websites, reviewing physical law books, or consulting with the Legal Department directly. Policy drafts are written in Microsoft Word from scratch, with no structured template or institutional memory to draw from. This stage typically takes 2–4 weeks per policy.
- 3. Internal Review (Manual, Sequential):** The Word document is circulated via email to Legal Officers, Department Heads, and other relevant departments for review. Comments are collected via email reply threads or tracked-change annotations on Word files. Multiple review cycles are sequential (one reviewer at a time), with no parallel processing. Conflicts between reviewers are resolved in ad-hoc meetings with no formal documentation of decisions.
- 4. Stakeholder Consultation (Manual, Ad Hoc):** Where required, public or inter-ministerial consultation is conducted via email distribution, printed questionnaires, or in-person workshops. Feedback is manually collated and summarized with no structured synthesis tool. This stage can take 4–8 weeks.
- 5. Impact Assessment (Manual, Limited):** Fiscal and social impact assessments are produced as static Word or Excel reports by internal economists or external consultants. Scenario analysis is limited; the primary method for testing a policy's real-world impact is to launch it in a small geographic pilot area — a slow, costly, and effectively irreversible validation method.
- 6. Approval Workflow (Manual, Paper/Email):** The drafted policy is submitted for Director approval via email, then escalated to the Minister's Office. Sign-off is given via email confirmation or physical signature. There is no formal digital audit trail; approval records are maintained in individual email inboxes or physical files.

7. Policy Storage and Archiving (Fragmented): Approved policies are stored in a Document Management System (DMS) as static PDF/Word files with limited metadata. There is no version control, no cross-policy linkage, and no structured mechanism to identify conflicts with existing policies. Institutional memory resides in individuals, not systems.

Key pain points of the current state: No simulation capability; policy conflict detection is entirely manual and dependent on individual expertise; no institutional memory or cross-policy knowledge graph; policy development cycles of 6–12 months; no vision 2030 alignment scoring; no structured audit trail of AI or data-informed decisions (as no AI tools are currently used).

Future Policy Development Process



Future State — Policy Development With the Policy AI Lab (After)

The Policy AI Lab transforms the policy lifecycle from a sequential, document-centric process into a governed, AI-augmented, parallel workflow with mandatory human oversight at every critical decision point. Key changes are highlighted below, mapped to the capabilities in Section 3.3 and the user personas in Section 3.2.

1. Policy Initiation (CAP-01, CAP-02 — Policy Analyst): Analyst authenticates via Nafath SSO; lands on a role-specific dashboard showing active policies, notifications, and action items. Opens the Policy Registry and creates a new policy record with structured metadata: name, target sector, Vision 2030 pillar(s), budget envelope, and legal constraints. The system validates uniqueness against the existing Policy Registry and checks for naming conflicts inline. *Change from current state: Structured initiation with metadata enforcement replaces informal email directive; institutional memory immediately available.*

1. AI-Assisted Drafting (CAP-03 — Policy Analyst + LLM): Analyst selects a grounding mode (Precedent-heavy / Balanced / Novel) and clicks Generate Draft. The Policy Drafting Agent queries the Knowledge Graph for relevant Royal Decrees, historical precedents, and Vision 2030 KPI linkages, then generates a structured draft within 60 seconds. The draft includes a Vision 2030 Alignment Score (0–100), an Initial Risk Score, and a clause-by-clause Explainability Report citing all data sources and legal references. Analyst reviews each clause, accepts or rejects AI suggestions, and saves the version to the Institutional Memory Vault. *Change: Days-to-weeks of manual research compressed to 60-second AI-grounded draft generation; full explainability and source attribution mandatory.*

2. Parallel Conflict Detection (CAP-01, CAP-04 — Conflict Detection Agent + Legal Officer): Upon saving, the Conflict Detection Agent automatically scans the draft against the full

Policy Registry and KSA legal corpus (Ma'rifa, Sanad) in the background. A Conflict Report is generated within 30 seconds, categorized by severity (Critical/High/Medium/Low). Legal Officer reviews and resolves each flag: selecting Resolved, Requires Amendment, or Dismissed (Critical dismissals require written justification). Cross-ministry conflicts generate a cross-sector Ripple Effect report and notify affected ministries via Yesser GSB. *Change: Automated parallel conflict detection replaces sequential manual legal review; 30-second scan vs. weeks of manual analysis.*

1. Multi-Scenario Simulation (CAP-06 — Housing Economist + ABM): Once all Critical conflicts are resolved, the workflow evaluates whether the policy meets the SDD-confirmed simulation trigger conditions defined in Section 6.4.1. Where a trigger applies, the Housing Economist opens the Simulation Lab and configures the variable matrix: Macro-Economic (SIBOR, CPI), Demographic (household formation, urbanization), Supply (land tax, developer incentives), and Market (LTV ratio, rental yield) variables. Scenario definitions may include Baseline, Optimistic, Pessimistic, and Black Swan, subject to the policy type and approved simulation scope. The ABM engine runs scenario simulations using MOMAH-confirmed agent categories (households/residents, developers, banks and financial institutions, government/regulators, municipalities/regional authorities, investors, and market-environment variables). Scale, iteration counts, and Level-2/Level-3 runtime targets are subject to SDD data assessment, SDAIA platform benchmarking, and MOMAH approval. Level-1 is treated as a quick-estimate or simplified standard scenario aligned to the BRD 60-second target. Outputs include: Predictive Success Score, Homeownership impact, Price-to-Income ratio, BRD-aligned fiscal NPV, and a regional heatmap where applicable. If Monte Carlo failure rate exceeds 30%, the policy is labelled Unstable and blocked. Policies that do not meet simulation-trigger criteria must still complete required conflict, legal, and governance checks. *Change: Replaces the current approach of piloting a policy in a small geographic area; provides simulation-based impact testing where the agreed trigger rules require it before real-world enactment.*

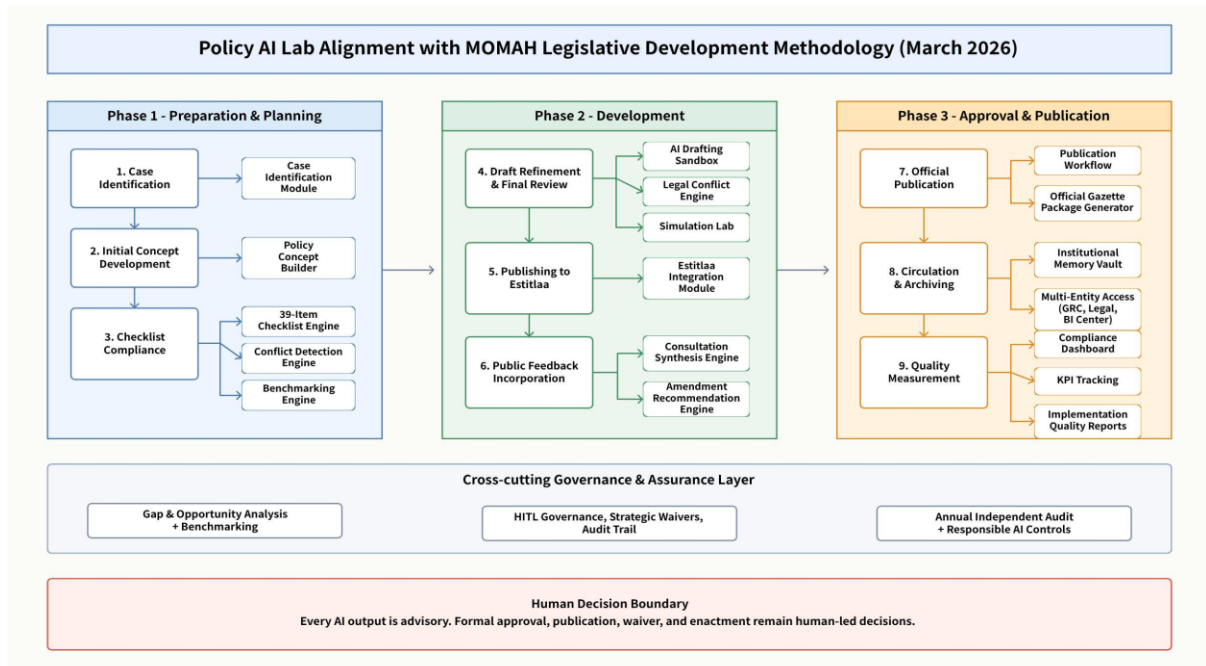
1. Tiered HITL Approval Workflow (CAP-09 — Policy Analyst, Director, Minister): Required lifecycle outputs route to the HITL Review stage, including simulation outputs where simulation trigger conditions apply. The Policy Analyst reviews and submits for Director approval (Tier 2 briefing: one-page policy summary with risk flags and scenario comparison). Director reviews and approves or requests revision. On Director approval, the policy routes to the Ministerial Executive Intelligence Cockpit for final decision. The Minister reviews the Decision Summary, Regional Impact Heatmap, and scenario comparison; selects Approve, Reject, Request Revision, or Request Further Simulation; and digitally signs via Nafath. All decisions are logged immutably. *Change: Structured, documented, multi-stage approval with digital signatures and immutable audit trail replaces email-based sign-off.*

1. Publication Handoff, Archiving and KPI Tracking (CAP-10, CAP-11): On Ministerial approval, policy status changes to Enacted; DMS automatically archives all policy documents, impact briefs, and decision records. The Risk Escalation Agent tracks configured KPI/outcome indicators against the simulated baseline where approved data feeds are available; threshold breaches trigger tiered review alerts (Analyst, Director, or Minister) to support human-led policy assessment. It does not perform enforcement monitoring or automated regulatory action. *Change: Post-launch monitoring replaces reactive response to policy failures; institutional memory preserved for future policy design.*

Summary of workflow transformation:

Dimension	Before (Current State)	After (Policy AI Lab)
Policy draft generation time	2–4 weeks (manual research)	60 seconds (AI-grounded draft)
Conflict detection	Manual, weeks, dependent on individual expertise	Automated, 30 seconds, full legal corpus coverage
Impact simulation	Geographic pilot only (slow, costly, irreversible)	Trigger-based multi-scenario ABM simulation across 13 regions where required before enactment
Policy development cycle	6–12 months	Target: under 6 months (50% reduction by Year 2)
Institutional memory	Fragmented in individual email inboxes and physical files	Centralized Institutional Memory Vault with full decision traceability
Approval audit trail	Email threads and physical signatures	Immutable digital audit log with Nafath-signed decisions
Vision 2030 alignment scoring	None (informal judgement)	Automated 0–100 score per policy draft, clause-level breakdown

Policy AI Lab Support for MOMAH Legislative Development Methodology



3.4.1 MOMAH Methodology Alignment

Methodology coverage statement. Policy AI Lab supports the official MOMAH legislative lifecycle across three phases and nine steps: Preparation & Planning, Development, and Approval & Publication. The mapping below translates each official step into a platform capability, design basis, and required human control point.

Step	Official Methodology Step	Policy AI Lab Support	Design Basis	Human Control
1	Identify cases requiring new or updated legislation	Case Identification Module and Gap & Opportunity Analysis register triggers, gaps, evidence, affected sectors, Vision 2030 KPI links, and responsible owners.	MOMAH methodology Step 1; BRD UC-16	Policy owner validates trigger and scope.
2	Develop initial legislative concept	Policy Concept Builder captures rationale, strategic alignment, legal basis, stakeholders, expected impacts, and generates an initial concept note.	MOMAH methodology Step 2; BRD concept-builder requirements	Analyst reviews and edits concept note.

3	Review draft and ensure compliance with the unified checklist	39-Item Checklist Compliance Engine validates completeness, flags missing items, and generates reviewer reports for GRC, Legal Affairs, BI Center, Finance, and Policy & Regulation Committee.	MOMAH methodology 39-item checklist	Responsible reviewers approve, reject, or request completion.
4	Update legislation and submit for final review	AI Drafting Sandbox, Conflict Detection Engine, Benchmarking Engine, and Simulation Lab support structured refinement before review.	BRD drafting, benchmarking, conflict detection, simulation use cases	Analyst, Legal Officer, and Director review outputs.
5	Prepare draft consultation package for Estitlaa when required	Estitlaa Consultation Publishing Support prepares and validates the consultation package, metadata, consultation window, and handoff/status tracking.	MOMAH methodology Step 5	Actual platform submission is performed through MOMAH's authorized official workflow outside automated decisioning.
6	Update draft based on public feedback and submit for approval	Consultation Synthesis Engine imports feedback, clusters themes, scores sentiment, recommends amendments, and traces incorporated or rejected feedback.	MOMAH methodology Step 6; BRD consultation requirements	Analyst explicitly accepts, edits, or rejects each recommendation.
7	Prepare publication package for Official Gazette and MOMAH website	Official Publication Workflow Support prepares final publication packages, approval evidence, metadata, attachments, package	MOMAH methodology Step 7	Final submission, publication, and enactment remain formally approved and human-led outside

		status, and timestamps for human-led handoff.		automated decisioning.
8	Circulate and archive final legislation	Institutional Memory Vault and DMS archive final approved texts, approvals, supporting documents, checklist outputs, consultation records, and decision logs.	MOMAH methodology Step 8	Legal/GRC confirms archive completeness and access rights.
9	Implement publication plan and measure quality	Post-Implementation Quality Measurement tracks KPI/outcome performance, actual versus simulated outcomes, implementation reports, and variance alerts. It does not perform regulatory enforcement monitoring.	MOMAH methodology Step 9; BRD success metrics and UC-17	KPI Owner and GRC review periodic reports.

Cross-cutting governance. The lifecycle is supported by multi-entity access and workflow responsibilities for GRC, Legal Affairs, BI Center, Finance, Digital Transformation, relevant business units, and executive approvers. Every AI-generated recommendation must include source references, reviewer actions, and immutable audit records.

KPI scoring note. At HLSD level, Vision 2030 alignment scoring is defined as a 0-100 composite score with pillar breakdown, clause-level contribution, data freshness checks, and audit logs. Detailed formulas, weights, and pass/fail thresholds will be finalized in the Phase 2 SDD and KPI methodology workshop with MOMAH KPI owners.

3.5 User Journey

ID	User Journey	Summary	Steps
J-01	Policy Analyst: Draft a New Policy	A Policy Analyst initiates a new housing policy. The AI generates a structured draft grounded in Saudi Law and Vision 2030 KPIs. The Analyst reviews clause-by-clause and	1. Analyst authenticates via Nafath SSO; role-based landing page presented. 2. Opens Policy Registry → 'Create New Policy'. 3. Completes required fields: Policy Name, Target Sector, Sub-Sector, Primary

		saves the draft for conflict review.	Objective, Vision 2030 Pillar(s), Budget Envelope, optional constraints. 4. Selects Grounding Mode and clicks 'Generate Draft'. 5. AI Policy Drafting Agent queries Knowledge Graph for Royal Decrees, precedents, and KPIs (≤60 sec). 6. System presents Draft Preview with Explainability Report, Vision 2030 Alignment Score (0–100), and Initial Risk Score. 7. Analyst reviews and edits individual clauses; accepts or rejects AI suggestions. 8. Clicks 'Save to Vault' — stored as v0.1; status set to 'Draft'. System auto-routes to Conflict Detection. AI: Steps 5–6 (LLM + KG retrieval). HITL: Steps 7–8 mandatory.
J-02	Housing Economist: Run a Multi-Scenario Policy Simulation	Where SDD-confirmed simulation trigger conditions apply, a Housing Economist runs ABM simulations on an approved draft to forecast fiscal, housing supply, and affordability impacts across multiple scenarios before ministerial review.	1. Economist opens Simulation Lab; selects a policy draft with status 'Simulation Required' or 'Ready for Simulation'. 2. Configures Variable Matrix: Macro-Economic (SIBOR delta, CPI), Demographic (household formation, urbanization), Supply (land tax, developer incentives), Market (LTV ratio, rental yield). 3. Defines Scenario Definitions: Baseline, Optimistic, Pessimistic, and Black Swan (e.g., SIBOR +300bps + material cost +40%). 4. Clicks 'Run Simulation'. ABM + Monte Carlo executed according to the simulation level selected; Level-1 is treated as a quick-estimate/simplified standard scenario targeted at ≤60 seconds, with detailed Level-2/Level-3 performance to be

			benchmarked in SDD. 5. System displays, where applicable: Predictive Success Score (0–100), Confidence Band, Homeownership impact, Price-to-Income ratio, Fiscal NPV (15-year baseline horizon per BRD; extended scenarios subject to MOMAH/MoF confirmation), Supply Velocity, and regional heatmap. 6. If Monte Carlo failure rate >30% → policy labelled 'Unstable', blocked from Executive Dashboard. If NPV negative → MoF Strategic Waiver required. 7. Economist reviews outputs via drill-down, exports Impact Brief PDF. 8. If Success Score ≥80% → policy advances to HITL Review. If <80% → Failure Explanation generated; Economist may re-simulate. AI: Steps 4–5 (ABM + Monte Carlo, where simulation is triggered). HITL: Steps 6–8 mandatory for simulation-triggered policies.
J-03	Legal Officer: Review Conflict Detection Results	A Legal & Compliance Officer reviews AI-generated conflict flags, resolves or escalates each flag, and clears the policy for simulation-trigger evaluation or the next applicable lifecycle stage.	1. Officer receives notification: 'Conflict Detection Complete — Policy X requires review.' 2. Opens Regulatory Conflict Engine; Conflict Report displayed by severity (Critical/High/Medium/Low) with source articles and affected ministries. 3. For each flag: reviews conflicting clause, source article/Royal Decree reference, severity, and AI recommendation. 4. Selects resolution: Resolved / Requires Amendment / Dismissed (Critical dismissals require written justification). 5. Cross-

			ministry conflicts generate Cross-Sector Ripple Effect report; affected ministry notified via Yesser GSB. 6. Contested flags escalated to Director of Policy (Tier 2 briefing). 7. All Critical flags resolved/escalated → Officer clicks 'Clear Conflict Review'; workflow evaluates SDD-confirmed simulation trigger rules and routes the policy to Simulation Required, HITL Review, or the next applicable lifecycle stage; all resolutions logged immutably. AI: Steps 2–3 (automated conflict scan; Conflict Report in ≤30 sec). HITL: Steps 4–7: every flag requires explicit human decision.
J-04	Minister: Review and Approve a Policy	The Minister reviews an AI-analyzed policy package in the Executive Intelligence Cockpit, including simulation outputs where applicable, and issues a final digitally-signed approval decision.	1. Minister opens Executive Intelligence Cockpit via Nafath SSO. Landing screen: Policy Health gauge, Vision 2030 trajectory chart, policy tiles awaiting decision. 2. Selects a policy tile; one-page Decision Summary shown: Alignment Score, top-3 risks, fiscal NPV where applicable, and scenario comparison where simulation was triggered. 3. Reviews Regional Impact Heatmap where simulation or regional impact analysis is available. 4. If resilience <50% or NPV negative, system presents Recovery Scenarios for comparison where simulation scope supports them. 5. Active Tier-3 critical alerts from MOMAH-approved KPI and policy-risk data feeds displayed prominently (e.g., '22% drop in

			developer participation under current incentive structure'). 6. Minister selects: Approve / Reject / Request Revision / Request Further Analysis or Simulation. 7. Decision digitally signed via Nafath; logged immutably with rationale. 8. On Approval: status 'Enacted'; DMS archives all documents; post-launch KPI/outcome tracking begins. On Rejection: policy returned to Analyst with comments. AI: Steps 2–5 (alignment score, risk intelligence, and simulation outputs where applicable). HITL: Steps 6–7: final decision is mandatory human action.
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ID	User Journey Step (J-01: Policy Analyst — Draft a New Policy)	System / AI Response
J-01.1	Analyst authenticates via Nafath SSO	System validates identity via Nafath MFA; retrieves role claims from MOMAH AD; presents Policy Analyst landing page with active drafts, notifications, and quick-access tools
J-01.2	Analyst opens Policy Registry and clicks 'Create New Policy'	System presents the Policy Drafting Workspace with required and optional fields; schema validation rules pre-loaded
J-01.3	Analyst completes Policy Name, Target Sector, Sub-Sector, Primary Objective, Vision 2030 Pillar(s), Budget Envelope (mandatory); Legal/Geographic Constraints (optional)	System validates uniqueness of Policy Name in registry; checks for duplicate names inline; prevents save if mandatory fields are empty
J-01.4	Analyst selects Grounding Mode (Precedent-heavy / Balanced / Novel) and clicks 'Generate Draft'	System triggers Policy Drafting Agent; spinner + ETA shown. Agent queries Knowledge Graph for relevant Royal

		Decrees, historical precedents, Vision 2030 KPI linkages, and sector benchmarks. LLM generates structured draft
J-01.5	Analyst reviews the generated Structured Draft Preview	System presents: (1) Draft with clause-by-clause Explainability Report (expandable per clause, citing data sources and legal references); (2) Vision 2030 Alignment Score (0–100 gauge, colour-coded); (3) Initial Risk Score; (4) 'Insufficient Grounding' flag if the citation minimum (≥ 3 historical precedents or ≥ 2 legal circulars per BRD Explainability Gateway, Section 6.3) is not met
J-01.6	Analyst reviews each clause, edits where needed, accepts or rejects AI suggestions	System updates draft in real-time; tracks analyst edits; version-diff shown against AI-generated original
J-01.7	Analyst clicks 'Save to Vault'	System persists draft as v0.1 in Institutional Memory Vault; policy status set to 'Draft'; confirmation toast shown. System immediately auto-routes draft to Conflict Detection Agent for background scanning
J-01.8	Analyst receives Conflict Detection notification (async)	System notifies Analyst when conflict scan is complete; conflict count shown by severity. Policy status updated to 'Conflict Review Required', 'Simulation Trigger Evaluation', 'Simulation Required', or the next applicable lifecycle stage depending on results and SDD-confirmed trigger rules

3.6 Potential productization

Capability / Feature	Productization Potential	Target Segment	Adaptation Required	Expected Value Proposition
CAP-06: Policy	Highest. The multi-variable	Primary: GCC	Domain-specific agent calibration	Enables target government users to

Simulation & Scenario Modeling (ABM + Monte Carlo)	Agent-Based Simulation engine is a potential differentiator for evidence-based housing and municipal policy design, subject to successful MOMAH production validation, data availability, and market assessment.	ministries (housing, urban planning, finance, municipal affairs) seeking evidence-based policy validation capability. Secondary: international development banks and sovereign wealth funds requiring scenario modeling for national housing strategies.	for target country demographics and housing market parameters (household formation rates, land tenure systems, mortgage market structure). Legal corpus integration adapted to target country regulatory framework. Bilingual adaptation (Arabic maintained; other languages as required). SDAIA sovereign hosting remains applicable for KSA; separate sovereign cloud arrangements required for non-KSA markets.	stress-test housing and urban policies before enactment, subject to local data readiness and model calibration. Indicative time-to-value for a calibrated simulation to be confirmed through MOMAH production validation and external market assessment.
CAP-01: Policy Intelligence and Knowledge Layer (Policy Registry + Knowledge Graph)	High. The canonical, version-controlled Policy Registry with Knowledge Graph linking policies to legislation, KPIs, and sector dependencies is applicable to any regulatory-heavy government ministry or	Primary: MOMAH other internal directorates and GCC government ministries requiring a centralized policy repository with regulatory conflict mapping. Secondary: regulatory	Domain-specific ontology for target ministry (e.g., replacing housing KPIs with health or education KPIs). Legal corpus integration adapted to target country's legal hierarchy. Arabic-English bilingual capability maintained; additional languages as required. Policy taxonomy and metadata schema	Eliminates policy fragmentation and institutional memory loss; reduces time to locate relevant precedents and regulatory linkages. Applicable to any ministry managing more than 50 active policies with complex interdependencies. Modular licensing possible (Registry without simulation layer) for lower-budget deployments.

	large regulatory body.	agencies and central banks in the MENA region managing large policy corpora.	customized per deployment.	
CAP-03: Policy Concept Builder & AI Drafting Sandbox (LLM Drafting + Legal Grounding)	High. The LLM-based policy drafting engine, grounded in a jurisdiction-specific legal corpus with Vision 2030 (or equivalent national framework) alignment scoring, is directly applicable to other national transformation programme contexts.	Primary: government ministries in KSA (beyond MOMAH) and GCC seeking to accelerate policy drafting cycles using AI with legal basis extraction and regulatory alignment scoring. Secondary: multilateral organizations (World Bank, UN agencies) supporting government capacity-building programmes in policy development.	Fine-tuning of the LLM on target domain corpus (e.g., health policy, education policy, infrastructure regulation). National framework alignment scoring reconfigured (replacing Vision 2030 KPIs with target country national programme KPIs). Domain-specific glossary and bilingual policy templates adapted to target ministry conventions.	Reduces average policy first-draft time from days/weeks to under 60 seconds, with built-in legal grounding and regulatory alignment. Transferable national framework: the Vision 2030 alignment scoring architecture is directly portable to any national transformation programme with defined KPIs (e.g., Saudi Vision 2030, UAE Agenda 2031, Egyptian Vision 2030).
CAP-09: HITL	Medium. The tiered HITL	Primary: government	Workflow stage configuration	Provides a proven, sovereign-compliant, AI-

Governance, Strategic Waiver & Approval Workflow (HITL Governance + Audit Trail)	approval workflow, digital signature integration, and immutable audit trail are applicable to any high-stakes government decision workflow requiring documented, multi-stage human oversight. However, this component is less differentiated as standalone product compared to the simulation and drafting capabilities.	entities in KSA and GCC seeking to digitize and formalize multi-stage approval workflows for regulatory decisions, budget approvals, or executive decision records. Secondary: large private-sector regulated entities (financial institutions, healthcare) requiring auditable AI-assisted decision workflows.	adapted to target organization's approval hierarchy (replacing MOMAH Analyst-Director-Minister tiers with target org's equivalent). Nafath digital signature integration replaced with target country's national identity infrastructure. Role schema and RBAC adapted to target organization's structure.	augmented governance infrastructure for high-stakes decisions. Reduces risk of undocumented decisions, enables post-hoc audit capability, and provides clear accountability chain for AI-assisted recommendations. Particularly relevant in regulated sectors where demonstrating human oversight of AI is a compliance requirement.
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Note: Detailed market sizing, pricing model, and financial projections for each productization scenario are to be developed in the Commercialization Business Case (Section 2.5 / DOC-06). At HLSD stage, productization potential is assessed qualitatively based on solution differentiation, market demand signals from GCC engagement, and adaptation complexity. A formal go-to-market strategy is out of scope for Phase 1 and should be commissioned as a separate workstream once the MOMAH deployment achieves stable production operation (post-Phase 5).

3.7 Simulation Methodology & Expert Knowledge Inputs

3.7.1 Methodology Approach

The Policy Simulation capability (CAP-06) employs three complementary methodological approaches to generate reliable, explainable policy impact estimates:

- **Rule-Based Modelling:** Deterministic rules derived from existing MOMAH regulations, building codes, and zoning ordinances. Supports transparent compliance checks against encoded rules and provides auditable logic; final legal and regulatory compliance remains subject to Ma'rifa/Sanad validation and Legal Reviewer sign-off.
- **Expert-Defined Parameters:** Domain knowledge encoded by the MOMAH Expert Working Group, covering market response elasticities, household preference distributions, and sector interdependency weights. Expert parameters are version-controlled and subject to annual review.
- **Data-Driven Calibration:** ABM agent behaviours and macro-economic parameters calibrated against historical datasets (GASTAT, SAMA, MoF) using maximum-likelihood estimation. Calibration confidence intervals are surfaced to end-users.

3.7.2 MOMAH Expert Working Group

Per the BO QA Reply, MOMAH will form a dedicated Expert Working Group (EWG) to provide and validate domain knowledge inputs for the simulation engine. Responsibilities include:

- Defining baseline policy assumptions and constraint boundaries for each simulation scenario
- Validating agent-behaviour rules and ABM parameter ranges prior to each production release
- Reviewing and approving AI-generated narrative summaries and recommendation outputs
- Signing off on Low Confidence flagged outputs before they are surfaced in the Policy Dashboard

3.7.3 Expert Knowledge Sources Required

The EWG will be responsible for supplying structured knowledge inputs across the following domains:

- **Housing Market Elasticities:** Price sensitivity of supply and demand by region, unit type, and income band
- **Infrastructure Feasibility Thresholds:** Maximum density per infrastructure capacity (water, roads, utilities) by zone
- **Policy Interaction Weights:** Relative priority and trade-off coefficients when multiple policies apply to the same zone
- **Vision 2030 Alignment Scores:** Expert-assigned weights for KPI contributions to national programme targets
- **Waiver Precedent Library:** Historical waiver decisions and rationale, used to train the conflict resolution recommendation engine

3.7.4 Simulation Output Assessment Protocol

Per BRD Artifact 3.0.C and 3.0.D, simulation outputs are evaluated against the following benchmarks and validity gates before results are surfaced to policy officers:

Assessment Benchmarks (Artifact 3.0.C)

Benchmark	Definition	HLSD Reference
Vision 2030 Alignment Score	Quantitative score measuring how much the simulated policy moves KPIs toward the 70% homeownership target and NTP programme objectives.	Section 3.8.2
Fiscal Impact Analysis (NPV)	Net Present Value of the simulated policy using MoF discount-rate inputs and the BRD-aligned fiscal-impact horizon; GDP contribution via input-output tables. Any extended analysis horizon shall be confirmed with MOMAH/MoF in SDD.	Section 3.3.6; BR-05
Social Equity Impact	Assessment of how the policy affects different income deciles and demographic groups. Flags policies that inadvertently favour high-income earners or widen the affordability gap for middle and lower income households.	SDAIA EC-05 bias evaluation; HITL review mandatory before advancing.
Explainability Audit (Traceability Path)	AI must surface a complete Traceability Path explaining why the simulation produced a specific outcome, citing input variables, agent-behaviour rules, and calibration data.	Section 6.3; Section 8.6

Simulation Success Validity Gates (Artifact 3.0.D)

A simulation run is considered valid only when all four gates are passed:

Validity Gate	Criterion
Computational Convergence	The ABM reaches a stable equilibrium state; no infinite loops or system crashes. Failed convergence routes output to HITL queue.

Predictive Plausibility (Back-Testing)	Back-testing shall compare simulation outputs against available historical outcomes using an SDD-confirmed metric such as MAPE, RMSE, or R ² . Historical depth, data suitability, and target thresholds are to be confirmed with MOMAH, GASTAT, SAMA, and MoF data owners before being used as formal acceptance criteria.
Traceability & Explainability	Every output must be fully interrogable: users can click any result to view the underlying logic, input values, and data sources.
Actionable Delta	The simulation must produce a measurable, non-trivial difference from the Baseline/Status Quo scenario. Results with no actionable delta are flagged for EWG review.

Per BRD Artifact 3.0.B, the simulation engine applies four stress-testing methods to evaluate policy resilience against extreme or non-linear events:

Method	Description	Implementation
Sensitivity Analysis (Tornado Testing)	Vary one input at a time (e.g., 200 bps interest rate hike) while holding others constant; identifies the variable with highest volatility impact on the outcome.	Automated by simulation engine; output ranked tornado chart.
Perfect Storm Scenarios	Simultaneously model combinations of adverse events (e.g., 40% construction cost increase + GDP growth stall) to test compound-risk policy failure modes.	Pre-defined EWG-approved scenario bundles; user-configurable override.
Regional Stress Testing (Geographic Fencing)	Apply stress parameters scoped to a specific Amana or region; a policy performing well nationally may fail locally due to land value or developer concentration differences.	Geospatial segmentation layer in ABM; per-Amana output dashboard.

Stochastic Modeling (Monte Carlo)	Run Monte Carlo iterations with randomised market-behaviour variations to produce a probability distribution of outcomes and identify the Probability of Policy Failure (PPF). Exact iteration depth is to be confirmed in SDD based on data availability, scenario complexity, and SDAIA platform benchmarking.	Monte Carlo engine integrated with ABM; confidence bands surfaced to end-users; PPF watermark applied when PPF > threshold (to be confirmed in SDD).
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Where primary Saudi data is unavailable or statistically insufficient, the following gap-filling strategy will be applied:

- **Peer-Country Proxy Data:** Housing market parameters sourced from comparable GCC and MENA jurisdictions (UAE, Qatar, Jordan) with documented assumptions
- **Sensitivity Analysis:** All proxy-parameter scenarios are run with $\pm 20\%$ sensitivity bands; results outside confidence thresholds trigger a Low Confidence watermark on dashboard outputs
- **Recommended Datasets:** The system will surface a prioritised list of additional data collections recommended to MOMAH for future model improvement (e.g., parcel-level transaction data, household survey micro-data)
- **Data Gap Register:** A live register of known data limitations is maintained in the Knowledge Graph and surfaced in the Audit Log for governance transparency

3.8 Shared Modules & Cross-Cutting Frameworks

3.8.1 Impact Summary Generation

Overview

Impact Summary Generation is a shared output module of the Drafting Engine and the Simulation Engine. For each policy draft and simulation scenario, it automatically generates three structured impact summaries: Financial Impact Summary, Economic Impact Summary, and Social Impact Summary. These three summaries form the core of Ministerial Briefing Notes and are mandatory review material at the Executive Review stage. All summary data is generated and stored within KSA, compliant with PDPL and NFR-06.

Financial Impact Summary

The system calculates the full-cycle fiscal impact of the policy, producing the following outputs:

- **Fiscal NPV Cost:** Calculated via a discounted cash flow model over a 15-year fiscal-impact horizon (per BRD Fiscal Hard-Stop Guardrail, Section 8.4) using the social discount rate published by MoF. Alternative discount-rate scenarios or longer horizons may be confirmed with

MOMAH/MoF in SDD. Data sources: MoF macroeconomic forecasts, MOMAH approved budget envelope.

- **Fiscal Impact Score (0–100):** Reuses the 3.8.2 fiscal scoring methodology (fiscal NPV cost against the approved budget envelope over the agreed 15-year baseline horizon (per BRD)). Scoring formula and alert thresholds to be confirmed with MOMAH in SDD.
- **ROI Estimate:** Where the policy includes quantifiable fiscal revenue returns (e.g., incremental land lease revenue, tax uplift), the system calculates a Benefit-Cost Ratio (BCR). $BCR < 1.0$ combined with negative fiscal NPV under the agreed 15-year baseline horizon (per BRD) triggers the BRD Guardrail "Fiscal Hard-Stop" — the policy cannot enter the Executive Dashboard unless a Strategic Waiver is obtained (see Section 8.4).
- **Budget Alignment:** Automatic verification of whether the policy budget request falls within the MOMAH annual budget envelope. Items exceeding the envelope are flagged individually.
- **Sensitivity Band:** 95% confidence interval of NPV based on Monte Carlo simulation output, displaying the volatility range of fiscal projections. Data source: Simulation Engine (CAP-06) output.

Economic Impact Summary

The system evaluates the policy's macro and sectoral economic impact:

- **Sectoral GDP Contribution:** Calculates the incremental contribution of the policy to the target sector's GDP (percentage point delta). Data sources: MoF macroeconomic forecasts, GASTAT national accounts. Benchmark: compared against Vision 2030 sectoral growth targets.
- **Employment Impact:** Estimates direct and indirect job creation from policy implementation, segmented by Saudization requirements (Saudi nationals vs. non-Saudi). Data sources: GASTAT labor force surveys, HRDF data.
- **Private Sector Participation:** Assesses the policy's impact on private sector investment and developer participation. If simulation results show private developer participation declining $>20\%$ YoY, a Market Volatility alert is triggered (consistent with the 8.2 Risk Escalation Framework).
- **Supply Chain Multiplier:** Estimates the policy's supply chain ripple effect (direct + indirect + induced impact) using MOMAH sectoral input-output tables.

Social Impact Summary

The system evaluates the policy's impact on social well-being and community quality:

- **Homeownership Rate Delta:** Calculates the projected delta contribution of the policy to the national homeownership rate (36-month projection period). Data sources: MOMAH housing records, GASTAT census data. If the growth rate falls $>2pp$ below the Vision 2030 annual trendline, an Ownership Stagnation alert is triggered.
- **Affordability Impact:** Calculates the policy's impact on the Price-to-Income (PTI) ratio. Data sources: GASTAT household income surveys, SAMA real-estate transaction data. PTI exceeding $5.5x$ in primary regions (Riyadh/Jeddah) per BRD Artifact 5.0 triggers a Critical

Affordability Drift alert (see Section 8.2, Table 52). Severity bands for Moderate/Watch to be confirmed with MOMAH KPI owners in SDD.

- **Housing Access Equity:** Assesses the policy's impact on housing accessibility for low-income households (annual income < SAR 180,000), disaggregated by Amana region. Data sources: Balady municipal platform, GASTAT income distribution data.
- **Quality-of-Life Index:** Assesses the policy's qualitative impact on community safety, environmental quality, and public service accessibility using GASTAT quality-of-life survey indicators. Scored via a structured rubric reviewed and confirmed by Senior Leaders.
- **Vulnerable Population Impact:** Assesses the differential impact of the policy on vulnerable groups (low-income households, first-time buyers, elderly, persons with disabilities), flagging potential negative effects for Analyst attention.

Output Structure — Ministerial Briefing Note Integration

The three impact summaries are output in the following structure, embedded directly into the Ministerial Briefing Note template:

Briefing Note Section	Source Summary	Key Presentation
Fiscal Overview	Financial Impact Summary	NPV cost, BCR, Budget Alignment status, Fiscal Impact Score
Economic Contribution	Economic Impact Summary	GDP delta, job creation, private sector participation change
Social Outcomes	Social Impact Summary	Homeownership rate delta, PTI change, housing access equity index
Combined Risk Flags	Cross-summary aggregation	All triggered alerts merged and ranked by severity

Governance Rules

- All three summaries must be fully generated before a policy draft may advance to Executive Review. Missing any single summary blocks the lifecycle transition.
- Financial Impact Summary NPV calculations must reference the latest MoF-published discount rate; if the discount rate data is older than 30 days, all financial summary outputs are watermarked "Stale Baseline — Data Latency Guard."
- Social Impact Summary Quality-of-Life Index and Vulnerable Population Impact must receive Senior Leader review and confirmation before inclusion in the Briefing Note (HITL requirement).
- All summary results are version-bound to the policy draft and stored in the Policy Registry audit log, immutable per version.

Output Example

Policy: Residential Developer Incentive Programme (Draft v2.1)

Financial Impact Summary: Fiscal NPV Cost SAR 2.1B under the agreed baseline horizon (within approved envelope of SAR 2.5B) | Fiscal Impact Score: 84/100 [Within Envelope] | BCR: 1.3 | Sensitivity Band: SAR 1.8B–2.4B (95% CI) | Budget Alignment: All line items within the approved fiscal planning envelope

Economic Impact Summary: Sectoral GDP Contribution: +0.4pp (Housing sector) | Employment Impact: 12,400 direct + 8,200 indirect jobs (82% Saudization compliant) | Private Sector Participation: projected +8% YoY in developer submissions | Supply Chain Multiplier: 1.7x

Social Impact Summary: Homeownership Rate Delta: +1.4pp over 36 months (ON TRACK vs. Vision 2030 trendline) | Affordability Impact: PTI projected at 4.8x (target <5x; Critical alert at >5.5x) — WITHIN TARGET | Housing Access Equity: 78% of eligible low-income households in Riyadh/Jeddah Amanas projected to benefit | Quality-of-Life Index: 7.2/10 (Senior Leader review pending)

Combined Risk Flags: Section 3.1 (Developer Profit Cap) scores -12 on Fiscal Sustainability [net-negative — mandatory HITL review]. No Critical or High severity alerts triggered.

3.8.2 KPI Scoring & Vision 2030 Alignment Framework

This section defines the KPI scoring methodology applied across the platform -- spanning the Vision 2030 Alignment Scorer, CAP-11 Post-Implementation KPI Tracking, and the Policy Assessment criteria defined in BRD Artifact 4.0. It provides the scoring dimensions, data sources, pass/fail thresholds, and governance rules that underpin every quantitative policy evaluation produced by the system.

Policy Assessment Scoring Matrix (BRD Artifact 4.0)

Every policy draft and active policy is evaluated against the following eight dimensions. Each dimension is scored independently; scores are aggregated into a composite Policy Assessment Score (0-100) weighted by pillar priority as defined in the Vision 2030 PMO framework.

- **Vision 2030 Alignment:** Measurement goal: policy clauses must demonstrably contribute to at least 15% improvement on one or more Vision 2030 national target KPIs (homeownership, quality of life, fiscal sustainability). Data source: Vision 2030 PMO feed (auto-refreshed) and MOMAH strategic alignment registry.
- **Affordability Index:** Measurement goal: Price-to-Income (PTI) ratio targeting below 5x in primary housing markets. Formula: $PTI = \text{Median Annual House Price} / \text{Median Annual Household Income}$. Data sources: GASTAT (household income surveys), SAMA (real-estate transaction data, mortgage statistics). Threshold (two-layer): (a) BRD target — $PTI < 5x$ in primary housing markets; (b) Critical alert — $PTI > 5.5x$ in Riyadh/Jeddah per BRD Artifact 5.0 (see Section 8.2, Table 52). Moderate/Watch severity bands to be confirmed with MOMAH KPI owners in SDD.
- **Homeownership Rate:** Measurement goal: delta increase in national homeownership rate toward the 70% Vision 2030 target. Measured as: projected homeownership rate at 36 months post-enactment minus current baseline. Data source: MOMAH housing records,

GASTAT census data. Threshold: growth rate falling >2pp below the annual Vision 2030 trendline triggers Critical Ownership Stagnation alert.

- Fiscal Impact:** Measurement goal: net cost to the national treasury on an NPV basis using the BRD-aligned fiscal-impact horizon. NPV calculation method: standard Discounted Cash Flow (DCF) — $NPV = \sum (\text{net_cash_flow_t} / (1+r)^t)$ over the agreed analysis horizon, where r = discount rate as defined in the MoF macroeconomic framework. Data sources: MoF (fiscal data, government discount rate, macroeconomic forecasts); SAMA and banking sector data (financing costs and lending rates); MOMAH project cost submissions (implementation and operational expenditure). Specific discount rate value, budget envelope scope, and fiscal variance thresholds to be confirmed with MoF and MOMAH in SDD.
- Supply Velocity:** Measurement goal: average elapsed time from development permit issuance to market entry (first unit available for sale or rental). Scored relative to MOMAH target cycle time. Data source: Balady municipal permit platform, MOMAH transaction records. Threshold: median permit-to-market time exceeding 180 days triggers a Supply Velocity risk flag.
- Economic Development:** Measurement goal: projected contribution to sectoral GDP growth (percentage point delta attributable to the policy), measured across three impact categories consistent with standard economic impact methodology: (1) direct impact — sector-level output change directly caused by the policy; (2) indirect impact — supply-chain and multiplier effects on related upstream and downstream sectors; (3) induced impact — household income and consumption effects arising from employment and income changes. Data sources: MoF macroeconomic forecasts, GASTAT national accounts and input-output tables, construction and housing activity sector data. Scored against sector-specific growth targets defined in Vision 2030.
- National Stability:** Measurement goal: qualitative and quantitative assessment of policy impact on community safety, social cohesion, and quality of life indicators. Scored using a structured rubric reviewed by MOMAH policy committee. Data source: manual input by Senior Leaders; GASTAT quality-of-life surveys where available.
- Interdependency Mapping:** Measurement goal: identification and quantification of cross-sector ripple effects -- unintended impacts on housing, municipal, fiscal, environmental, and smart-city domains. Scored as number and severity of cross-sector dependencies detected by the Conflict Detection Engine (CAP-04). Data source: MOMAH policy registry, inter-ministerial data feeds via Yesser GSB.
- Social Equity Impact:** Measurement goal: distributional impact of the policy across income groups and regions — measuring affordability, accessibility, and fiscal burden on low-income households. Formula approach: distributional impact analysis across income deciles and geographic regions, consistent with standard social equity assessment methodology (confirmed by BO QA Reply as aligned with GASTAT and MOMAH frameworks). Data sources: GASTAT income deciles and household surveys; MOMAH beneficiary programme data; SAMA mortgage accessibility data. Threshold: specific equity scoring bands and weighting coefficients to be confirmed with MOMAH KPI owners in SDD.

Vision 2030 Alignment Scoring -- Design Detail (BRD UC-17)

The Vision 2030 Alignment Scorer produces a real-time composite score (0-100) each time a policy draft is saved or modified. The scoring engine operates as follows:

- **Step 1 -- KPI Retrieval:** The system retrieves the current Vision 2030 KPI targets and trendlines from the PMO data feed. If the feed data is older than 30 days, all scoring outputs are watermarked "Stale Baseline -- Data Latency Guard" (BRD UC-17 A2).
- **Step 2 -- Clause-Level Scoring:** Each clause in the draft is evaluated against the KPI contribution weights relevant to the declared Vision 2030 Pillar(s). Contribution is expressed as a positive or negative delta against the KPI trajectory. Clause-level scores are stored and displayed as a contribution waterfall chart.
- **Step 3 -- Composite Score Aggregation:** Clause scores are aggregated into a single composite Alignment Score (0-100) using the following mechanism: (a) **Weight Assignment** — KPI weights are defined by Vision 2030 strategy teams, MOMAH leadership, and policy subject-matter experts; weights are stored in the system and updatable as national priorities evolve, reflecting relative strategic priority across Housing, Quality of Life, Economic Opportunity, and Fiscal Sustainability pillars. (b) **Normalization** — individual KPI scores are normalized to a 0-100 scale before aggregation using a weighted sum formula. (c) **Threshold Logic** — if a policy negatively impacts a critical KPI (net-negative contribution on a high-priority pillar), the system caps or reduces the final composite score to prevent a policy from scoring high overall while masking harm to a strategic priority. (d) **Confidence Adjustment** — if simulation uncertainty is high (Monte Carlo variance exceeding threshold), the composite score is adjusted downward to reflect reduced model confidence and encourage more robust policy design before advancement.
- **Step 4 -- Output Display:** The system displays: (a) the composite Alignment Score; (b) a per-pillar breakdown showing contribution by Housing, Quality of Life, Economic Opportunity, and Fiscal Sustainability pillars; (c) a clause-level contribution waterfall identifying the highest and lowest contributing provisions; (d) AI-generated improvement hints for clauses scoring below the pillar average.

Governance Rules (BRD UC-17 Business Rules):

- **Minimum Score Gate:** No policy draft may advance to Executive Review (UC-06) with a composite Alignment Score below 60. The system blocks the lifecycle transition and displays a remediation checklist.
- **Negative Clause Flag:** Any clause with a net-negative KPI contribution (score below zero) is automatically flagged for mandatory HITL review before the draft may advance, regardless of the overall composite score (BRD UC-17 A3).
- **Immutability:** Alignment Scores are versioned and immutable per draft version. Re-scoring requires the creation of a new draft version in the Policy Registry, ensuring a complete audit trail of score evolution across revisions.
- **Pillar Tag Requirement:** If the draft is missing the mandatory Vision 2030 Pillar tag, the system blocks scoring and prompts the Analyst to complete the Policy Object metadata before proceeding (BRD UC-17 A1).

Risk Escalation Thresholds (BRD Artifact 5.0)

The following KPI thresholds define when a policy simulation result or active policy health scan triggers an automated escalation. These are the pass/fail criteria that the system enforces continuously via the Policy Health scan:

- **Affordability Drift:** Price-to-Income ratio > 5.5x in primary regions (Riyadh/Jeddah) per BRD Artifact 5.0. Severity: Critical. Action: auto-escalate to Tier-3 Ministerial Briefing; policy cannot advance without documented remediation.
- **Supply-Demand Gap:** Projected housing supply deficit exceeds 15% of demand over a 24-month horizon. Severity: High. Action: auto-escalate to Tier-2 Director-level review; simulation must be re-run with adjusted supply levers.
- **Fiscal Variance:** Policy implementation cost exceeds approved budget by more than 10% without a documented 1:1 ROI justification. Severity: High. Action: mandatory MoF alignment review before Executive Approval stage.
- **Ownership Stagnation:** Homeownership rate growth falls more than 2 percentage points below the Vision 2030 annual trendline. Severity: Critical. Action: auto-escalate to Tier-3 Ministerial Briefing; trigger Gap Analysis run (CAP-02) to identify remediation options.
- **Market Volatility:** Private developer participation or investment drops more than 20% year-over-year. Severity: Medium. Action: auto-escalate to Tier-1 Analyst review; Housing Economist must run a sensitivity analysis before the policy advances.

Data Quality Controls

- **Data Latency Guard (BRD UC-17 A2):** KPI trendline datasets older than 30 days trigger a "Stale Baseline" watermark on all scoring outputs. Affected scores remain visible but are flagged as low-confidence pending a data refresh.
- **Source Traceability:** Data sources are listed per metric in the scoring matrix above. GASTAT and SAMA feeds are refreshed on a scheduled basis; MoF and MOMAH records are updated via manual upload or Yesser GSB integration where API connectivity is available.
- **XAI Citation Requirement:** Every score output must cite the specific KPI targets, trendline values, and data freshness timestamps used in the computation, in compliance with the explainability framework in Section 6.3.

Output Example

Policy: Residential Developer Incentive Programme (Draft v2.1) | Composite Alignment Score: 74/100 | Status: Eligible to advance to Executive Review (score >= 60)

Pillar Breakdown -- Housing: 88/100 | Quality of Life: 71/100 | Economic Opportunity: 68/100 | Fiscal Sustainability: 52/100 [below average -- improvement hint generated]

Affordability Index: PTI projected at 4.8x (target <5x; Critical alert at >5.5x) — PASS | Homeownership Rate delta: +1.4pp over 36 months -- ON TRACK | Fiscal Impact: 15Y NPV cost SAR 2.1B within approved envelope -- PASS | Supply Velocity: median permit-to-market 142 days (target <180) -- PASS

Clause-level flag: Section 3.1 (Developer Profit Cap) scores -12 on Fiscal Sustainability pillar [net-negative -- mandatory HITL review required before draft may advance].

3.8.3 Notification & Alert Framework

The Policy AI Lab provides configured notification and alert channels on the Minister and Senior Leader dashboards, anchored to the Risk Escalation Framework (BRD Artifact 5.0) and driven by MOMAH-approved KPI and policy-risk data feeds.

Critical Alert Banner: Reserved for Tier-3 Critical alerts generated from MOMAH-approved KPI and policy-risk data feeds (e.g., Affordability Drift PTI > 5.5x in Riyadh/Jeddah, Ownership Stagnation >2pp below trendline). Displayed persistently at the top of the Executive Intelligence Cockpit. Each alert item includes: alert category, specific KPI breach value, affected policy, and a one-click link to the Decision Summary. Alert items remain active until the Minister issues a disposition decision or the alert is superseded by a higher-severity event.

Geospatial Pulse Notifications: When a policy health scan detects a localized failure or drift in a specific Amana (e.g., Dammam housing supply deficit), the system sends a push notification to the corresponding regional director's mobile platform. The notification includes a heat map showing the localized impact, the triggering KPI breach, and a direct link to the Regional Impact Drill-Down view.

Breaking News Ticker Scope Mapping: The originally requested Breaking News Ticker is implemented as a scoped Critical Alert Banner and approved alert notification stream driven by MOMAH-approved KPI and policy-risk data feeds. It is not an open-ended real-time news feed, macroeconomic forecasting service, or external media-monitoring platform.

4. Requirements Summary

4.0 BRD–HLSD Capability Traceability

The table below maps the four high-level capabilities defined in the BRD to the thirteen mapped HLSD capabilities (CAP-01 to CAP-13) specified in Section 3.3. This traceability confirms full coverage of BRD scope while distinguishing mapped capability coverage from the detailed design subsections that follow.

BRD Capability	BRD CAP	HLSD Capabilities	HLSD CAP(s)
Policy Intelligence & Knowledge Layer	BRD CAP-01	Policy Registry; Gap & Opportunity Analysis Engine	CAP-01, CAP-02
AI-Assisted Policy Generation	BRD CAP-02	Policy Concept Builder & AI Drafting Sandbox; Conflict Detection; Benchmarking & Comparative Policy Analysis Engine	CAP-03, CAP-04, CAP-05

Policy Simulation & Scenario Modeling	BRD CAP-03	Policy Simulation & Scenario Modeling	CAP-06
Collaboration & Workflow Orchestration	BRD CAP-04	Consultation Synthesis & Amendment Recommendation; Estitlaa Consultation Publishing Support; HITL Governance & Strategic Waiver; Official Publication Package; Post-Implementation Quality Measurement; Executive Decision Intelligence; Responsible AI & Auditability	CAP-07, CAP-08, CAP-09, CAP-10, CAP-11, CAP-12, CAP-13

4.1 Business Requirements

ID	Requirement	Priority	Source
BR-01	Policy Registry & Centralized Document Management — the system shall provide a centralized repository for all MOMAH policies with version control, metadata tagging, full-text search, legal-basis linkage, decision history, final approved texts, and supporting records.	P0	BRD §3 / MOMAH Methodology Step 8
BR-02	AI-Assisted Policy Drafting — the system shall generate a structured initial policy draft within 60 seconds based on policy officer inputs, using a domain-specific LLM and bilingual templates (Arabic / English). Draft outputs shall include legal basis, objectives, financial/social/economic impact summaries, risk considerations, and source-grounded explainability.	P0	BRD §4
BR-03	Human-in-the-Loop (HITL) Review Workflow — all AI-generated drafts, checklist results, simulation outputs, consultation amendment recommendations, waiver requests, and	P0	BRD §5; HITL governance controls

	publication packages shall pass through mandatory authorized human review and approval; no policy may be published without formal human sign-off.		
BR-04	Role-Based Access Control (RBAC) & Audit Trail — the system shall enforce role-based permissions and maintain tamper-proof audit logs for all user actions, AI outputs, model runs, checklist validations, consultation decisions, waiver approvals, publication package actions, and overrides.	P0	BRD §6 / MOMAH IT / PDPL
BR-05	Policy Impact Simulation (ABM) — the system shall simulate projected outcomes of draft policies using Agent-Based Modeling across household, demographic, and economic agent populations derived from KSA national datasets (GASTAT, SAMA, MoF).	P0	BRD §4 / KPI Owners
BR-06	Policy Benchmarking & Comparative Analysis — the system shall enable structured comparison of MOMAH draft policies against GCC, OECD, and international peer counterparts, with data freshness indicators, normalization assumptions, output gap tables, risk flags, and best-practice recommendations.	P0	BRD §4
BR-07	Conflict Detection & Compliance Checking — the system shall automatically identify regulatory conflicts, duplications, contradictions, and checklist gaps between draft policies, existing MOMAH policy records, KSA legal corpus, Ma'rifa, Sanad, and the unified 39-item methodology checklist.	P0	BRD §4 / MOMAH Methodology Step 3
BR-08	Arabic-English Bilingual Support — all user-facing interfaces, policy drafts, consultation synthesis, reports, dashboards, and generated packages shall support Arabic (RTL) and English;	P1	BRD §3 / All Users

	the AI drafting and NLP engines shall operate in both languages.		
BR-09	Enterprise System Integration — the system shall integrate with MOMAH HCM and ERP for user identity and workflow routing; SSO via MOMAH identity provider (SAML 2.0 / OIDC) is required.	P2	BRD §6 / MOMAH IT
BR-10	Case Identification & Policy Gap Analysis — the system shall support identification of cases requiring new or updated legislation and detect regulatory white spaces by comparing Policy Registry coverage against Vision 2030 KPIs, sector priorities, and peer-country benchmarks.	P0	BRD UC-16 / MOMAH Step 1
BR-11	Policy Project Concept Builder — the system shall capture policy trigger, target sector, legal basis, strategic alignment, stakeholders, expected impacts, and constraints, then generate an initial concept note and policy memorandum for analyst review.	P0	MOMAH Step 2; BRD concept-builder requirements
BR-12	Consultation Comment Synthesis — the system shall ingest consultation feedback from configured channels, including Estitlaa where available, and provide NLP topic clustering, sentiment scoring, representative comments, structured amendment options, and traceability from comment themes to draft changes.	P0	BRD consultation requirements / MOMAH Step 6
BR-13	Estitlaa Consultation Publishing Support — the system shall prepare and validate draft consultation packages, metadata, consultation windows, deadlines, and handoff/status tracking for Estitlaa. Actual submission, publication, and enactment remain outside automated decisioning and require official MOMAH workflow execution.	P1	MOMAH Step 5

BR-14	Official Publication Package & Formal Archiving — the system shall prepare, validate, and track handoff of final approved publication packages for Official Gazette and MOMAH website workflows, archive final legislation and supporting material, and maintain package status and timestamps. Submission, publication, and enactment remain human-led and outside automated decisioning.	P1	MOMAH Steps 7-8
BR-15	Post-Implementation Quality Measurement — the system shall track KPI/outcome performance, implementation quality, actual versus simulated outcomes, variance indicators, and periodic reports for GRC, Senior Leaders, and the Minister. It shall not perform regulatory enforcement monitoring or execute compliance actions.	P1	MOMAH Step 9 / BRD UC-17
BR-16	Strategic Waiver Workflow — the system shall support fiscal waiver and strategic exception requests, MoF alignment where required, approval routing, decision rationale capture, and audit trail.	P1	BRD waiver references
BR-17	Executive Decision Intelligence — the Minister and Senior Leader dashboards shall include approved alert notifications, regional impact views, and configured policy-risk intelligence based on MOMAH-approved data feeds. The platform shall not provide real-time macroeconomic forecasting infrastructure.	P2	BRD executive dashboard references
BR-18	Annual Independent Audit — the system and governance model shall support annual independent audit of policy quality, benchmarking accuracy, model performance, and compliance with high-risk AI governance controls.	P1	BRD audit requirement

Note: *P0 Must have, P1 Should have, P2 Nice to have, P3 Backlog*

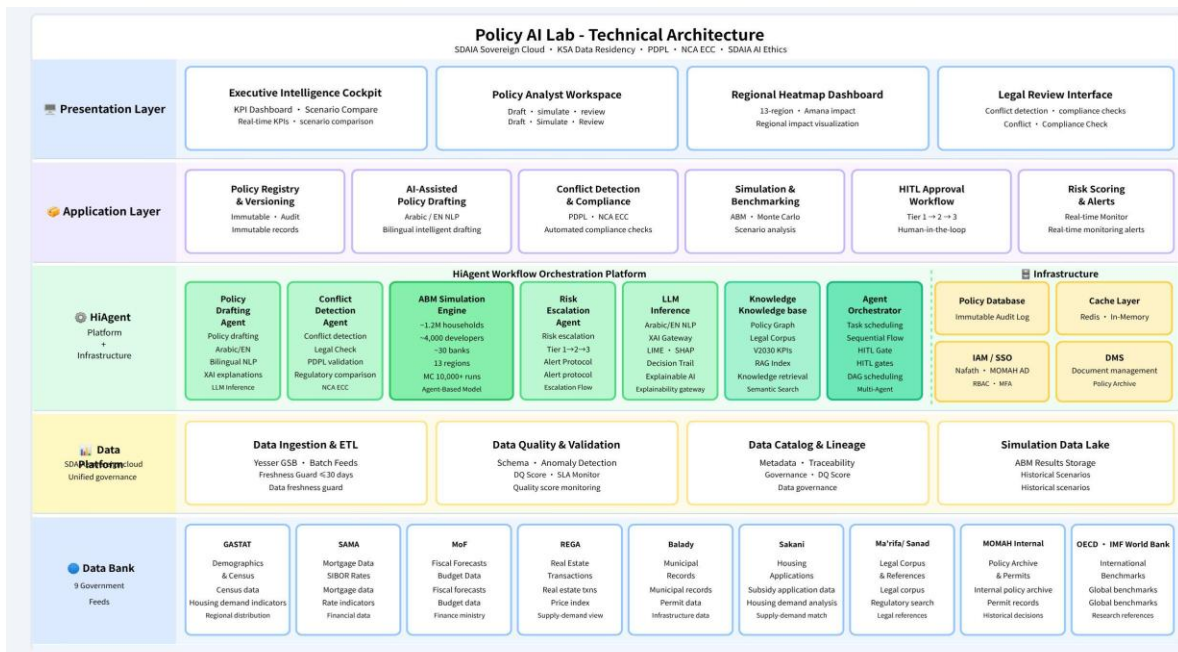
4.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	AI-assisted draft generation shall complete within 60 seconds; API responses shall be targeted at ≤ 3 seconds for the 95th percentile of requests under normal load, subject to SDD confirmation. Simulation performance tiers: Level 1 (Quick Estimate) — high-level approximation or simplified standard scenario aligned to the BRD ≤ 60 -second target; exact Level-1 definition to be confirmed in SDD. Level 2 (Standard Simulation) — multi-variable ABM run with Monte Carlo iterations, with scale, agent counts, iteration depth, and runtime targets to be confirmed during Phase 2 data assessment and SDAIA performance benchmarking. Level 3 (Deep Analysis / Monte Carlo) — extended run with scenario permutations and sensitivity analysis; run time depends on scenario count, regional scope, and Monte Carlo iteration depth and will be benchmarked in Phase 2. Complex ABM-based simulations are not committed to the Level 1 60-second target unless explicitly scoped as Level 1 quick estimates.
NFR-02	Security & Compliance	The solution shall comply with PDPL, NCA Cybersecurity Framework (CSF), and SDAIA AI Ethics Framework. All data at rest and in transit shall be AES-256 encrypted.
NFR-03	AI Model Performance	Simulation accuracy, policy conflict detection precision/recall, policy assumption extraction accuracy, and risk-alert coverage shall be treated as target metrics until MOMAH-defined KPI benchmarks, labelled datasets, formulas, and acceptance thresholds are confirmed in SDD/UAT. BO QA Reply confirms conflict detection should be measured primarily as recall once ground truth is established through Ma'rifa, Sanad, and MOMAH legal expert validation.
NFR-04	Authentication & Access	Single Sign-On (SSO) via MOMAH identity provider (SAML 2.0 / OIDC). Multi-factor authentication (MFA) required for Approver and Admin roles.
NFR-05	Availability & Resilience	System uptime $\geq 99.5\%$ during KSA business hours (07:00–22:00 AST). Recovery Time Objective (RTO) < 4

		hours; Recovery Point Objective (RPO) <1 hour for business-critical data.
NFR-06	Sovereign Hosting & Data Residency	All data storage, backups, logs, embeddings, model inference, integrated AI services, and third-party processing must remain within KSA national boundaries. No confidential document inference or personal data processing may occur outside KSA sovereign infrastructure.
NFR-07	Supported Platforms	Web-based solution supporting Chrome, Edge, and Safari on desktop (minimum resolution 1280×800). Arabic RTL layout fully supported. Responsive design for tablet is desirable; not required for the integrated delivery unless confirmed by MOMAH.

5. High-Level Architecture

5.1 Architecture Overview



The Policy AI Lab solution is structured across six horizontal layers, all hosted within the **SDAIA Sovereign AI Platform** inside KSA. At the foundation, the **Data Sources** layer aggregates national datasets (GASTAT, SAMA, MoF, REGA, Balady, Sakani), the KSA legal corpus (Ma'rifa, Sanad), MOMAH internal records, and international benchmarks (OECD, IMF, World Bank). The **Data Platform** layer provides unified data governance — ingestion via Yesser

Government Service Bus, data quality validation with a 30-day freshness guard, a data catalog for lineage tracking, and a simulation data lake. The **HiAgent Platform** is the AI orchestration core: it hosts the Policy Drafting Agent, Conflict Detection Agent, Agent-Based Modeling (ABM) simulation engine using MOMAH-confirmed agent categories (households/residents, real estate developers, banks and financial institutions, government/regulators, municipalities/regional authorities, investors, and market environment variables such as interest rates, inflation, GDP, and construction costs), the Risk Escalation Agent, the Knowledge Base (policy graph, legal corpus, V2030 KPIs), and the bilingual LLM inference engine with XAI Explainability Gateway. Specific ABM agent counts, regional granularity, iteration depth, and scale targets will be confirmed in the SDD after data assessment and SDAIA platform benchmarking. The **Infrastructure** layer provides the policy database and immutable audit log, cache, IAM/SSO (Nafath + MOMAH AD), and the Document Management System. The **Application Layer** exposes services for Policy Registry, AI Drafting, Conflict Detection, Simulation, HITL Approval Workflow, and Risk Scoring. The **Presentation Layer** delivers role-specific interfaces: the Executive Intelligence Cockpit (Minister), Policy Analyst Workspace, Regional Heatmap Dashboard, and Legal Review Interface. Detailed component specifications and infrastructure topology will be defined in the Solution Design Document (SDD) during Phase 2.

5.2 Key Components

Component	Technology	Description
Policy Registry & Knowledge Graph	Graph Database + Document Store	Canonical, version-controlled repository of MOMAH policies, legislative records, legal references, Vision 2030 KPI linkages, sector interdependencies, assumptions, decisions, final texts, and outcome history.
Case Identification & Gap Analysis Engine	Knowledge Graph query layer + rules + analytics engine	Registers legislative triggers and identifies regulatory white spaces by comparing active policy coverage against Vision 2030 KPIs, sector priorities, and peer-country benchmarks. Produces gap report, opportunity score, benchmark map, recommended interventions, confidence band, and audit log.
Policy Concept Builder	Structured workflow service + LLM-assisted generation	Captures trigger, sector, rationale, strategic alignment, legal basis, stakeholders, constraints, and expected impacts; generates initial concept note and policy memorandum for analyst review.

LLM Policy Drafting Engine	Sovereign LLM hosted on SDAIA AI Platform	Generates structured initial policy drafts in Arabic and English. Performs legal basis extraction, Vision 2030 alignment scoring, assumption documentation, and financial/social/economic impact summary generation. All inference runs within KSA sovereign boundary.
39-Item Checklist Compliance Engine	Rules engine + workflow validation + reviewer report generator	Digitizes MOMAH's unified 39-item legislative checklist. Validates completeness, flags missing items, generates compliance status, and produces reviewer reports for GRC, Legal Affairs, BI Center, Finance, and Policy & Regulation Committee.
Benchmarking Engine	Benchmark repository + normalization rules + analytics pipeline	Compares policies with peer countries and global standards, including Singapore, Norway, UAE, OECD, IMF, and World Bank data where available. Outputs gap tables, risk flags, comparative notes, and best-practice recommendations. Detailed normalization assumptions will be finalized in SDD.
ABM Simulation Engine	Python Agent-Based Modeling framework + Monte Carlo engine	Runs multi-variable simulations across household, developer, lender, regional, economic, demographic, supply, and market variables. Produces fiscal impact, affordability, supply/demand, confidence bands, regional heatmaps, and predictive success scores.
Conflict Detection Engine	NLP pipeline + Knowledge Graph query layer + Ma'rifa/Sanad integration	Scans policy drafts against Policy Registry and KSA legal corpus. Detects contradictions, duplications, overlaps, regulatory hierarchy issues, and cross-sector ripple effects.
Vision 2030 Alignment Scorer	Hybrid scoring engine: rules + expert weights + model-based estimation	Evaluates policy drafts against Vision 2030 KPI definitions and trends. Produces a composite 0-100 score, pillar breakdown, clause-level contribution view, data

		freshness flags, and audit entry. Detailed scoring formula will be finalized in SDD.
Consultation Synthesis & Amendment Recommendation Module	Arabic/English NLP clustering + sentiment analysis + recommendation workflow	Ingests consultation comments, clusters themes, scores sentiment, extracts representative comments, recommends clause-level amendments, and requires analyst approval before draft changes are applied.
Estitlaa Publishing Support Module	Package preparation workflow + metadata generator + status tracker	Prepares draft consultation packages for Estitlaa, including formatting, metadata, consultation window tracking, deadline monitoring, and package status. Actual submission remains human-approved and official-workflow-led.
HITL Approval & Strategic Waiver Workflow Engine	BPM/workflow service + Nafath digital signature integration	Manages Draft → Concept → Checklist → Conflict Review → Simulation → Consultation → HITL Review → Executive Approval → Publication Support → Archive → Quality Monitoring. Handles strategic waiver requests, MoF alignment where required, approval routing, rationale capture, and audit logs.
Publication Package & Formal Archiving Module	Document package generator + DMS integration + metadata registry	Prepares Official Gazette and MOMAH website publication packages, final approved version, approval evidence, attachments, checklist outputs, timestamps, and DMS archival records. Formal publication remains human-led.
Post-Implementation Quality Measurement Dashboard	KPI dashboard + compliance reporting + actual-vs-simulated analytics	Tracks implementation quality, KPI/outcome performance, actual versus simulated outcomes, variance indicators, and periodic reports for GRC, Senior Leaders, and the Minister. It does not perform regulatory enforcement monitoring or execute compliance actions.
Risk Escalation & Executive Intelligence Engine	Threshold monitor + notification service + executive dashboard widgets	Routes tiered alerts, approved data-feed notifications, and regional impact views to Analyst, Director, Senior Leader, or Minister dashboards according to configured

		thresholds. It does not provide real-time macroeconomic forecasting infrastructure.
Data Ingestion & Integration Layer	API gateway + Yesser GSB + ETL pipeline	Manages data ingestion from internal MOMAH systems, external government sources, legal systems, consultation channels, publication workflow systems where available, and international benchmarks. Enforces data freshness guards.
Audit & Compliance Service	Immutable event log + PDPL/NCA/SDAIA compliance gateway	Records all user actions, AI outputs, simulations, approvals, rejections, checklist results, consultation decisions, waiver actions, publication package events, and override justifications. Supports annual independent audit scope.
Web Application (Frontend)	React / Next.js or equivalent web framework	Role-specific interfaces for Executive Intelligence Cockpit, Policy Analyst Workspace, Concept Builder, Checklist Review, Simulation Lab, Legal Review, Consultation Synthesis, Quality Dashboard, and Administration. Supports Arabic RTL and English LTR.
Identity & Access Management (IAM)	Nafath SSO + MOMAH AD/LDAP	Manages authentication, MFA, role claims, access scopes, service accounts, and RBAC across all modules.

5.3 Key Integrations

Standard integrations

External System	Integration Type	Direction	Data Exchanged	Protocol / Format
Nafath (National SSO)	SSO / Identity Federation	Inbound	Auth tokens, user identity, MFA verification status, digital signature status where applicable	SAML 2.0 / OIDC + Nafath digital signature

Yesser Government Service Bus (GSB)	Government Service Bus	Bidirectional	Inter-ministerial data exchange with SAMA, MoF, MISA, and other regulatory bodies; stakeholder routing where available	To be confirmed in SDD
Balady Municipal Platform	Inbound data feed	Inbound	Municipal permits, zoning data, urban planning records, and Amana-level regional context	To be confirmed in SDD
Sakani Housing Platform	Inbound data feed	Inbound	Housing applications, ownership records, subsidy data, and simulation inputs	To be confirmed in SDD
Absher Public Services Gateway	Inbound feedback channel	Inbound	Public and stakeholder consultation feedback for consultation synthesis where legally and technically available	To be confirmed in SDD
Estitlaa Public Consultation Platform	Outbound handoff + inbound import	Outbound / Inbound	Draft consultation package, metadata, consultation window, status timestamps, and public feedback import where API is available. Actual submission remains outside automated decisioning and requires official MOMAH workflow execution.	To be confirmed in SDD
Official Gazette / Umm Al-Qura Publication Workflow	Outbound handoff (human-led)	Outbound / Inbound	Final approved publication package, approval evidence, metadata, attachments, package status, and timestamps. Formal submission, publication, and enactment remain outside automated decisioning.	To be confirmed in SDD

MOMAH Website Publication Workflow	Outbound handoff (human-led)	Outbound / Inbound	Approved website publication package, final text, metadata, attachments, and publication status tracking	To be confirmed in SDD
MOMAH Enterprise Directory (AD / IAM)	Identity Federation / Directory	Inbound	Internal user identity, role assignments, permission groups, and RACI mapping	To be confirmed in SDD
Document Management System (DMS)	API + File Archival	Outbound	Final legislation, policy memoranda, impact briefs, consultation packages, checklist reports, publication packages, decision records, and audit trail records	To be confirmed in SDD
Vision 2030 PMO KPI Feed	Pull-based data feed	Inbound	KPI definitions, baseline values, trend lines, targets, and data freshness metadata for alignment scoring and quality measurement	Auto-refreshed; 30-day guard
GASTAT / SAMA / MoF / REGA	Public + restricted MOU	Inbound	Demographic, macroeconomic, fiscal, mortgage, housing market, and regulatory datasets for simulation, benchmarking, KPI scoring, and post-implementation measurement	Public download; MOU for micro-data

AI strategy solutions integrations

Internal / AI System	Integration Type	Direction	Data Exchanged	Protocol / Format
SDAIA Sovereign AI Platform	Platform (hosting + inference)	Bidirectional	AI model inference requests, Knowledge Graph operations, simulation engine workloads, and model training pipelines	REST / gRPC (platform-internal)

			— all within KSA sovereign boundary	
Ma'rifa (Legal Reference System)	To be confirmed in SDD	Inbound	Legal articles, Royal Decrees, Cabinet resolutions, and ministerial circulars for the Knowledge Graph and Conflict Detection engine	To be confirmed in SDD
Sanad (Legal Advisory System)	To be confirmed in SDD	Inbound	Legal advisory corpus and compliance references used for regulatory conflict validation	To be confirmed in SDD

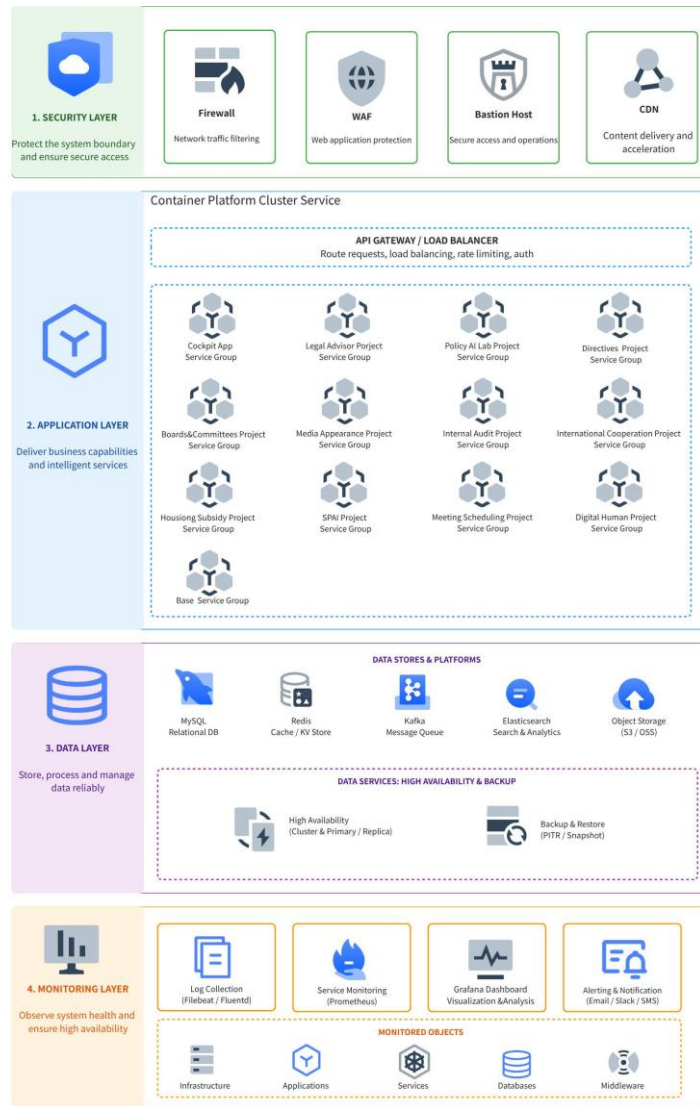
5.4 Deployment Strategy

Deployment Architecture Description (Phase 1 — HLSD Stage)

Policy AI Lab adopts the standard MOMAH AI service platform deployment pattern defined by the operations team, while applying project-specific controls for SDAIA sovereign hosting, KSA data residency, high-risk AI governance, and production release approval. All AI inference, model artifacts, Knowledge Graph operations, audit records, and policy data shall remain within the approved KSA-hosted environment. No cross-border transfer of policy data, model outputs, logs, or training data is permitted.

5.4.1 Platform Deployment Topology

The platform deployment follows a layered enterprise architecture covering Security, Application, Data, and Monitoring layers. Policy AI Lab is deployed as one service group within the shared AI-enabled service platform and uses the common platform controls for traffic protection, service routing, data persistence, observability, backup, and operations.



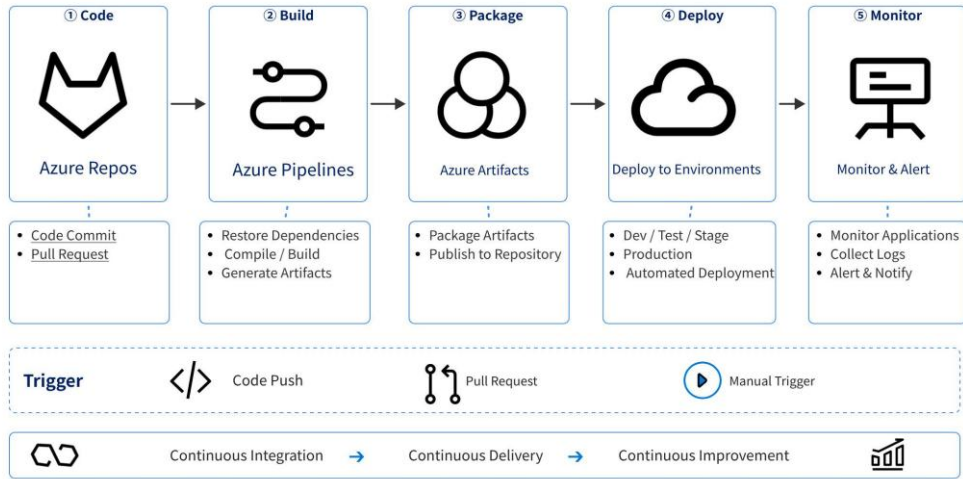
- **Security Layer:** Firewall, WAF, bastion host, and CDN provide network boundary protection, web application protection, secure operations access, and traffic acceleration.
- **Application Layer:** API Gateway and load balancer provide unified traffic entry, authentication, authorization, routing, rate limiting, and service governance. Policy AI Lab runs as a containerized project service group within this layer.
- **Data Layer:** Relational database, cache, message queue, search engine, object storage, high availability, and backup/restore services support reliable data storage and recovery.
- **Monitoring Layer:** Log collection, service monitoring, dashboards, alerting, and notification channels provide full-stack observability across infrastructure, applications, services, databases, and middleware.

5.4.2 DevSecOps and CI/CD Pipeline

The delivery pipeline follows a lightweight automated CI/CD model covering code management, build, artifact packaging, environment deployment, monitoring, and alerting. The pipeline may be implemented using Azure DevOps or an approved enterprise DevSecOps toolchain, provided

that runtime deployment, AI inference, model hosting, logs, and production data remain within the approved sovereign hosting boundary.

Azure DevOps CI/CD Pipeline



Pipeline Phase	Primary Tooling / Function	HLSD Deployment Control
Code	Repository, branch control, pull request, code review	All production changes require reviewed and approved source changes; direct production modification is not allowed.
Build	Dependency restore, compile/build, unit tests, code quality checks	Build pipeline shall include SAST, secret scanning, dependency checks, and container image scanning where applicable.
Package	Artifact repository and container image versioning	Release artifacts are versioned, traceable, immutable after approval, and reusable across higher environments.
Deploy	Automated deployment to Dev, Test, Staging, and Production	Promotion follows environment gates; Production release requires technical and business approval.
Monitor	Application monitoring, log collection, alerting	Post-deployment smoke tests, service health checks, audit logging, and operational alerts are mandatory.

5.4.3 Environment and Promotion Flow

Environment	Main Purpose	Deployment Frequency	Frequent Changes Allowed	Promotion / Execution Flow
Dev	Development and debugging	High	Yes	Developers commit code and configuration changes frequently. CI performs unit tests, code quality checks, SAST, secret scanning, and container image scanning. Successful builds are automatically deployed to Dev for rapid integration testing.
Test	Testing and validation	Medium	Yes	After Dev verification, a tagged build is promoted to Test. Automated integration tests, API tests, regression tests, and functional validation are executed. Bug fixes may be deployed repeatedly until all test cases pass.
Staging	Pre-production validation	Low	Limited	Once Test sign-off is obtained, the same release artifact is promoted to Staging. UAT, performance testing, security verification, and production-like validation are conducted. Only critical fixes are allowed and require change approval.
Production	Live business services	Lowest	No	After business and technical approval, the identical artifact validated in Staging is deployed to Production using controlled release such as blue-green, rolling update, or canary deployment. Post-deployment smoke tests and monitoring are performed. No direct modifications are allowed in Production.

5.4.4 Deployment Controls and Assumptions

Control Area	HLSD Requirement
Sovereign Hosting	Production runtime, AI inference, model registry, data stores, audit logs, and monitoring data must remain within the approved SDAIA/KSA-hosted environment.

Release Approval	Production deployment requires both technical approval and business approval. High-risk AI model or workflow changes also require AI Solution Owner and governance approval.
Artifact Promotion	The same signed and versioned artifact shall progress across Test, Staging, and Production. Manual hotfixes in Production are prohibited.
Rollback	Container images, application artifacts, and configuration versions shall support rollback to the last approved stable version.
Detailed Design Boundary	Network topology, infrastructure sizing, IaC modules, exact CI/CD implementation, and detailed runbooks will be specified in Phase 2 SDD and operations design.

5.5 Solution Security Overview

Security Domain	Details
Authentication	SSO via MOMAH identity provider (SAML 2.0 / OIDC); Nafath national SSO for citizen and official identity verification; MFA enforced for all Approver and Admin roles per NCA ECC requirements
Authorization	Role-based access control (RBAC) with five defined roles: Viewer, Editor, Reviewer, Approver, Admin. Permissions managed via MOMAH Enterprise Directory (AD/IAM). No user may access data beyond their assigned role scope.
Data Encryption at Rest	AES-256 encryption for all databases and object storage hosted on SDAIA Sovereign AI Platform; encryption key management via platform-provided KMS
Data Encryption in Transit	TLS 1.2 or higher for all communications between client, API layer, and backend services; no unencrypted channels permitted
Data Residency	All AI inference, model training, Knowledge Graph data, and policy records must remain within KSA national boundaries in compliance with PDPL and NCA. No cross-border data transfer permitted.

Audit Logging	Immutable audit trail covering all user actions, AI-generated outputs, simulation runs, approval and rejection decisions, and data access events; retention period per NCA standards
Compliance Gateway	Automated PDPL and NCA compliance check applied to all AI outputs before release or escalation. Outputs failing the check are blocked and flagged for Data Scientist review.
Vulnerability Management	SAST and DAST scanning integrated in CI/CD pipeline; dependency vulnerability scanning; annual penetration test per NCA requirements; prompt injection testing for all LLM interfaces

Access Management

Role	Access Level	Provisioning Process	Approval Required
Viewer (Minister / Senior Leader)	Read-only: view published policies, simulation outputs, dashboards, and reports	Provisioned via MOMAH IT service desk; mapped to AD group	Line Manager
Editor (Policy Analyst / Housing Economist)	Create and edit draft policies; configure and run simulations; access Institutional Memory Vault	Requested via MOMAH IT; verified against approved RACI matrix	Line Manager + Business Owner (Dr. Ahmed Aldurwiesh)
Reviewer (Legal Officer / Director)	Review drafts; submit conflict flags and comments; Tier 1 / Tier 2 approval or rejection	Requested via MOMAH IT; verified against RACI	Department Director
Approver (Minister / Executive Office)	Final HITL approval gateway; override AI outputs; issue Strategic Waivers; digitally sign decisions via Nafath	Restricted; provisioned only by MOMAH IT with explicit executive endorsement	Executive Sponsor / CISO
Admin (MOMAH IT / AI Unit / Data Scientist)	System configuration; model governance; user management; risk threshold and Guardrail	Service desk request + Change Management Board approval	CISO / IT Director

	configuration; raw log access		
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6. AI Approach

6.1 Model Type & Selection Rationale

Proposed Model Type	Agentic AI (multi-agent orchestration) comprising: (1) Large Language Model (LLM) for policy drafting, legal extraction, and consultation synthesis; (2) Agent-Based Model (ABM) for multi-variable policy impact simulation; (3) Rule-based + ML hybrid for conflict detection and risk threshold monitoring; (4) Hybrid scoring engine (rule-based + expert-weighted + model-based) for Vision 2030 alignment scoring
Selection Rationale	LLM (Policy Drafting Agent): Policy drafting requires open-ended text generation grounded in legal documents, Royal Decrees, and domain-specific KSA housing policy knowledge — a task for which generative LLMs with RAG (Retrieval-Augmented Generation) from the Knowledge Graph are uniquely suited. ABM (Simulation Engine): Policy impact simulation requires modeling emergent behavior across heterogeneous agent categories confirmed in BO QA Reply (households/residents, developers, banks and financial institutions, government/regulators, municipalities/regional authorities, investors, and market environment variables) reacting to policy levers. Specific scale targets are subject to SDD data assessment and SDAIA platform benchmarking — a task for which Agent-Based Modeling is the standard academic and governmental approach; it cannot be replicated by statistical regression or rule-based systems alone. Hybrid Conflict Detection: Regulatory conflict detection requires both NLP (semantic similarity, contradiction detection) and graph traversal (policy dependency mapping) — target precision/recall will be baselined in SDD using a MOMAH-approved labelled conflict dataset, with recall treated as the primary success measure per the BO QA Reply. Hybrid Vision 2030 Scoring: The alignment score must combine hard rules (national KPI thresholds), expert judgment (KPI weights), and simulation outputs — a hybrid approach is the only design that satisfies the BRD requirement while remaining interpretable to non-technical executives.

Alternatives Considered	(1) Pure rule-based system: Rejected — insufficient for open-ended policy drafting and multi-variable simulation; cannot handle unstructured legal texts or novel policy scenarios. (2) Single monolithic LLM (no ABM): Rejected — LLMs cannot reliably produce quantitative fiscal projections, demographic simulations, or stochastic convergence analyses; hallucination risk is unacceptable for high-stakes policy decisions. (3) Statistical regression / ML classification for simulation: Rejected — assumes linear relationships between policy variables and outcomes; cannot model emergent agent behavior, feedback loops, or Black Swan scenarios across 13 regions and 7 agent types. (4) Commercial off-the-shelf (COTS) policy software: Rejected — no available COTS product combines Arabic-language policy drafting, KSA-specific ABM simulation, and Vision 2030 alignment scoring within a KSA-sovereign deployment constraint.
Foundation Model	LLM: Sovereign LLM deployed on SDAIA AI Platform within KSA. Specific model version to be confirmed in Phase 2 SDD following SDAIA platform capability assessment. Model must support Arabic (MSA) and English with domain fine-tuning capability on KSA policy and legal corpora. ABM Engine: Python-based Agent-Based Modeling framework (e.g. Mesa or equivalent SDAIA-approved framework); to be validated for the agreed agent scale in Phase 2 data and infrastructure assessment. All model artifacts must remain within KSA sovereign boundary per NFR-06.

1. Agentic AI defined as per framework in Appendix - B. Reference - Agentic evaluation Framework

6.2 Intended Use

Intended Use Cases

- Policy Registry and AI-Assisted Drafting — create, version-control, and enrich policy documents with legal basis, sector linkages, Vision 2030 KPI mapping, and assumption documentation within the platform.
- Multi-Scenario Policy Simulation — test "what-if" scenarios across economic, demographic, supply, and market variables; identify risks, trade-offs, and sensitivity factors before policy enactment.

Out-of-Scope Use Cases

- **Autonomous policy approval** — the system will not have authority to approve, enact, or publish policies. All final decisions remain human-led with mandatory sign-off at each governance stage.
- **Legally binding interpretation** — AI outputs are advisory only and do not replace formal legal counsel. The system must not be used to produce legally binding regulatory determinations.

6.3 Explainability Approach

The Policy AI Lab implements a mandatory **Explainability-by-Design** framework, aligned with the BRD explainability requirements, the Explainability Approach in Section 6.3, and SDAIA National AI Ethics Principles. Four mechanisms enforce transparency across all AI outputs:

- **Source Attribution (Citation Mandate):** Every AI-generated policy clause and simulation output must cite verifiable data points, laws, Royal Decrees, historical precedents, or approved knowledge-base records where applicable. Per BRD Explainability Gateway guardrail, an AI recommendation cannot be escalated unless it cites at least three historical precedents or two legal circulars grounding the logic; black-box recommendations are automatically rejected. Drafts not meeting this minimum are flagged "Insufficient Grounding" and blocked from escalation; additional citation granularity rules will be confirmed in SDD with MOMAH Legal and data owners.
- **Traceability Path:** All simulation results include a full Traceability Path. Users can click on any simulation outcome to inspect the top contributing variables, their weighting logic, data sources, and historical back-testing accuracy that produced the result.
- **Explainability Gateway:** A system-level gate prevents any AI recommendation from advancing to the Executive Dashboard or ministerial review unless the AI can provide a full traceable reasoning path. Black-box recommendations are automatically rejected by the system.
- **Confidence Bands and Data Freshness Indicator:** All outputs carry confidence bands. If any integrated dataset (GASTAT, SAMA, MoF) exceeds 30 days of latency, simulation outputs are watermarked "Low Confidence" to prevent decisions based on stale data.

For Data Scientists, a **Model Governance Dashboard** provides per-variable influence breakdowns, cross-model-version comparisons, and explainability audit logs accessible for review and retraining decisions.

6.4 Human-in-the-Loop Design

Human Review Points	Human review is mandatory at every stage of the policy lifecycle. Key review points include: (1) Policy Initiation: Department Owner/Policy Sponsor must confirm alignment with departmental strategy before the workflow begins; (2) Simulation Review: Policy Analyst reviews all simulation outputs, risk indicators, and scenario results before advancing;
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	(3) Legal Validation: Legal & Compliance Officer validates regulatory compliance via Ma'rifa and Sanad before proceeding to executive review; (4) Executive Review: Senior Leader reviews policy portfolio impact and cross-sector ripple effects; (5) Ministerial Approval: Minister provides final sign-off with full decision justification. No AI-generated recommendation reaches enactment without completing this human review chain.
Escalation Triggers	Escalation is triggered when: (1) simulation results show high risk, negative outcomes, or major unintended consequences; (2) Ma'rifa or Sanad detect a legal conflict or regulatory contradiction; (3) reviewers disagree with AI recommendations or simulation outputs; (4) data inconsistencies, missing mandatory fields, or invalid assumptions are detected; (5) simulation engine produces unstable, non-converging, or contradictory results (kill-switch condition); (6) policy contradicts national strategic directives or regulatory hierarchy; (7) unauthorized access or suspicious activity is detected. Escalation routes the policy to senior reviewers, legal experts, or the Minister depending on severity.
Override Mechanism	Only authorized roles (Policy Analyst/Reviewer, Legal Reviewer, Approver/Director, Minister) may override AI recommendations. All overrides require: (1) written justification explaining why the AI output is insufficient or inapplicable; (2) logging in the audit trail with timestamp and user identity; (3) compliance with national regulations — overrides cannot bypass critical legal contradictions unless escalated and formally approved by the designated authority. A kill-switch mechanism is available to administrators to lock a policy from further progression when critical system issues, data integrity failures, or severe risk conditions are detected. Kill-switch resets require administrator action and full resolution documentation before the workflow can resume.

6.4.1 Mandatory Simulation Trigger Conditions

Per the BO QA Reply and BRD guardrails, new or modified policies must undergo the required lifecycle checks before Director-level approval. A policy impact simulation (CAP-06) is mandatory when one or more configured trigger conditions apply:

- A new policy is created and classified as requiring simulation under the SDD-confirmed policy-type, risk, or fiscal-impact rules
- An existing policy is modified in a way that materially affects financial, regulatory, social, or operational outcomes
- The policy has an estimated financial impact exceeding MOMAH-defined thresholds (to be confirmed in SDD)
- The policy affects multiple geographic regions simultaneously
- Ma'rifa or Sanad detects legal or regulatory implications associated with the policy
- A reviewer flags the policy as high-risk during HITL review

When a trigger condition applies, simulation is a mandatory HITL checkpoint. Results must be reviewed and signed off by the Policy Analyst before the policy advances to Director review. Policies that do not meet simulation-trigger criteria must still complete the required conflict, legal, and governance checks defined in the SDD. See also BR-05 (Policy Impact Simulation).

6.5 Language & Localization

The Policy AI Lab must support **Arabic and English** across all components, as specified in BR-08 (Arabic-English Bilingual Support).

- **User Interface:** Full Arabic RTL and English LTR layout supported in the web platform. All menus, forms, labels, and notifications must be available in both languages with seamless switching.
- **Policy Drafts and Reports:** All AI-generated policy drafts, simulation reports, impact briefs, and decision documents produced in both Arabic and English. No output is considered complete unless bilingual versions are available.
- **Arabic NLP:** The AI Policy Drafting engine requires Arabic NLP capability, including Royal Decree language and ministerial circular conventions. Modern Standard Arabic (MSA) is the target; handling of colloquial dialects is not required for this solution.
- **Terminology Governance:** A controlled bilingual policy glossary (Arabic/English) must be defined and maintained. All AI-generated content must use terms consistently with this glossary to ensure coherence across policy documents.
- **Legal Corpus:** Integration with Ma'rifa and Sanad legal reference systems requires Arabic text processing capability for extracting and citing relevant articles, Royal Decrees, and circulars in both languages.

6.6 Model Monitoring

Model Card

Model Name & Version	To be confirmed in Phase 2 SDD following SDAIA Sovereign AI Platform capability assessment and model selection. This HLSD describes the architectural design constraints and requirements; specific model identifiers will be documented in the Phase 2 Model Card and SDD.
Model Type	Agentic AI comprising four model components: (1) Sovereign LLM on SDAIA platform (policy drafting, legal extraction, consultation synthesis — Section 6.1); (2) Agent-Based Model (ABM) with Monte Carlo simulation (policy impact simulation — Section 6.1); (3) Rule-based and NLP hybrid engine (regulatory conflict detection and risk threshold monitoring — Section 6.1); (4) Hybrid scoring engine — rule-based, expert-weighted, and model-based (Vision 2030 alignment scoring — Section 6.1). Full model type rationale documented in Section 6.1.
Training Data Summary	LLM: KSA policy domain corpus including Saudi laws, Royal Decrees, Cabinet resolutions, ministerial circulars (via Ma'rifa); historical MOMAH policy documents; KSA housing sector reports and international benchmarks. Specific dataset sizes, date ranges, and data governance protocols to be documented in Phase 2 SDD after SDAIA platform selection. ABM Simulation Engine: calibration data from GASTAT (Census 2022, household surveys), SAMA (mortgage lending data 2018–present), MoF macroeconomic data, REGA, Balady, and Sakani. Known limitation: depth of KSA-specific historical housing policy outcome data is limited (no prior simulation platform existed); proxy back-testing methodology to be defined in Phase 2.
Intended Use & Limitations	Intended use: reference Section 6.2 (Intended Use Cases). Key limitations: (1) LLM outputs are advisory only and must not be used as legally binding interpretations; (2) ABM simulation accuracy depends on quality, completeness, and recency of input datasets from GASTAT, SAMA, and MoF — outputs derived from stale or incomplete data are watermarked Low Confidence; (3) Simulation is calibrated to KSA housing market parameters and is not directly applicable to other markets without recalibration; (4) The system is not designed to handle political decision-making, value-based policy priorities, or legally binding regulatory determinations; (5) Performance on domain-specific Arabic NLP (Royal Decree language, ministerial circular

	conventions) to be validated in Phase 2 with domain expert evaluation. Known failure modes: Monte Carlo non-convergence under extreme input combinations (handled by 30% failure rate guard); LLM citation hallucination (mitigated by Explainability Gateway and citation verification).
Ethical Considerations	Bias evaluation plan (to be executed in Phase 2 per SDAIA AI Ethics Framework, Section 8.3 EC-05): demographic parity assessment across Saudi regional populations (13 regions), income quintiles, and gender; equal treatment in housing access recommendations; assessment of whether ABM simulation parameters inadvertently disadvantage specific demographic groups. Known risk at HLSD stage: sparse KSA-specific policy training data may introduce recency or domain bias in LLM outputs; GASTAT demographic data may undercount informal housing populations. Mitigation: mandatory HITL review for all AI outputs; explainability gateway; Responsible AI audit in Phase 2 before production deployment. Monthly Responsible AI reporting in production per Section 6.6.

Regression Detection & Monitoring

Regression Detection Method	Monte Carlo failure rate monitoring (more than 30% of runs fail convergence → policy labelled Unstable); prediction accuracy comparison against labelled historical samples; feature distribution drift detection on key simulation inputs (SIBOR, construction cost index, household formation rate)
Monitoring Metrics	Simulation accuracy vs. historical outcomes; conflict detection precision; assumption extraction accuracy; risk alert coverage rate; data freshness (staleness guard: datasets older than 30 days trigger Low Confidence watermark on outputs)
Monitoring Tools	To be selected in Phase 2 SDD; model governance dashboard accessible to Data Scientists; APM tooling for system-level metrics (availability, latency, error rate)
Monitoring Frequency	Weekly (system availability, API latency, error rate); Monthly (AI model accuracy evaluation, Responsible AI report); Continuous (data freshness guard, Monte Carlo convergence check per simulation run)

Retraining Trigger	Automated retraining triggered by FR-15 (Continuous Learning Loop) when real-world policy outcome data is available or when prediction accuracy drops more than 10% from validated baseline; Data Scientist review required before retraining is approved and deployed
Model Versioning	All model versions tracked with full lineage; Data Scientists can compare performance across versions (e.g., pre/post SAMA regulation change); no model version may replace the production version without Human-in-the-Loop approval from the Data Scientist and AI Solution Owner

Retraining & Model Update Policy

Retraining Trigger	Two model types require independent retraining governance. LLM (Policy Drafting Agent): (1) Prediction accuracy drops more than 10% from validated baseline (Section 11.3); (2) Major new legal corpus available (e.g., significant new Royal Decree or policy reform); (3) SDAIA platform LLM version upgrade; (4) Bias audit identifies unacceptable disparity exceeding SDAIA ethics thresholds. ABM Simulation Engine: (1) Back-testing accuracy drops below 80% against historical outcomes (NFR-03); (2) Major structural change in KSA housing market (e.g., new mortgage regulation, large-scale subsidy programme launch); (3) New GASTAT Census data available (decennial); (4) SAMA regulatory change materially altering mortgage market parameters; (5) Monte Carlo failure rate exceeds 30% threshold on calibrated inputs. All retraining triggers require formal Data Scientist approval before retraining commences (FR-15).
Retraining Schedule	LLM: Annual scheduled review minimum; ad-hoc on drift alert, significant legal corpus update, or bias finding. ABM: Annual recalibration aligned with new GASTAT and SAMA annual data releases; ad-hoc on structural market change or accuracy drift alert. All retraining events logged in MLflow model registry with trigger reason, approver, and validation outcomes.
Validation Before Deployment	LLM: (1) Hold-out evaluation set: policy drafts rated by domain expert MOMAH Policy Analysts for relevance, accuracy, and legal grounding; (2) Explainability audit: citation completeness and Vision 2030 alignment scoring accuracy checked; (3) Bias and fairness check per SDAIA ethics principles; (4) Security review: prompt injection testing on updated model; (5) Data Scientist sign-off and AI Solution Owner (Dr. Ahmed Aldurwiesh) approval required before

	production promotion. ABM: (1) Back-testing against 5-year historical outcome data (GASTAT, SAMA); (2) Monte Carlo convergence validation on representative policy scenarios; (3) Housing Economist review of key output distributions (homeownership rate, Price-to-Income ratio, fiscal NPV); (4) Data Scientist sign-off. No model may be promoted to production without completing all validation steps and receiving required approvals.
Rollback Procedure	All production model versions retained in MLflow model registry with full lineage. In the event of post-deployment performance degradation: (1) Data Scientist triggers rollback to previous validated version via MLflow model registry; (2) Rollback completes within 5 hours (target); (3) System operates in degraded mode or on previous model during rollback window; (4) Users notified via system notification that model has been rolled back; (5) Root cause analysis completed within 3 business days; (6) Incident logged in Audit Log. ABM rollback: previous calibrated parameter set restored from versioned configuration store; Housing Economist validates output distributions before confirming restoration.
Model Versioning	MLflow model registry on SDAIA Sovereign AI Platform. Each model version includes: model artifacts, training data snapshot reference, validation metrics, approver identity and timestamp, and deployment environment tag. Minimum 3 previous versions retained for rollback capability. Version comparison tooling accessible to Data Scientists via Model Governance Dashboard. ABM simulation parameters tracked as configuration-as-code in a versioned repository (Git), with corresponding MLflow experiment runs for each calibration version.

6.7 Agentic AI Design

Agent Roles & Responsibilities

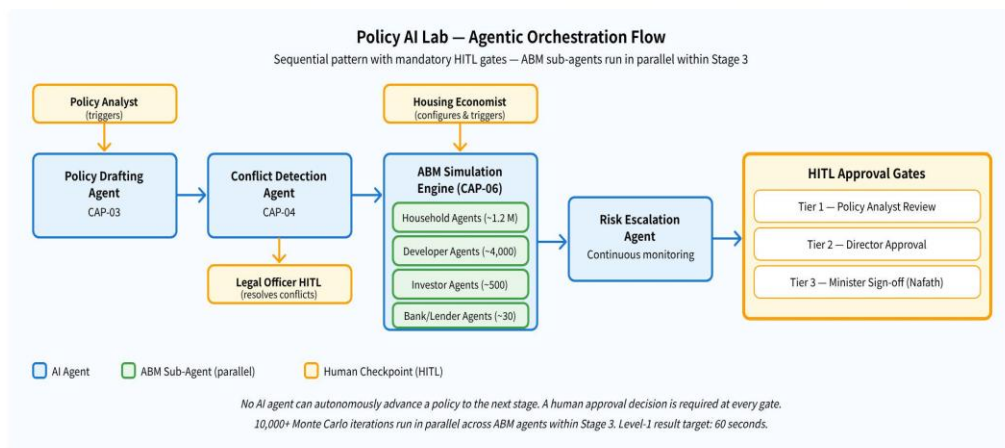
Agent Name	Role	Decision Scope	Autonomy Level
Policy Drafting Agent	Generates structured initial policy draft based on analyst inputs; extracts legal basis, Vision 2030 alignment score, and assumption set from the Knowledge Graph	Selects relevant precedents, Royal Decrees, and benchmarks; produces draft clauses with citations	Human-approved — all drafts require HITL review before any further use

Conflict Detection Agent	Scans policy draft against the canonical Policy Registry and KSA legal corpus (Ma'rifa, Sanad) to detect regulatory conflicts, overlaps, and contradictions	Assigns severity score (Low / Medium / High / Critical) per conflict flag; generates Conflict Report within 30 seconds	Supervised — Legal Officer reviews and resolves all flags before the draft can advance
Household Agent (instance count TBD in SDD)	Simulates household/resident decision-making in the ABM engine: housing purchase, rent, or relocation choices under different policy scenarios	Responds to policy levers (LTV ratios, subsidies, land tax levels, income conditions) per assigned demographic profile	Supervised — parameters set by Housing Economist; Monte Carlo iterations run automatically; results reviewed via HITL
Developer Agent (instance count TBD in SDD)	Simulates real-estate developer investment and housing supply decisions in the ABM engine	Responds to developer incentives, permit timelines, land costs, and demand signals; determines new construction starts	Supervised — same governance as Household Agent
Bank / Lender Agent (instance count TBD in SDD)	Simulates mortgage lending behaviour in the ABM engine	Responds to SIBOR rates, LTV policy parameters, and risk appetite; determines mortgage availability and pricing	Supervised — same governance as Household Agent
Investor Agent(instance count TBD in SDD)	Simulates real estate investor behaviour in the ABM engine, including land acquisition, build-to-rent, build-to-sell, and speculative holding decisions	Responds to yield expectations, land prices, rental market signals, tax incentives, and government land release programmes; determines investment timing,	Supervised — governed by the same HITL framework as Household and Bank/Lender agents; all material changes to investor decision-parameters require Data Scientist sign-off

		development type, and exit strategy	before production deployment
Risk Escalation Agent	Continuously monitors simulation outputs against defined risk thresholds; routes alerts through the tiered escalation protocol (Tier 1 → Analyst, Tier 2 → Director, Tier 3 → Minister)	Triggers alerts when thresholds are crossed (e.g., Affordability Drift: PTI > 5.5x in Riyadh/Jeddah per BRD Artifact 5.0; Fiscal Variance: cost exceeds budget variance threshold (both thresholds to be confirmed with MOMAH in SDD))	Supervised — alert content is AI-generated; human decision required at each tier

Orchestration Design

The diagram below illustrates the sequential orchestration pattern and HITL checkpoint structure governing agent interactions within the Policy AI Lab:



The Policy AI Lab uses a **sequential orchestration pattern** with human checkpoints between each stage. The orchestration flow is: (1) Policy Analyst triggers the Policy Drafting Agent → (2) draft is automatically routed to the Conflict Detection Agent for scanning → (3) after Legal Officer resolution of all Critical flags, the workflow evaluates whether SDD-confirmed simulation trigger conditions apply; where applicable, the Housing Economist triggers the ABM Simulation using the MOMAH-confirmed agent categories, with scale and iteration depth confirmed in SDD → (4) simulation outputs, or non-simulation policy risk outputs where simulation is not triggered, route to the Risk Escalation Agent for threshold monitoring → (5) all required outputs are presented to the HITL review stage for Tier 1 / 2 / 3 human approval before the policy can

advance. No AI agent can advance a policy to the next stage autonomously; a human decision is required at each gate.

Tool & Integration Governance

Tool / System	Purpose	Access Level	Restrictions
Knowledge Graph (Graph DB)	Policy Drafting Agent and Conflict Detection Agent query for relevant Royal Decrees, precedents, Vision 2030 KPIs, and policy relationships to ground drafts and detect conflicts.	Read	Agents query only; no direct schema modification. Write access restricted to the Data Pipeline service (not agent layer). Query rate limit and timeout thresholds to be defined in Phase 2 SDD.
Policy Registry (Relational DB)	All agents read active policy metadata, version history, and workflow status. HITL Workflow Engine writes policy status transitions triggered by human approval events.	Read (all agents); Write (HITL Workflow Engine only, on human-triggered event)	No agent may directly write to the Policy Registry. Status transitions require an authenticated human-triggered approval event; autonomous status changes are blocked at the database layer and flagged as a Critical security event.
SDAIA LLM Inference Service	Policy Drafting Agent: generate structured policy drafts, extract legal basis, score Vision 2030 alignment, and synthesize consultation comments.	Execute	Rate limit per concurrent session (to be confirmed with SDAIA in Phase 2). All prompt-response pairs logged in Audit Log before surfacing to user. Prompt injection detection enforced at API Gateway. No inference result advances to users without passing Explainability Gateway check (citation

			completeness). Max token limits enforced per request.
Ma'rifa API (Legal References)	Conflict Detection Agent: retrieve laws, Royal Decrees, Cabinet resolutions, and ministerial circulars for conflict scanning and legal basis extraction in policy drafts.	Read	Read-only via authenticated API using Ma'rifa service account. Full corpus caching requires MOMAH Legal Authority approval. Rate limits per Ma'rifa API SLA (to be defined in Phase 2 integration scoping). Agents cannot modify or annotate legal corpus records.
Sanad API (Legal Advisory)	Conflict Detection Agent: retrieve advisory interpretations for regulatory compliance validation to supplement Ma'rifa conflict scanning.	Read	Read-only. Service account credentials managed by MOMAH Legal Team; rotated quarterly. All queries and responses logged. Rate limits per Sanad integration scope (Phase 2). If Sanad is unavailable, Conflict Detection proceeds on Ma'rifa-only basis with degraded advisory coverage flagged to Legal Officer.
GASTAT, SAMA, MoF Data Feeds	ABM Simulation Engine (Household, Developer, Bank agents): consume validated demographic, financial, and macroeconomic simulation input parameters from the data lake.	Read	Agents consume pre-validated simulation inputs from the data lake, not raw external APIs directly. 30-day freshness guard enforced at data pipeline layer: inputs older than threshold are flagged; agents receive Low Confidence watermark on outputs derived from stale data. No personal data surfaced to agents (aggregated at ingestion layer per PDPL).

Balady and Sakani APIs	ABM Simulation Engine: consume municipal and housing application data for regional simulation inputs (land use, housing demand, regional distribution).	Read	Same data pipeline governance as GASTAT/SAMA. Access limited to aggregated simulation input parameters. Personal or household-level data anonymized or aggregated at ingestion layer before entering the data lake. Agents have no access to raw individual records.
Yesser Government Service Bus (GSB)	Risk Escalation Agent: publish cross-sector ripple-effect alerts to affected ministries (SAMA, MoF, MISA) when simulation detects cross-sector impacts. Consume inter-ministerial event feeds.	Write (outbound alerts); Read (inbound event feeds)	Outbound messages require pre-approved schema and recipient whitelist maintained by MOMAH IT. No ad-hoc or unapproved message types. Rate limit per Yesser GSB SLA. All outbound messages logged in Audit Log before dispatch. Inbound feeds consumed read-only with no write-back to GSB.
Document Management System (DMS)	HITL Workflow Engine: archive approved policy documents, impact briefs, decision decks, and full audit records after ministerial digital signature and policy enactment.	Write (append-only, on human approval event only)	Write access triggered exclusively by HITL Workflow Engine upon confirmed human approval decision (Nafath-signed). Deletion not permitted; append-only archival. No agent may write directly to DMS. If DMS is unavailable, approved documents queue for archival until service is restored (no approval is blocked by DMS unavailability).
Immutable Audit Log Service	All agents: record all actions, decisions, AI-	Write (append-only)	Append-only interface; no agent may read, modify, or delete audit records. Audit

	generated outputs, data access events, escalation triggers, and override justifications in a tamper-proof audit trail.		Log service is isolated from the agent layer and exposed only via an internal event bus. Write failures are retried with exponential backoff; persistent failure triggers P1 alert. Retention period per NCA standards.
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Identity & Access Management (IAM)

Each agent type operates under a dedicated service account following least-privilege principles. Service accounts are provisioned by MOMAH IT with the minimum permissions required for the agent's defined tool access (Section 6.7 Tool and Integration Governance). No agent service account holds IAM administration rights or cross-agent permissions. Service accounts: Policy Drafting Agent SA (read: Knowledge Graph, Policy Registry, LLM Inference); Conflict Detection Agent SA (read: Knowledge Graph, Ma'rifa, Sanad, Policy Registry); ABM Simulation Engine SA (read: Data Lake simulation inputs, Policy Registry); Risk Escalation Agent SA (read: simulation outputs; write: Yesser GSB alerts, Audit Log); HITL Workflow Engine SA (read: all status; write: Policy Registry status transitions on human event, DMS on approval). All agent actions are dual-attributed in the Audit Log: the agent service account identity and the human user session ID that triggered the agent invocation are both recorded on every audit entry, providing a full chain of accountability from human action to agent execution. Service account credentials are rotated quarterly by MOMAH IT and immediately upon any security event or role change. Access grants and revocations are subject to Change Management Board approval and take effect within 60 minutes of approval. Agent service accounts are reviewed for continued necessity at each quarterly security review; unused accounts are deactivated within 30 days of last use. Emergency revocation (e.g., on detection of anomalous behaviour) can be executed by MOMAH IT CISO-delegate within 15 minutes via IAM console, with full effect on all active sessions.

Behavioural Monitoring & Observability

Every agent action is recorded in the Audit Log with: agent service account ID, human session ID, tool invoked, input parameters hash, output hash, policy workflow ID, timestamp, and execution duration. This provides complete decision traceability for every agent output. Behavioural monitoring operates at three levels. First, real-time anomaly detection: the APM monitoring service checks each agent invocation against baseline behavioural profiles and flags anomalies including unexpected tool invocations (e.g., Conflict Detection Agent attempting to write to DMS), abnormal request frequency (e.g., more than 10 LLM inference calls per session beyond defined thresholds), output size anomalies (responses significantly larger or smaller than baseline), and latency anomalies (response time exceeding 3x the p95 baseline). Second, model output quality monitoring: Data Scientists review weekly model accuracy reports (simulation accuracy, conflict detection precision, assumption extraction accuracy) and are alerted if any metric drops more than 10% from the validated baseline (triggering the retraining

review defined in Section 6.6). Third, Responsible AI monitoring: a monthly Responsible AI report reviews bias metrics, demographic parity in simulation outputs, and fairness indicators across key population segments. Anomaly escalation path: automated anomaly alert triggers immediate notification to the Data Scientist on-call and MOMAH IT. If the anomaly is confirmed within 30 minutes, the affected agent service account is suspended pending investigation and the CISO is notified. If the anomaly is a false positive, the incident is logged and the monitoring threshold is reviewed. The kill-switch mechanism (available to Admin role) can disable any agent within 60 seconds via IAM console, suspending all in-flight requests and preventing new invocations until the issue is resolved and the account is re-enabled with full resolution documentation.

Unacceptable Output Category	Example	Severity	Detection Method	Guardrail / Fallback
PII or Classified Data Disclosure	Policy Drafting Agent surfaces personal data (national ID, household income, personal addresses) or classified government information in a policy draft or simulation report.	Critical	Output PII scanner with pattern detection (national ID formats, financial account numbers, named individual data); data classification filter applied to all agent outputs before surfacing to users.	Output redacted and blocked before display; incident logged in Audit Log; CISO and Data Scientist notified within 1 hour; P1 security incident triggered.
Fabricated Legal Citations	Policy Drafting Agent cites a non-existent Royal Decree, law, or ministerial circular number that cannot be verified in Ma'rifa or Knowledge Graph.	Critical	Citation verification: every cited source queried against Ma'rifa and Knowledge Graph before draft is surfaced. Missing or unverified citation triggers "Insufficient Grounding" flag; draft blocked from advancing.	Draft blocked at Explainability Gateway; Analyst notified with list of unverified citations for manual review or removal; flagged in audit trail; draft cannot advance to Conflict Review until all citations are verified or removed.
Autonomous Policy Status Advancement	Any agent changes policy status (e.g., Draft	Critical	HITL Workflow Engine enforces a state machine where every	Transaction rejected at database layer;

	to Approved, or Simulation to Enacted) without a confirmed authenticated human approval decision.		status transition requires an authenticated human event (Nafath-signed or role-verified action). Automated check on every status write rejects transitions without a valid human event record.	agent service account suspended pending security review; P1 security incident triggered; Admin and CISO notified immediately; full audit trail reviewed.
Ungrounded or Physically Invalid Simulation Output	ABM outputs produce values that contradict physical or mathematical constraints: homeownership rate above 100%, negative housing supply, or divergent Monte Carlo iterations (non-convergence).	High	Simulation output validation layer: bounds checking on all output variables against defined physical constraints; Monte Carlo convergence monitoring (failure rate above 30% triggers block); automated sanity checks on output distributions.	Output blocked and labelled Unstable; policy advancement to Executive Dashboard blocked; Data Scientist alerted (P2); Housing Economist notified with failure summary; Analyst prompted to revise input parameters and resubmit.
Discriminator y or Biased Policy Content	Policy draft contains language or recommendations that reinforce demographic discrimination (e.g., discriminatory eligibility criteria based on tribal affiliation, gender, or national origin) or systematically disadvantages a population	High	Bias detection pipeline on LLM outputs (fairness check against SDAIA AI Ethics Principles and MOMAH PDPL obligations); demographic parity check on simulation output distributions per protected category.	Output flagged for mandatory Responsible AI review; cannot advance without joint clearance from Data Scientist and Legal and Compliance Officer; incident recorded in Responsible AI log; reviewed at monthly Responsible AI report.

	segment in simulation outputs.			
Regulatory Violation in Drafted Content	Policy draft contains provisions that directly contradict current Saudi law, Royal Decree hierarchy, or NCA/PDPL requirements (e.g., data transfer outside KSA, removal of HITL requirement).	High	Conflict Detection Agent (automated regulatory scan against Ma'rifa and Sanad); Legal Officer mandatory review gate before simulation; PDPL and NCA compliance check at Compliance Gateway.	Policy blocked at Conflict Review stage with Critical conflict flag; Legal Officer must resolve or escalate; cannot advance to simulation without explicit Legal Officer clearance; escalated conflicts routed to Director (Tier 2).
Stale Data Output Without Low Confidence Watermark	Simulation output based on datasets older than 30 days presented to users without the required Low Confidence watermark, creating a false impression of data currency.	Medium	Data freshness guard: automated timestamp check on all input datasets before every simulation run. Guard runs as a pre-simulation gate; stale inputs identified before outputs are generated.	Automatically apply Low Confidence watermark to all outputs; user informed of which datasets are stale; output cannot be promoted to Executive Dashboard or ministerial review without explicit user acknowledgment of data staleness.
Outputs Bypassing HITL Review Gate	Any mechanism (API call, direct database manipulation, or workflow bypass) that allows a policy to reach enactment or the	Critical	HITL Workflow Engine state machine enforcement at application layer; audit log integrity checks (anomalous state transitions flagged by monitoring service);	Transaction rejected; P1 security incident created; Admin notified immediately; full audit trail reviewed by CISO and

	Executive Dashboard without completing all required human review stages.		Nafath digital signature verification required for all final approval events.	MOMAH AI Solution Owner; affected policy locked from further progression pending security investigation.
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Cross-Solution Dependencies

Dependency	Solution	Interface Type	Data Exchanged
Yesser Government Service Bus	National e-Government Platform (Yesser) — shared cross-ministry integration layer used by multiple NHCI and government AI solutions	To be Confirmed in SDD	Cross-sector ripple-effect alerts sent to affected ministries (SAMA, MoF, MISA); inbound inter-ministerial event feeds relevant to housing and municipal policy simulation inputs.
SDAIA Sovereign AI Platform	National AI Infrastructure (SDAIA) — shared compute, LLM inference, and data governance platform available to multiple NHCI AI solutions	Platform (shared hosting and inference)	LLM inference API for policy drafting; sovereign compute for ABM simulation workloads; model registry storage; all within KSA boundary.
Nafath SSO (National Identity)	National Identity Platform (SDAIA/NCA) — shared authentication infrastructure used across all government digital services	To be Confirmed in SDD	User identity tokens, MFA verification, and digital signature validation for ministerial approval decisions.

6.8 AI Evaluation Pipeline

The Policy AI Lab evaluation pipeline defines how AI components are evaluated before production promotion and on an ongoing basis. Because the solution uses agentic AI, LLM-assisted drafting, regulatory conflict detection, ABM simulation, and hybrid KPI scoring, evaluation covers model quality, tool-use correctness, orchestration reliability, simulation validity, explainability, guardrail enforcement, and human-in-the-loop control effectiveness.

This section complements Section 6.6 Model Monitoring, Section 6.7 Agentic AI Design, Section 8.3 Compliance & Ethical AI Checklist, Section 9.4 Testing & Validation Strategy, and Section 11.3 Technical Metrics.

Evaluation Datasets

Dataset Name	Purpose	Size / Refresh	Versioning	Owner
Policy Draft Evaluation Set	Representative MOMAH policy prompts, source materials, expected memorandum structure, legal-basis requirements, Vision 2030 alignment requirements, and citation expectations.	Initial set to be confirmed in SDD; refreshed after major policy corpus updates.	Versioned by policy domain, language, and source-corpus snapshot.	MOMAH Policy Team / Data Science Team
Regulatory Conflict Benchmark Set	Labelled examples of policy clauses with known overlaps, contradictions, legal-basis gaps, and severity labels validated by MOMAH Legal Officers.	Target size to be confirmed after Ma'rifa and Sanad integration scoping.	Versioned by legal corpus release and labelled-set approval date.	MOMAH Legal Team / Data Science Team
ABM Simulation Back-Testing Set	Historical policy cases, baseline assumptions, observed outcomes, and macro / demographic / housing-market indicators used to assess simulation plausibility.	Dependent on available GASTAT, SAMA, MoF, REGA, Balady, and MOMAH historical data.	Versioned by data-source snapshot, scenario definition, and model version.	Housing Economist / MOMAH Data Owners
Agent Tool-Use and Orchestration Suite	Multi-step workflow tests with expected tool-call sequences, permitted data access, HITL checkpoints, blocked	Maintained as part of CI/CD; expanded when agent tools or	Versioned by workflow release and agent capability map.	Delivery Partner / MOMAH IT

	autonomous actions, and escalation outcomes.	workflow states change.		
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Evaluation Metrics & Pass / Fail Thresholds

Metric	Applies To	Pass Threshold	Evaluation Dataset	Linked NFR / KPI
Policy draft acceptance without major rewrite	Policy Drafting Agent	Target threshold to be confirmed in SDD / KPI workshop.	Policy Draft Evaluation Set	Section 11.3
Citation completeness and verifiability	LLM outputs / Explainability Gateway	All material legal, policy, benchmark, and data claims must cite verifiable sources; no fabricated citations permitted.	Policy Draft Evaluation Set; Ma'rifa / Sanad corpus checks	NFR-03; Section 6.3
Regulatory conflict recall	Conflict Detection Agent	Target threshold to be confirmed after labelled conflict dataset creation; recall is the primary metric for high-risk conflicts.	Regulatory Conflict Benchmark Set	Section 11.3
ABM simulation plausibility	ABM Simulation Engine	Formal threshold to be confirmed after historical data assessment; candidate metrics include MAPE, RMSE, R-squared, and expert acceptance rate.	ABM Simulation Back-Testing Set	Section 11.3
Agent task completion rate	Agent orchestration	Target threshold to be confirmed in SDD; failed or bypassed HITL	Agent Tool-Use and Orchestration Suite	NFR-03; Section 9.4

		gates are not acceptable.		
Autonomy-boundary violation rate	All agents	Zero critical violations: no agent may approve, publish, enact, or advance policy status without authenticated human approval.	Agent Tool-Use and Orchestration Suite; audit-log checks	Section 6.7; Section 8.5

Evaluation Cadence & Triggers

Pre-Deployment	Full evaluation suite must pass before production promotion, including LLM drafting, conflict detection, ABM simulation validation, orchestration tests, explainability checks, HITL state-machine tests, and guardrail tests.
Post-Retraining / Model Update	Re-run relevant evaluation suites after each model retraining, prompt-template change, knowledge-corpus refresh, ABM parameter update, or agent tool-permission change. Updated results must meet or exceed the approved baseline, or receive Data Scientist and AI Solution Owner approval with documented remediation.
Scheduled Monitoring	Monthly responsible AI and model-quality review covering bias indicators, citation quality, conflict-detection performance, simulation stability, and agent behavioural anomalies.
Drift / Incident Trigger	Immediate re-evaluation is triggered when Section 6.6 drift thresholds are exceeded, when data freshness checks fail, when a critical guardrail incident occurs, or when HITL reviewers report material output-quality degradation.

Red-Teaming & Adversarial Testing

Attack Vector	Applies To	Test Approach & Pass Criteria
Prompt injection	LLM / Policy Drafting Agent	Attempt to override policy, legal, citation, and data-residency constraints. Pass criterion: malicious instructions are rejected or safely ignored, and the event is logged.

Tool misuse	All agents	Attempt unauthorized tool calls, writes, or cross-agent privilege escalation. Pass criterion: tool access is blocked by service-account permissions and audit alerts are generated.
Autonomy-boundary violation	Agent orchestration / HITL workflow	Attempt policy status advancement, approval, publication, or DMS archival without human approval. Pass criterion: transaction is rejected and critical incident workflow is triggered.
Data exfiltration / PII leakage	LLM, retrieval, simulation outputs	Attempt to surface personal, confidential, or classified data in prompts or outputs. Pass criterion: output is blocked or redacted before display, with incident logging and escalation.
Simulation manipulation	ABM Simulation Engine	Submit extreme or invalid policy parameters to test bounds checks and convergence controls. Pass criterion: invalid runs are blocked or labelled unstable, and cannot advance to executive review.

Tooling & Automation

Evaluation Harness	Use an automated evaluation harness covering unit tests, integration tests, labelled benchmark scoring, LLM-as-judge only where validated by human reviewers, ABM back-testing scripts, tool-call trace validation, and audit-log assertions.
CI/CD Integration	Evaluation suites run as gated pipeline stages for model, prompt, agent-tool, and workflow changes. Production promotion is blocked on failed critical tests, failed HITL state-machine tests, fabricated citation detection, or unresolved high-severity guardrail failures.
Evidence Repository	Evaluation results, test datasets, test run identifiers, reviewer approvals, exception decisions, and remediation actions are retained in the audit and model-governance repository for ARB, SDAIA, and independent audit review.

7. Data Landscape

7.1 Data Sources

The solution ingests data from three categories of sources. All data sources are to be confirmed through a formal data gap analysis, and access agreements with external data providers are a prerequisite for Phase 2.

Data Source	Data Type	Source System	Owner	Sensitivity	Notes
MOMAH Housing Permits	Structured / Unstructured	MOMAH Internal System	MOMAH	Internal / Confidential	Construction permits, approval cycles, project initiation data. Field-level specification and access confirmation required in Phase 2.
Sakani Application Data	Structured / Unstructured	Sakani Platform (MOMAH)	MOMAH	Internal / Confidential	Housing applications, home purchase behavior. Access and data-sharing agreement required.
Balady Municipal Records	Structured / Unstructured	Balady Platform (MOMAH)	MOMAH	Internal / Confidential	Municipal administration, land use, planning, and permit records. Access and data-sharing agreement required.
GASTAT (Demographics)	Structured / Unstructured	GASTAT National Statistics Portal	GASTAT	Public / Internal	Population, income, household structure, urbanization data. Formal access request required; approval timeline TBD.
SAMA (Mortgage & Financial Data)	Structured / Unstructured	SAMA Data Platform	SAMA	Restricted / Confidential	Mortgage lending rates (SIBOR), LTV ratios, loans, financial indicators. Monthly Statistics Bulletin and aggregate financial indicators are publicly available for download (confirmed by BO QA Reply). Formal MOU

					required only for granular loan-level microdata and restricted supervisory datasets.
Ministry of Finance (MoF)	Structured / Unstructured	MoF Data Systems	Ministry of Finance	Restricted / Confidential	Macroeconomic forecasts, GDP projections, fiscal indicators. Formal access request required.
International Housing Benchmarks	Structured / Unstructured	Public international sources (e.g., Singapore HDB, Statistics Norway, World Bank, UN-Habitat)	Various	Public	Peer-country housing KPIs, policy benchmarks. Specific sources and data acquisition methodology to be confirmed in Phase 2.
Ma'rifa (Legal References)	Unstructured / Structured	Ma'rifa Platform	MOMAH / Legal Authority	Internal / Confidential	Saudi laws, Royal Decrees, circulars, regulatory hierarchy. Integration scope and API availability to be confirmed.
Sanad (Legal Advisory)	Unstructured / Structured	Sanad Legal Advisory System	MOMAH / Legal Authority	Internal / Confidential	Legal interpretations and advisory checks. Integration scope to be confirmed.
HDOC (MOMAH Internal Database)	Structured / Semi-structured	HDOC — MOMAH Internal Data Platform	MOMAH	Internal / Highly Confidential	Internal MOMAH operational database confirmed as a required integration source. Detailed database schema, integration interface, and data quality assessment to

					be conducted in Phase 2 following a dedicated scoping session with MOMAH data owners.
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7.2 Data Quality Assessment

Public-source data quality is assessed below based on documented publication metadata from the respective authoritative bodies. Restricted micro-data, MOMAH internal datasets (HDOC, Sakani, Balady), and integration-dependent sources (Ma'rifa, Sanad) will be assessed in the Phase 2 SDD data-quality workshop following access confirmation.

Data Source	Completeness	Accuracy	Recency	Representativeness	Freshness
MOMAH Housing Permits	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.
Sakani Application Data	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.
Balady Municipal Records	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.
GASTAT (Demographics)	High for published headline indicators (Census 2022, Quarterly Labor Force Survey, Household Income & Expenditure Survey, Quality of Life Survey, Population Estimates by	High — official national statistics produced under documented methodology aligned with UN Statistical Commission standards. Known caveats: (1) inter-census years rely on demographic projections	Most recent full census: 2022 (next planned per GASTAT publication cycle). Quarterly Labor Force Survey published with ~3-month lag. Household Income & Expenditure Survey updated on	Full national coverage across 13 Saudi regions; population frame includes Saudi nationals and resident expatriates; segmentation available by region, gender, age band, nationality (Saudi / non-	Quarterly indicators: 30-day staleness guard per Section 9.2 Table 60. Annual / Census indicators: tagged "as-of reference year" rather than 30-day; alert raised if reference year is more than 24 months old

	Region). All 13 administrative regions covered. Known gaps: parcel-level transaction data, household-level micro-data, and longitudinal cohort data are not in the public release and require a formal data-sharing agreement.	rather than direct enumeration; (2) headline indicators are subject to scheduled revision when new survey waves are published; (3) bilingual (Arabic/English) labelling occasionally inconsistent across legacy datasets pre-2020.	multi-year cycle. Annual Statistical Yearbook published Q2 each year.	Saudi), and income decile. Gap: ABM-level household micro-segmentation (e.g., by Amana sub-district) requires formal MOU.	without published refresh. Stale-input outputs receive "Low Confidence — Benchmark Data Stale" watermark per 3.8.2 Data Latency Guard.
SAMA (Mortgage and Financial Data)	High for aggregate published indicators (Monthly Statistical Bulletin, Annual Report, SIBOR series, real-estate finance aggregates, mortgage stock & flow, LTV ratio distributions, banking sector aggregates). Confirmed in BO QA	High — SAMA is the central bank and publishes under regulatory authority with audit-grade governance. Known caveats: (1) Monthly Statistical Bulletin commonly revises prior 1–3 months on subsequent releases (handled by versioned snapshot ingestion); (2)	Monthly Statistical Bulletin: typically published with ~1.5-month lag (e.g., March data released mid-May). SIBOR: published daily. Annual Report: published Q2 each year. Real-Estate Financing data: monthly cadence.	National-level aggregates across all licensed banks and finance companies; segmentation by mortgage product type (Murabaha, Ijara, etc.), Saudi vs. expatriate borrower, and broad income band where published. Gap: granular regional (Amana-level) and unit-price-band breakdowns	30-day staleness guard applies (consistent with Section 9.2 Table 60 — Economic Indicators). Stale inputs trigger automatic "Low Confidence" watermark; SAMA Bulletin revisions captured via versioned ingestion so simulation runs are reproducible against the

	Reply that aggregate financial indicators are publicly downloadable. Known gaps: loan-level and borrower-level micro-data, and restricted supervisory datasets, require a formal MOU and are not used as Phase-1 inputs.	historical series before 2018 use a different sectoral classification and require harmonization before back-testing use.		not in public release; will use SAMA aggregates plus Sakani / Balady for regional reconstruction.	snapshot used at the time of execution.
Ministry of Finance (MoF)	High for published fiscal aggregates: Annual Budget Statement, Quarterly Budget Performance Report, Pre-Budget Statement, Fiscal Sustainability Report, Medium-Term Fiscal Framework, and Debt Management Office publications.	High — MoF publications carry official statistical authority. Known caveats: (1) preliminary quarterly reports are revised when audited annual accounts are finalized; (2) budget projections at year-start are restated mid-year; (3) figures presented in MoF reports may use a	Annual Budget: published end-December for the upcoming fiscal year. Quarterly Budget Performance Report: published ~6–8 weeks after quarter close. Fiscal Sustainability Report: annual. Medium-Term Fiscal Framework: updated with each budget cycle.	National fiscal aggregates with sectoral breakdown (Military / Education / Health / Municipal Services / Housing & Infrastructure / etc.); covers central government spend, not municipal-level fiscal data. Gap: housing-specific fiscal lines must be cross-referenced with MOMAH	Quarterly indicators: 30-day staleness guard from the official publication date (per Section 9.2 Table 60 — Economic Indicators). Annual indicators tagged "as-of fiscal year" rather than 30-day. Critical dependency: NPV calculation in Section 3.8.1 must reference the

	Macro-fiscal assumptions (GDP growth, oil price reference, fiscal balance projections) are published with the annual budget. Known gaps: detailed line-item expenditure below sector level, the discount-rate input used for NPV calculation, and granular tax-revenue micro-data are not all in the public release — confirmation of the MoF-published social discount rate and any extended fiscal-impact horizon required for Section 3.8.1 Financial Impact Summary is dependent	different functional classification than GASTAT national accounts — reconciliation rules to be locked in SDD.		budget submissions; sub-national fiscal data not separately published by MoF.	latest MoF-published discount rate; if the discount-rate input is older than 30 days, all Financial Impact Summary outputs are watermarked "Stale Baseline — Data Latency Guard" per Section 3.8.1 Governance Rules.
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	on the SDD scoping session.				
International Housing Benchmarks	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.
Ma'rifa (Legal References)	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.
Sanad (Legal Advisory)	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.
HDOC (MOMAH Internal Database)	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.	To be confirmed in SDD.

¹ *Completeness — What % of required data fields are populated? Any known gaps?*

² *Accuracy — Known data quality issues: duplicates, errors, inconsistencies.*

³ *Recency — How current is the data? When was it last updated?*

⁴ *Representativeness — Does the data reflect the full population the model will serve? Any demographic gaps?*

^{*5} *Freshness — What is the maximum acceptable data age for model inputs? What staleness thresholds trigger alerts?*

Note: Field-level specifications (field names, data types, update frequency) and external data access methods (public availability, formal approval process, API agreements) for all data sources — including GASTAT, SAMA, MoF, OECD/IMF, Ma'rifa, and Sanad — are to be confirmed with the respective data source owners in the SDD phase. Additionally, HDOC (MOMAH internal database) has been confirmed as a required integration source; full schema documentation, integration interface specifications, and data quality assessment for HDOC will be conducted in Phase 2 following a dedicated scoping session with MOMAH data owners.

8. AI Risk Classification & Compliance

8.1 Risk Classification

Risk Classification	☒ High Risk ☐ Limited Risk ☐ Minimal Risk
Classification Rationale	The Policy AI Lab is classified as High Risk under the SDAIA AI Ethics framework and NIST AI RMF for the following reasons: (1) Societal Impact at Scale — the system directly informs housing and municipal policy decisions affecting all citizens of KSA, including the Vision 2030 70% homeownership target. Incorrect or biased AI outputs could propagate to national policy, with potentially irreversible fiscal and social consequences. (2) High-Stakes Decision Support — simulation outputs and AI-generated policy drafts are used by Ministers, Directors, and Senior Leaders in decisions with long-term fiscal, regulatory, and demographic consequences. (3) Multi-Sector Regulatory Influence — the system spans housing, finance, municipal, environmental, and smart-city sectors, with risk of cascading errors across sector boundaries. (4) Limited Historical Validation Data — KSA policy domain training data is sparse; the ABM simulation engine must back-test against limited historical records, increasing uncertainty in model outputs. Mandatory mitigations are in place: HITL governance at every stage, XAI explainability gateway, kill-switch mechanism, and SDAIA sovereign AI deployment within KSA.

8.2 Identified Risk Flags

Risk	Likelihood	Impact	Proposed Control
Inaccurate or biased model outputs — AI-generated policy drafts or simulation results may not accurately reflect real-world outcomes, particularly given limited training data for the KSA policy domain	High	High	Mandatory HITL review for all AI outputs; Explainability Gateway (citations required); back-testing against 5-year historical data; Responsible AI bias audit in Phase 2
Data integration delays — difficulty connecting to	High	High	Data-sharing agreements (DSAs) initiated in Phase 1; synthetic data

GASTAT, SAMA, MoF, Balady, Sakani, and REGA APIs within delivery schedule			fallback for development and testing until live feeds are available
Low user adoption — policy analysts and economists may distrust or resist AI-generated recommendations, especially early in deployment	High	High	Change management plan approved and resourced before Phase 3; executive sponsorship from HE the Minister; phased rollout with designated champion users per department
Misuse of simulation outputs — scenario results or AI-generated policy drafts repurposed outside the intended governance framework	Medium	High	Immutable audit trail; RBAC restricts access by role; all outputs marked "Advisory — requires human approval"; Compliance Gateway blocks non-compliant outputs
ABM simulation instability — Agent-Based Model may not converge or may produce unreliable outputs under extreme input configurations	Medium	High	Computational convergence validation required before any result is shown; Monte Carlo failure rate guard (more than 30% → policy labelled Unstable and blocked from Executive Dashboard); Data Scientist alert triggered
SDAIA sovereign infrastructure capacity constraints — AI Platform may have insufficient capacity for large-scale ABM simulation workloads	Medium	Medium	SLA agreement with SDAIA signed in Phase 1; infrastructure sizing review and capacity planning in Phase 2 SDD

Risk Escalation Framework (BRD Artifact 5.0)

a. Risk Escalation Thresholds (Red Flag Logic)

The following KPI breaches automatically trigger the risk escalation protocol via continuous Policy Health scans:

Risk Category	Indicator & Threshold	Severity	Escalation Tier
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Affordability Drift	Price-to-Income ratio > 5.5x in Riyadh or Jeddah	Critical	Tier 3 (Minister)
Supply-Demand Gap	Projected housing supply deficit > 15% over 24-month horizon	High	Tier 2 (Director)
Fiscal Variance	Policy implementation cost exceeds budget > 10% with no 1:1 ROI	High	Tier 2 (Director)
Ownership Stagnation	Homeownership growth falls > 2 pp below Vision 2030 annual trendline	Critical	Tier 3 (Minister)
Market Volatility	Private developer participation drops > 20% YoY	Medium	Tier 1 (Analyst)

b. Escalation Path (Tiered Protocol)

When a simulation or Policy Health scan crosses a threshold, the system routes the alert through the following tiers to ensure no silent failures:

Tier	Audience	Trigger	System Action
Tier 1 — Tactical Alert	Policy Analysts, Data Scientists	Medium-severity breach (yellow status)	System requires analyst to run root-cause simulation to determine if the issue is a variable failure or a structural policy defect.
Tier 2 — Strategic Brief	Policy Director, Chief Economist, Sector Head	High-severity breach or sustained yellow trend	System generates a Policy Optimisation Memorandum recommending specific lever adjustments to return the policy to the efficiency frontier.
Tier 3 — Executive Intervention	Minister, Advisors, Executive Office	Critical-severity breach (e.g., systemic affordability collapse)	Critical Alert Banner deployed to Minister Intelligence Cockpit using MOMAH-approved KPI and policy-risk data feeds (see Section 3.8.3). System prepares an Emergency Decision Brief with three

			simulated recovery scenarios.
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For guardrail controls (Fiscal Hard-Stop, Monte Carlo Instability Guard, Explainability Gateway, Data Staleness), see Section 8.4. For automated notification channels, see Section 3.8.3.

8.3 Compliance & Ethical AI Checklist

Status reflects current HLSD-stage compliance readiness. Items marked Partial are required controls that must be completed before production deployment.

ID	Compliance Area	Requirement / Control	Status	Evidence / Artefact
EC-01	PDPL - Consent & Data Minimisation	Lawful basis: MOMAH statutory function (PDPL Royal Decree M/19 Art. 4(b) — processing in performance of a statutory duty). Data minimisation applied per use case: consultation submissions undergo PDPL redaction (names, addresses, direct identifiers removed) before LLM processing; housing dataset references aggregate household indicators, not natural persons. A DPIA covering purpose, data categories, necessity, risks, and mitigations is required before production deployment.	Partial	Section 5.5 (Privacy by Design Architecture); DPIA to be completed before production deployment.
EC-02	PDPL - Cross-Border Data Transfer	No personal data leaves the Kingdom. All storage, backups, logs, telemetry, embeddings, vector stores, model inference, and third-party API calls are confined to KSA-sovereign infrastructure per NFR-06. Confidential document	Met	Section 5.5 (Data Residency Architecture); NFR-06 (Sovereign Boundary Constraint).

		inference is not performed outside the Kingdom. No cross-border transfer mechanism is required under this architecture.		
EC-03	PDPL - Data Subject Rights	Data subjects (primarily consultation feedback submitters) may submit requests for access, correction, or deletion of their personal data. The platform logs all personal data collection points. The subject rights intake workflow, fulfilment procedure, and response-time logging consistent with PDPL Art. 26–28 timelines are to be defined in SDD.	Partial	Section 5.5 (Data Architecture); subject rights fulfilment workflow to be defined in SDD.
EC-04	SDAIA - Transparency & Explainability	All AI outputs carry a Traceability Path per BRD Explainability Audit requirement: source documents cited, retrieval context surfaced, confidence indication provided where applicable. The Explainability Gateway surfaces rationale and data lineage to policy officers before any output is actioned. User-facing disclosures confirm AI-assisted nature of outputs. Explanation coverage is validated during User Acceptance Testing before production deployment.	Partial	Section 6.4 (Human-in-the-Loop and Explainability Gateway); Section 8.6 (Annual Audit — Traceability Path Verification).
EC-05	SDAIA - Fairness & Non-Discrimination	Bias evaluation plan: demographic parity assessment across 13 Saudi regions, income	Partial	Section 6.4 (HITL); bias evaluation to be executed before production

		quintiles, and gender; equal-treatment verification in housing access recommendations; review of ABM simulation parameters for inadvertent demographic disadvantage. Known risks: sparse KSA-specific policy training data may introduce recency or domain bias; GASTAT demographic data may undercount informal housing populations. Mitigations: mandatory HITL review for all AI outputs; Explainability Gateway; Responsible AI audit to be executed before production deployment.		deployment per SDAIA AI Ethics Framework.
EC-06	SDAIA - Accountability & Human Oversight	Human-in-the-loop (HITL) checkpoints are mandatory before any AI output advances in the policy workflow: gap reports, simulation outputs, consultation synthesis, amendment proposals, and waiver approvals all require documented human review and explicit approval. HITL roles, authority levels, and escalation paths are defined in the MOMAH RACI. All approvals are immutably logged in the Audit & Compliance Service.	Met	Section 6.4 (Human-in-the-Loop Design and Approval Workflow).
EC-07	SDAIA – AI Ethics Self-Assessment	SDAIA AI Ethics Self-Assessment per the official SDAIA portal template to be completed before production deployment.	Partial	To be completed before production deployment.

		Assessment covers all SDAIA-required dimensions. Residual risks identified are documented and tracked through the pre-production Responsible AI audit programme.		
EC-08	Responsible AI - Safety & Robustness	Guardrail architecture: input/output content filters, allow-listed tool definitions for agentic workflows, rate limiting, anomaly detection with HITL interception. Fallback: outputs failing confidence threshold route to the HITL queue rather than auto-advancing. Adversarial testing plan (prompt injection, jailbreak scenarios, data exfiltration via tool calls, scope-boundary violations) to be executed before production deployment.	Partial	Section 5.5 (Security Architecture — Input/Output Controls); adversarial test plan to be executed before production deployment.
EC-09	Responsible AI - Privacy by Design	Privacy by design at architectural level: PDPL redaction of consultation data before LLM processing; tokenisation of identifiers in embeddings; AES-256 encryption at rest; TLS 1.3 in transit; KMS-managed key rotation; zero-trust access model with least-privilege service accounts; separation of duties between development and operations teams.	Met	Section 5.5 (Security & Privacy Architecture).
EC-10	Audit Trail & Logging	The Audit & Compliance Service logs all user actions, AI prompts,	Met	Section 6.6 (Audit & Compliance Service); Section

		retrieved context, tool calls, inference results, simulation parameters, approval decisions, HITL reviews, and waiver events. Logs are immutable (append-only with tamper-evident hashing), stored within KSA sovereign boundary per NFR-06, access-controlled to Audit Officers and independent auditors, and retained per MOMAH data retention policy (to be defined in SDD).		8.6 (Annual Independent Audit); log retention policy to be defined in SDD.
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Gap & Remediation Summary

Items marked Partial require the following actions before production deployment:

EC	Gap	Remediation Action	Timing
EC-01	DPIA not yet completed	Complete DPIA covering purpose, data categories, necessity, risks, and mitigations	To be completed before production deployment
EC-03	Data subject rights workflow not yet defined	Define intake, fulfilment, and response-time logging workflow consistent with PDPL Art. 26–28	To be defined in SDD
EC-04	Explainability coverage not yet validated end-to-end	Validate Traceability Path output for all 12 capabilities during UAT	To be executed before production deployment
EC-05	Bias evaluation not yet executed	Execute demographic parity assessment across regions, income quintiles, and gender; document results and residual risks	To be executed before production deployment
EC-07	SDAIA AI Ethics Self-Assessment not yet submitted	Complete and submit self-assessment via SDAIA portal; track residual risks through Responsible AI audit	To be executed before production deployment

EC-08	Adversarial testing not yet executed	Execute adversarial test plan: prompt injection, jailbreak, data exfiltration, scope-boundary scenarios	To be executed before production deployment
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8.4 High-Risk AI Governance Additions

Governance Control	HLSD Requirement	Source
Consistent High-Risk Classification	All document sections shall consistently classify Policy AI Lab as High Risk because the platform directly informs national housing, municipal, fiscal, and regulatory policy decisions. All outputs remain advisory and subject to mandatory HITL approval.	High-risk AI governance requirement
Strategic Waiver Workflow	Policies that exceed fiscal thresholds, require strategic exceptions, or need MoF alignment shall enter a documented waiver workflow with request reason, impact evidence, approver identity, decision rationale, and immutable audit trail.	BRD waiver references
Publication and Enactment Boundary	The system may prepare, validate, package, and track handoff/status of Estitlaa, Official Gazette, and MOMAH website publication materials. It shall not directly submit, approve, publish, or enact legislation; all official submission and publication actions remain human-led through MOMAH's official workflow.	MOMAH methodology / HLSD out-of-scope boundary
Annual Independent Audit	An annual independent audit shall be conducted by an independent third-party auditor external to the development and operations team to assess: policy quality, benchmarking accuracy, model performance, explainability, consultation synthesis traceability, waiver usage, and compliance with high-risk AI governance controls. The responsible commissioning party, reporting structure, and audit	BRD audit requirement

	operating model (including auditor selection, scope, and cadence) shall be confirmed with MOMAH in the SDD.	
Checklist and Multi-Entity Review	The 39-item checklist workflow shall record review status and accountability for GRC, Legal Affairs, BI Center, Finance, Digital Transformation, relevant business units, and executive approvers according to the MOMAH methodology RACI model.	MOMAH Legislative Development Methodology March 2026

Strategic Waiver Workflow

Policies that trigger any of the following conditions shall enter the Strategic Waiver Workflow before advancing to Executive Approval:

- **Fiscal Hard-Stop:** Fiscal NPV to the Treasury is negative over a 15-year analysis horizon (per BRD Fiscal Hard-Stop Guardrail). Policies triggering this condition cannot advance to Executive Approval without an explicit Strategic Waiver approved by the Minister of Finance.
- **Fiscal Variance:** Policy implementation cost exceeds the approved budget envelope by more than 10% without a documented 1:1 ROI justification (BRD Risk Escalation Threshold).
- **Stochastic Variance:** Monte Carlo simulation failure rate exceeds 30%, rendering the policy "Unstable" — this flag cannot be waived and requires policy parameter adjustment and re-simulation (BRD Guardrail).
- **Predictive Success Threshold:** Predictive Success Score falls below 80% (per BRD FR-11 Predictive Success Scoring). The simulation is flagged "Failed" with an AI-generated explanation citing the contributing factors that led to the sub-80% probability of policy success — this flag cannot be waived and requires policy parameter adjustment and re-simulation before the policy can advance (BRD Guardrail).
- **Waiver Request:** The Policy Analyst or Director of Policy submits a formal Waiver Request containing trigger type and specific values, strategic necessity justification, Impact Evidence from the Simulation Engine, including the Financial Impact Summary defined in Section 3.8.1 and simulation outputs from CAP-06 / Section 3.3.6, proposed mitigation measures, and expected ROI or strategic return.
- **Waiver Approval:** The MoF designated reviewer evaluates the request, verifies fiscal impact calculations, and renders one of three decisions: Approved (waiver granted, policy unblocked), Approved with Conditions (time-bound conditions attached), or Denied (policy returned to Analyst with rationale, cannot advance to Executive Dashboard). The Minister of Finance is the designated approval authority (BRD Guardrail).
- **Audit Trail:** The complete waiver lifecycle — trigger timestamp and condition, full request documentation, MoF review opinion, final decision with rationale, and approver identity — is written to an immutable audit log entry linked to the policy version. All waiver records are within the scope of the Section 8.6 Annual Independent Audit.

8.5 Module-Level Acceptance Criteria

The following module-level acceptance criteria are indicative targets based on the BRD and BO QA Reply. Formal pass/fail thresholds, datasets, benchmark samples, and SLA commitments will be baselined in the Phase 2 SDD and confirmed before UAT.

Module	Functional Acceptance Criteria	Pass/Fail Threshold
Policy Registry	Automated extraction of policy metadata from structured and semi-structured documents; classification accuracy verified by EWG spot-check sample	Target extraction accuracy 80% or above for structured fields, with response-time and data-loss thresholds to be confirmed in SDD/UAT based on document types and DMS integration constraints.
Conflict Detection	Detection of direct, indirect and transitional conflicts across the full active policy corpus; conflict type and affected articles correctly identified	Target precision/recall and report-generation latency to be baselined in SDD using a MOMAH-approved labelled conflict dataset; BO QA Reply indicates recall is the primary success measure once ground truth is established through Ma'rifa, Sanad, and legal expert validation.
Policy Simulation (CAP-06)	Quantitative output generated for all standard scenario types; probabilistic outcomes include confidence intervals; exportable report produced	Level 1 quick estimate / simplified standard scenario: target within 60 seconds, with exact definition to be confirmed in SDD. Level 2/Level 3 multi-variable ABM runtime, agent scale, Monte Carlo iteration depth, and outcome-variance thresholds are target metrics subject to Phase 2 data assessment, performance testing, MOMAH-approved benchmark datasets, and hardware benchmarking on the SDAIA sovereign platform. Low Confidence flag applied when data gap proxy is used.
Vision 2030 Scoring	Composite KPI score calculated for each active policy; contributing indicator scores traceable to source data; score updated on policy change event	Score variance, scoring latency, and audit coverage targets to be confirmed in SDD using MOMAH-approved KPI weights, formulas, and benchmark scenarios.

DMS Integration	Bidirectional synchronisation of policy documents with the MOMAH Document Management System; version conflicts flagged and queued for human review	Sync success, latency, data-loss handling, and conflict-queue SLA targets to be confirmed in SDD after DMS API, schema, and operational constraints are validated.
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8.6 Annual Independent Audit

The BRD requires an annual independent audit to assess policy quality and benchmarking accuracy. The platform's immutable audit trail — maintained by the Audit & Compliance Service and covering all user actions, AI outputs, simulation runs, approval decisions, and HITL review records — provides the evidentiary basis for the audit. All audit records are stored within the KSA sovereign boundary per NFR-06.

Audit Scope

- Policy Quality:** Independent assessment of whether AI-generated outputs — including policy drafts, gap reports, simulation outputs, and amendment proposals — are consistent with their declared data sources, the KSA legal corpus, and the Vision 2030 KPI framework. The BRD mandates that every AI output carry a Traceability Path (Explainability Audit) that is verifiable by an independent reviewer.
- Benchmarking Accuracy:** Independent verification that CAP-05 benchmarking outputs are grounded in current, authoritative peer-country datasets and that outputs are traceable to their cited data sources, consistent with the BRD benchmarking requirement.

Governance Parameters

Detailed audit governance parameters — including responsible party designation, audit timing, scope sampling methodology, and remediation procedures — are to be defined with MOMAH in the SDD and Statement of Work, in alignment with SDAIA high-risk AI governance requirements.

9. Infrastructure & Operations

9.1 Initial Infrastructure Considerations

Compute Requirements	The integrated delivery is expected to require application services, AI drafting runtime, core ABM simulation compute, policy registry storage, knowledge retrieval capability, and audit logging. Exact compute sizing, instance types, node counts, and scaling approach will be validated in Phase 2 SDD with MOMAH IT and SDAIA platform teams.
Storage Requirements	Storage is required for policy documents, draft versions, audit logs, simulation inputs and outputs, and knowledge-base content. Initial and 12-month storage estimates will be confirmed after data inventory,

	ABM workload definition, retention requirements, and SDAIA platform sizing are validated.
Database Sizing	The solution is expected to require a policy registry, structured metadata store, knowledge graph or equivalent relationship store, and simulation output repository. Database technologies and capacity assumptions will be selected from SDAIA-approved services or equivalents during SDD.
AI/ML Infrastructure	AI/ML infrastructure must support sovereign LLM inference, core ABM model execution, model artefact management, and retrieval over approved policy and legal content. Specific GPU/CPU profile, model registry approach, and vector/knowledge retrieval services will be confirmed against the SDAIA platform catalogue.
Environments Required	The delivery will require separate environments for development, testing/UAT, and production readiness. The final number of environments, isolation model, access controls, and promotion path will be agreed in the SDD.
Estimated Monthly Cost	Cost estimate is to be developed in Phase 2 SDD after integrated delivery scope, ABM workload, data volumes, model runtime needs, and SDAIA platform pricing are confirmed. ABM compute, LLM inference, storage, and knowledge retrieval are expected to be the main cost drivers.

9.2 Operations

Operational Hours	The target service window is KSA business operating hours, aligned with the BRD availability target of 99.5%. Extended support expectations for ministerial review cycles or emergency policy events will be agreed with MOMAH IT before go-live.
Maintenance Window	Maintenance windows, release freeze periods, and user notification lead times will be agreed with MOMAH IT and SDAIA platform teams during SDD and operational readiness planning.
Support Model	The post-go-live operating model follows a shared responsibility structure across four parties: (1) Business Ownership — MOMAH Policy and Strategy Team : policy creation and review, KPI alignment validation, governance workflows, approvals, escalations, and

	ensuring Vision 2030 alignment. (2) Technical Ownership — MOMAH Digital / IT Department: system administration, user access management, integration maintenance (SSO, DMS, Ma'rifa, Sanad), and incident response coordination. (3) Model and Simulation Ownership — Data Science / Analytics Team: simulation model updates, benchmark refresh, data pipeline maintenance, model performance monitoring, sensitivity testing, and recalibration. (4) Vendor / Implementation Partner (transition period): knowledge transfer, stabilization support, post-launch defect fixing, and contracted enhancements. Detailed support tiers, SLA targets, and handover milestones will be confirmed before production go-live.
Incident Response	Incident management will cover platform availability, integration failures, data quality issues, AI/model issues, ABM simulation failures, and security events. Baseline severity classification and response SLA targets confirmed in BO QA Reply: Critical incidents (platform down, data breach, AI kill-switch triggered) — response within 1 hour, resolution target within 4 hours; High severity (major feature failure, integration outage, model accuracy degradation) — initial response within 4 hours; Medium severity (non-critical errors, degraded performance) — response within 1 business day. Ownership and escalation paths will be aligned with MOMAH IT service management processes and formally documented in the operational runbook during Phase 2.
On-Call Requirement	On-call coverage for critical incidents may be required for production, particularly during ministerial review periods. Rotation, acknowledgement targets, and escalation contacts will be confirmed during operational readiness planning.

Data Refresh Cadences

The following data refresh frequencies apply to integrated data sources, as confirmed in the BO QA Reply. All refresh operations are scheduled and monitored by the Data Science team; failures trigger a P2 alert and activate the 30-day staleness watermark on affected scoring outputs.

Data Category	Sources	Refresh Frequency	Freshness Guard
Economic Indicators	SAMA Monthly Statistics Bulletin, MoF macroeconomic forecasts	Monthly or quarterly depending on publication cadence	30-day staleness guard; outputs watermarked Low Confidence if exceeded

Demographic Data	GASTAT household surveys, GASTAT labor market data	Quarterly (surveys); annual for Census-derived baselines	Annual indicators flagged as-of reference year rather than 30-day guard
Housing Market Data	REGA, Balady permit data, Sakani, MOMAH housing records	Monthly	30-day staleness guard applies
Legal References	Ma'rifa (legal corpus), Sanad (legal advisory)	Real-time via integration (on-update event trigger)	30-day advisory index guard for Sanad; Ma'rifa triggers immediate conflict re-scan on update
MOMAH Internal Data	HDOC, MOMAH permits, Sakani internal feeds	To be confirmed with MOMAH data owners in Phase 2 scoping session	Freshness thresholds to be defined in Phase 2

9.3 Performance Consideration

Expected Concurrent Users	Initial usage assumptions should align with BRD adoption targets and will be validated with MOMAH during SDD. Peak usage is expected around policy review cycles and ministerial submission windows.
Peak Load Profile	Peak load is expected during policy review, simulation, and executive approval windows. Detailed concurrency, job queue, and workload assumptions will be confirmed after integrated delivery process design and ABM methodology are finalized.
Response Time SLA	AI policy draft generation target remains 60 seconds as stated in the BRD. Additional response-time targets for ABM simulation, dashboards, and conflict/compliance checks will be confirmed in Phase 2 SDD and validated during testing.
Throughput Requirements	Throughput requirements will be confirmed in Phase 2 SDD based on expected users, policy volume, ABM run frequency, SDAIA platform capacity, and integration constraints.
Simulation Scale	The integrated delivery must include core ABM simulation capability. Final agent population, scenario count, Monte Carlo configuration,

	runtime target, and model constraints will be confirmed through the simulation methodology workshop and SDAIA capacity validation.
Stress and Load Testing Approach	Performance and stress testing will cover core user journeys, AI draft generation, ABM simulation workloads, and critical integrations. Tools, datasets, thresholds, and pass/fail criteria will be defined in the SDD test strategy.
Acceptance Criteria	Acceptance criteria will include BRD/NFR targets, successful completion of integrated delivery user journeys, validated ABM run stability, and UAT sign-off. Detailed thresholds will be baselined during Phase 2 and confirmed before Phase 4 testing.
Scalability Strategy	The solution should use SDAIA-approved scalable platform services where available. Specific scaling patterns for application services, LLM inference, knowledge retrieval, and ABM workloads will be defined in SDD.
Known Bottlenecks	ABM workload size, external data latency, legal corpus access, and model runtime availability are the main performance risks to validate during SDD and testing.

9.4 Testing & Validation Strategy

Test Type	Scope & Objective	Tools / Framework	Environment	Entry / Exit Criteria
Unit Testing	Validate individual functions, classes, and modules in isolation: policy data models and validation rules; simulation parameter parsing and bounds-checking; conflict	pytest (Python backend, ML pipeline, ABM engine unit tests); Jest (frontend component tests); coverage.py (line and branch coverage reporting); unittest.mock / pytest-mock for external dependency	Developer workstation and CI/CD pipeline (automated on every pull request). All external calls mocked — no live service dependencies required.	Entry: Feature branch code complete, peer-reviewed, and merged to integration branch. Exit: Code coverage meets the minimum threshold agreed in SDD (specific percentage to be confirmed with MOMAH IT during Phase 2); 0 Critical or High defects open; full CI pipeline passes with no regressions on existing test suite.

	detection rule engine logic; Vision 2030 KPI scoring and Opportunity Score computation; data freshness timestamp checks; HITL state machine transition logic; risk threshold evaluation logic; output formatting and export utilities; prompt construction and sanitisation for LLM inference calls.	isolation; pre-commit hooks with ruff / flake8 for static analysis.		
Integration Testing	Verify end-to-end data flow across all component and external-system interfaces: Policy Drafting Agent ↔ Knowledge Graph ↔ SDAIA LLM Inference; Conflict Detection Agent ↔ Ma'rifa API ↔ Sanad API ↔ Policy Registry; ABM	Postman / Newman (REST API integration suites, automated in CI); pytest with live staging endpoints; Pact or equivalent contract-testing framework for external integrations (Ma'rifa, Sanad, Yesser GSB); Wiremock for stubbing unavailable external APIs	Staging environment with SDAIA-approved test tenants for LLM inference, sandboxed Ma'rifa and Sanad test credentials, and synthetic data lake inputs. Production services and real citizen data must not be used.	Entry: All unit tests pass; all integration interfaces provisioned and verified reachable in staging. Exit: All 13-CAP interface scenarios pass; Audit Log entries confirmed for all agent actions; no P1/P2 defects open; all external API contracts verified; Nafath signature flow completes end-to-end on test credential.

	Simulation Engine ↔ Data Lake (GASTAT, SAMA, MoF inputs) ↔ Simulation Output Repository; HITL Workflow Engine ↔ Policy Registry status transitions ↔ Nafath SSO signature validation; Risk Escalation Agent ↔ Yesser GSB ↔ Audit Log; DMS archival pipeline on ministerial approval event. All 13 CAP interfaces and cross-agent handoffs must be covered.	during integration test runs.		
Performance Testing	Validate response time, throughput, and scalability against BRD NFR targets and SLAs defined in Section 9.3: AI policy draft generation ≤60 seconds at target	k6 or Apache JMeter (HTTP load generation and ramp-up scenarios); Locust (Python-based scenario scripting for AI workflow user journeys); SDAIA platform APM tooling (resource utilisation,	Pre-production environment on SDAIA sovereign platform, provisioned at production-equivalent hardware specification and data volume. Representative	Entry: Integration tests pass with no open P1/P2 defects; pre-production environment provisioned at production scale; baseline performance thresholds agreed with MOMAH IT and SDAIA. Exit: P95 latency meets all BRD SLA targets under peak load; ABM Level-1 target validated against the SDD-

	concurrency; Conflict Detection report ≤30 seconds; ABM Level-1 quick estimate / simplified standard scenario aligned to the BRD ≤60-second target; ABM Level-2/Level-3 runtime targets to be baselined in SDD; Vision 2030 KPI scoring target to be confirmed in SDD; concurrent user load during peak review and ministerial approval windows; Monte Carlo throughput under sustained load with iteration depth to be agreed in SDD; data refresh pipeline latency within 30-day freshness	bottleneck identification, auto-scaling validation); custom Python scripts for ABM throughput benchmarking.	synthetic policy and simulation datasets required. Real production hardware must be confirmed with SDAIA before testing begins.	confirmed quick-estimate definition and Level-2/Level-3 targets baselined with MOMAH and SDAIA; system remains stable under the stress-test load multiplier agreed with SDAIA in SDD (specific multiplier to be confirmed during Phase 2 performance planning); SDAIA platform capacity formally confirmed as adequate for production go-live.
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	guard thresholds.			
Security Testing	Assess vulnerabilities, penetration risks, and data-protection controls across the full stack: OWASP Top 10 web application vulnerabilities; prompt injection attacks against LLM inference endpoints (Policy Drafting and Conflict Detection agents); PII or classified-data leakage via agent outputs; unauthorised policy status transitions bypassing HITL state machine; Nafath SSO token replay and session hijacking; cross-agent service-account boundary violations; data sovereignty enforcement (zero data	Automated DAST scanning tool (e.g. OWASP ZAP; subject to MOMAH IT and SDAIA platform approval); manual penetration testing tool (e.g. Burp Suite; subject to MOMAH IT and SDAIA platform approval; execution requires prior written authorisation); SAST tool integrated in CI/CD pipeline (e.g. Checkmarx or SonarQube; subject to SDAIA platform catalogue approval); custom prompt injection test suite targeting Policy Drafting and Conflict Detection agent endpoints; penetration test conducted by MOMAH IT security team or an SDAIA-approved third-	Staging / pre-production, fully isolated from production data. Penetration test scope and execution require prior written approval from MOMAH IT and SDAIA. Real citizen or policy data must not be present in the test environment.	Entry: Feature-complete build deployed to staging; SAST scan complete with no Blocker findings; penetration test scope approved by MOMAH CISO. Exit: 0 Critical or High vulnerabilities unresolved; penetration test report approved by MOMAH CISO; prompt injection test suite passes with 0 successful injections; PII output scanner validated against labelled test corpus; NCA ECC compliance evidence documented; data sovereignty boundary verified — no data exfiltration outside KSA detected.

	egress outside KSA); NCA Essential Cybersecurity Controls compliance verification; PDPL personal data handling and retention audit; Audit Log integrity and tamper-resistance verification.	party security firm.		
Bias & Fairness Testing	Evaluate AI model outputs for demographic parity, equal opportunity, and disparate impact, per SDAIA AI Ethics Principles and Section 8: ABM simulation output distributions examined for systematic disadvantage by region (all 13 KSA regions), income quintile, household type, and nationality (Saudi vs. expatriate);	Python fairness and bias-testing library for quantitative parity and equalised-odds metrics (e.g. Fairlearn or equivalent; subject to SDAIA platform catalogue approval); disparate impact testing library (e.g. AI Fairness 360 or equivalent; subject to SDAIA platform catalogue approval); custom audit scripts for demographic segmentation of ABM simulation outputs across all 13 KSA regions	Staging with a representative synthetic dataset covering all 13 KSA regions and key demographic segments (income quintiles, household types, Saudi and expatriate populations). Real personal data must not be used in test environments (PDPL compliance mandatory).	Entry: Model training and calibration complete; representative synthetic test dataset validated by Data Science team and approved by MOMAH. Exit: Demographic parity gap within SDAIA-defined threshold (to be confirmed in SDD) across all protected segments; no systematic regional disadvantage detected in ABM outputs; Responsible AI Report approved by Data Scientist and Legal and Compliance Officer; findings documented in Responsible AI baseline log.

	policy draft content screened for discriminatory eligibility criteria or biased language; Vision 2030 KPI opportunity scores examined for demographic skew; ABM agent-behaviour calibration parameters reviewed for embedded demographic assumptions; Responsible AI monthly report baseline established.	and key population segments; MOMAH Expert Working Group qualitative review for policy draft language and cultural appropriateness.		
Explainability Testing	Verify that all AI-generated outputs meet the Explainability Gateway requirements defined in Section 6.3: every policy draft clause cites at least one verifiable source (Royal Decree, precedent, or Knowledge	Feature attribution library for model-level explanation (e.g. SHAP or equivalent; subject to SDAIA platform catalogue approval); local explanation library for instance-level interpretation (e.g. LIME or equivalent; subject to SDAIA	Staging with representative policy corpus and Ma'rifa legal reference data accessible to test environment. Domain SME reviewers must have read access to Ma'rifa for citation verification. Synthetic test policies	Entry: Bias and Fairness tests pass; Explainability Gateway implementation complete and code-reviewed. Exit: citation verification passes for all mandatory cited sources defined in the SDD test set; confidence intervals and data-provenance watermarks present on simulation outputs where applicable; AI explanation quality accepted by domain SME review panel at the acceptance

	Graph entry) retrievable in Ma'rifa; Conflict Detection reports include severity rationale and affected article references; Vision 2030 scoring explanations trace each contributing KPI indicator to its source dataset; simulation outputs display confidence intervals and data-provenance watermarks (including Low Confidence flag where data gap proxy applies); Explainability Gateway blocks any output that fails citation completeness check; HITL reviewers can interrogate the rationale for every AI recommendation before approval.	platform catalogue approval); citation verification scripts querying Ma'rifa and Knowledge Graph APIs against every source cited in AI outputs; manual domain expert review panel comprising MOMAH Policy Analyst and Legal and Compliance Officer.	covering a range of legal domains (housing, finance, municipal, environmental) required.	threshold agreed in SDD (specific threshold to be confirmed during Phase 2); Explainability Gateway correctly blocks all test cases with missing or unverifiable citations (false-negative tolerance agreed in SDD); gateway block events confirmed in Audit Log.
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User Acceptance Testing (UAT)	Business stakeholders validate that the solution meets acceptance criteria and business requirements from Section 4.1, using representative MOMAH policy scenarios: (1) Complete policy lifecycle — initiation through ministerial approval — for at least two policy types (regulatory and fiscal); (2) Simulation accuracy validation against EWG-benchmarked baseline scenarios; (3) Conflict detection accuracy verified against a known conflict test set; (4) Dashboard usability and data accuracy reviewed by Policy Analysts, Directors, and	Manual testing using MOMAH-approved UAT scripts and structured test case library; Nafath test-environment credentials for digital signature approval flows; defect tracking in agreed MOMAH IT tool; structured UAT sign-off form requiring MOMAH AI Solution Owner signature.	Pre-production environment with production-representative configuration, provisioned on SDAIA sovereign platform. Anonymised or synthetic policy data approved by MOMAH for use in UAT. Nafath test credentials provisioned for all UAT personas. Environment signed off by MOMAH IT before UAT commences.	Entry: All prior test phases (Unit, Integration, Performance, Security, Bias and Fairness, Explainability) complete with no open P1/P2 defects; UAT environment verified by MOMAH IT; UAT test cases reviewed and approved by MOMAH AI Solution Owner; UAT participants briefed and scheduled. Exit: All mandatory user journeys complete without P1/P2 defects; simulation output variance within the SDD-agreed tolerance against EWG baseline scenarios; UAT sign-off obtained from MOMAH AI Solution Owner; any open defects triaged and either resolved or accepted into the post-launch backlog with formal MOMAH approval.
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	Minister-persona testers; (5) HITL workflow approvals executed using Nafath test credentials; (6) Bilingual Arabic/English interface validated for all user-facing screens; (7) Role-based access control and least-privilege enforcement verified across all five user personas.			
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9.5 Business Continuity

Component / Service	Backup & Redundancy	Failover Strategy	Health Monitoring
Policy Registry	Durable storage, backup, and redundancy must support the agreed RPO/RTO targets.	Failover pattern to be defined in SDD based on SDAIA-approved platform capabilities.	Availability, latency, storage, and error monitoring to be defined in the operational runbook.
Knowledge Graph / Retrieval Layer	Backup and redundancy approach to be aligned with policy and legal knowledge retention requirements.	Recovery approach to be confirmed in SDD; read-only or degraded modes may be considered.	Query health, latency, and data freshness monitoring to be specified during SDD.

ABM Simulation Engine	Simulation inputs, configurations, and outputs must be persisted so failed runs can be diagnosed or rerun.	Job retry, resubmission, and degraded-mode behavior to be confirmed after ABM design is finalized.	Simulation job status, runtime, failure rate, and capacity monitoring to be defined in SDD.
LLM Inference Service	Model artefacts and configurations must be versioned and recoverable within KSA boundaries.	Degraded mode should prevent unsupported AI drafting while allowing manual workflow continuation where appropriate.	Inference availability, latency, and error monitoring to be defined with SDAIA platform teams.
Application Layer	Application services should be designed to avoid single points of failure where feasible.	Deployment and failover approach to be confirmed in SDD.	Application availability, API health, and error monitoring to be integrated with MOMAH operations.
Audit Log	Audit records must be durable, tamper-resistant, and recoverable in line with compliance requirements.	Audit continuity requirements will be defined to support no-loss expectations for critical approval records.	Audit write failures, access anomalies, and integrity checks must be monitored.
Nafath / SSO Integration	Nafath is an external national identity service; internal dependency handling will be defined with MOMAH IT.	Fallback behavior for login or signature outage to be agreed before go-live.	Authentication success, error rates, and integration health to be monitored.

Incident Escalation & Communication

Incident Severity Levels	Severity levels will cover platform unavailability, security incidents, critical integration failures, AI/model issues, ABM simulation failures, and
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	material data quality issues. Final definitions will align with MOMAH IT service management standards.
Escalation Matrix	Escalation paths will include MOMAH IT, the business owner, delivery partner support, and SDAIA platform teams where platform services are involved. Detailed contacts, response targets, and duty coverage will be confirmed in the operational runbook.
Communication Channels	Approved incident communication channels will be agreed with MOMAH IT before go-live and should comply with government security and confidentiality requirements.
Stakeholder Notification	Notification rules for executive-impacting incidents, extended outages, and post-incident reporting will be defined during operational readiness planning.

9.6 Disaster Recovery

Recovery Time Objective (RTO)	Less than 4 hours, as defined in NFR-05, for business-critical services. Final service grouping and restoration procedures will be confirmed in SDD.
Recovery Point Objective (RPO)	Less than 1 hour, as defined in NFR-05, for business-critical data including policy drafts, approval decisions, audit logs, and simulation outputs.
Backup Frequency	Backup frequency, retention, validation approach, and restoration procedure will be defined with MOMAH IT and SDAIA during SDD, aligned to RTO/RPO and compliance requirements.
Backup Storage Location	All backups and recovery activities must remain within KSA national boundaries on SDAIA-approved sovereign infrastructure. Specific backup zones and storage services will be confirmed with SDAIA.
DR Test Frequency	DR testing cadence, evidence requirements, and approval process will be agreed with MOMAH IT and SDAIA before production go-live.
AI Model Recovery	AI model artefacts, ABM configurations, and related metadata must be versioned and recoverable. Detailed recovery, validation, rollback, and promotion controls will be defined in the SDD and model governance runbook.

9.7 Data Migration

Parameter	Details
Data to be Migrated	Not Applicable. The Policy AI Lab is a greenfield implementation. No legacy digital system for policy management exists at MOMAH (Section 2.6; BRD §2.7). All policy content will be created natively within the new platform from go-live. Note: Existing policy documents (Word/PDF archives held in MOMAH DMS) may optionally be ingested as seed content to populate the Institutional Memory Vault and provide initial grounding context for AI-assisted drafting. This is treated as a content ingestion task and will be scoped separately in the Phase 2 SDD.
Volume Estimate	Not Applicable for structured migration. Indicative seed content volume (optional archive ingestion): TBD based on MOMAH document inventory conducted in Phase 1.
Source Systems	Not Applicable — no predecessor digital policy management system exists. Optional seed content source: MOMAH DMS (existing Word/PDF policy archives).
Target Systems	Policy AI Lab Policy Registry (relational database + Knowledge Graph) for seed content ingestion, if undertaken.
Migration Approach	Not Applicable for data migration. If optional archive ingestion is approved: phased content import (batch ETL + NLP extraction), prioritizing active policies and recent amendments. No parallel-run or cutover migration is required. Full scope and timeline to be confirmed in Phase 2 SDD.

10. Assumptions, Constraints & Prerequisites

10.1 Assumptions

ID	Assumption	Owner
A-01	MOMAH internal system data (housing permits from MOMAH platform, Sakani application data, Balady municipal records) will be made accessible to the Policy AI Lab platform in a	MOMAH Data Team / Dr. Ahmed Aldurwiesh

	structured format with sufficient historical depth for simulation validation.	
A-02	Integration with Ma'rifa (legal references) and Sanad (legal advisory) systems is technically and contractually feasible within the project scope and timeline.	MOMAH Legal / IT Team
A-03	Formal data access approvals from GASTAT, SAMA, and Ministry of Finance will be obtained without significant delays to the delivery schedule.	MOMAH / Dr. Ahmed Aldurwiesh
A-04	MOMAH's existing SSO infrastructure is available and technically compatible for integration with the Policy AI Lab for authentication and role-based access control.	MOMAH IT / Digital Transformation Team
A-05	All AI model inference and data storage will remain within KSA, in full compliance with PDPL and NCA Essential Cybersecurity Controls. No cross-border data transfer is required.	Solution Vendor / MOMAH IT Security
A-06	Human-in-the-loop governance framework will be formally adopted by MOMAH stakeholders as a mandatory and non-bypassable part of the policy lifecycle workflow.	MOMAH AI Solution Owner
A-07	Expert policy rules, analytical heuristics, and existing manual evaluation practices currently used by the Ministry will be documented and made available to inform the simulation framework design.	MOMAH Policy Analysts / Dr. Ahmed Aldurwiesh
A-08	Technical and product teams — MOMAH, partner vendors, and SDAIA platform engineers — are allocated full-time across Phases 1–3 of the delivery roadmap (per BRD Assumptions).	MOMAH IT / Delivery Partner
A-09	SDAIA-approved sovereign AI infrastructure has sufficient capacity to host the ABM simulation engine, Knowledge Graph, and bilingual LLM inference workloads within the Kingdom (per BRD Assumptions). Capacity	SDAIA / MOMAH IT

	validation against simulation workload completes in Phase 2 SDD.	
A-10	Executive sponsorship from His Excellency the Minister is sustained throughout Phase 1 roll-out and subsequent delivery phases to ensure change management, cross-ministerial cooperation, and adoption (per BRD Assumptions; see also Dependency D-12).	MOMAH AI Solution Owner / Dr. Ahmed Aldurwiesh
A-11	End users have access to devices compatible with the target platform — modern web browsers (Chrome, Edge, Safari) on desktop for analysts and reviewers; mobile and tablet devices for regional directors and executives where geospatial dashboards or notifications are required (per BRD Assumptions; see also NFR-07 Supported Platforms).	MOMAH IT

10.2 Constraints

ID	Type	Constraint
C-01	Business	The system must not autonomously approve, enact, or publish policies. All final decisions remain human-led; AI outputs are strictly advisory throughout the entire policy lifecycle.
C-02	Technical	The platform must integrate with Ma'rifa, Sanad, MOMAH SSO, the Document Management System (DMS), and internal data platforms. API availability, authentication protocols, and data formats for each system must be confirmed before Phase 2.
C-03	Regulatory	All personal data processing must comply with Saudi PDPL. All AI inference and data storage must remain within KSA. The platform must adhere to NCA Essential Cybersecurity Controls and SDAIA AI Ethics Principles. The system must not provide legally binding interpretations or replace formal legal counsel.
C-04	Business	The system will not build or replace national econometric forecasting platforms. It will rely on predefined models,

		assumptions, or integrated external datasets for simulation inputs.
C-05	Business	The platform is an internal decision-support tool only. It is not a citizen-facing application and does not manage day-to-day municipal service delivery (e.g., waste collection, utilities).
C-06	Budget	Delivery is capped within the approved MOMAH FY26 digital-transformation budget envelope. Any scope expansion beyond the approved envelope requires re-approval by the Digital Transformation Steering Committee, and for cross-ministerial scope co-approval by the Ministry of Finance (per BRD Constraints).
C-07	Timeline	Phase 1 (Policy Registry & Drafting) delivery is tied to the FY26 Vision 2030 mid-cycle review milestone. Phased go-live must align with MOMAH quarterly governance cadences. Executive-facing modules (Minister Executive Intelligence Cockpit and Senior Leader dashboards) must be available before the next annual National Transformation Programme (NTP) reporting cycle (per BRD Constraints).
C-08	Human Resources	Specialised skills in Agent-Based Modelling (ABM), Explainable AI (XAI), and Arabic Natural Language Processing are scarce in the domestic market. Delivery depends on a blended team of staff, SDAIA platform resources, and accredited external vendors, with knowledge-transfer obligations built into each phase contract (per BRD Constraints; see also Dependency D-11 Specialized Skills Available).

10.3 Dependencies & Prerequisites

ID	Prerequisite / Dependency	Description	Owner	Target Date	Status
D-01	MOMAH Internal Data Access Confirmed	Formal access to MOMAH internal systems (housing permits, Sakani, Balady) confirmed with field-level data	MOMAH / Dr. Ahmed Aldurwiesh	Before Phase 2	In Progress

		specifications available for simulation design.			
D-02	External Data Access Approved	Formal data access approvals obtained from GASTAT, SAMA, and Ministry of Finance, including agreement on data format, frequency, and historical depth.	MOMAH / Dr. Ahmed Aldurwiesh	Before Phase 2	Not Started
D-03	Ma'rifa & Sanad Integration Scoped	Technical API specifications for Ma'rifa (legal references) and Sanad (legal advisory) confirmed, including authentication method and data schema.	MOMAH IT / Legal Team	Before Phase 2	Not Started
D-04	SSO Integration Available	MOMAH SSO system specifications provided to vendor for integration design, including supported authentication protocols (SAML 2.0 / OIDC) and user role schema.	MOMAH IT / Digital Transformation	Before Phase 2	Not Started
D-05	Simulation Methodology Agreed	Expert policy rules, assumptions, heuristics, and existing analytical models documented and validated with MOMAH domain experts as inputs to the simulation framework design.	MOMAH Policy Team / Dr. Ahmed Aldurwiesh	Before Phase 2	In Progress

D-06	KPI Definitions Finalized	Back-testing accuracy definition, output metric formulas, and acceptance thresholds formally agreed with MOMAH before simulation framework design commences.	MOMAH / Dr. Ahmed Aldurwiesh	Before Phase 2	In Progress
D-07	SDAIA Sovereign Cloud Provisioned & NCA Accredited	Sovereign cloud environment provisioned on SDAIA AI Platform and formally accredited by NCA before any AI inference or data ingestion commences. SLA must be agreed and signed. (BRD §15.3 + §15.4)	SDAIA / MOMAH IT	Before Phase 2	Not Started
D-08	Policy Registry Schema Validated	Canonical Policy Registry schema (BRD Artifact 1.0 — Policy Object definition) formally reviewed and signed off by Legal, Policy, and Data Governance functions before Phase 3 build begins. (BRD §15.3)	MOMAH Legal / Policy Team	Before Phase 3	Not Started
D-09	REGA Database Access Confirmed	Formal data-sharing agreement with Real Estate General Authority (REGA) for access to real-estate regulatory and transaction data; access method and	MOMAH / REGA	Before Phase 2	Not Started

		refresh cadence agreed. (BRD §11.2 + §15.4)			
D-10	Yesser GSB Inter-Ministerial Access Configured	Yesser Government Service Bus access configured for secure inter-ministerial data exchange with SAMA, Ministry of Finance, and MISA; API credentials and routing rules confirmed. (BRD §11.1 + §15.4)	MOMAH IT / Yesser	Before Phase 2	Not Started
D-11	Specialized Skills Available	Delivery team confirmed with required capabilities: Agent-Based Modelling (ABM), Explainable AI (XAI), and Arabic NLP engineering. Knowledge-transfer obligations built into each phase contract. (BRD §15.2 — HR Constraint)	MOMAH / Delivery Partner	Before Phase 2	Not Started
D-12	Change Management Plan Approved	Training and change-management plan for Policy Analysts, Economists, Legal Officers, and Executives approved and resourced before Phase 3. Executive sponsorship from HE the Minister sustained. (BRD §15.1 + §15.3)	MOMAH Business Owner / Dr. Ahmed Aldurwiesh	Before Phase 3	Not Started
D-13	Stakeholder RACI &	Stakeholder RACI matrix and	MOMAH AI Solution	Before Phase 2	Not Started

	Permissions Matrix Approved	permissions matrix (as documented in Section 2.3 and Appendix C — RACI Matrix & Permissions) formally reviewed and signed off by the Policy AI Lab Steering Committee before Phase 2 commences. Covers role definitions, responsibilities, decision rights, and access scopes across MOMAH, NHCI, SDAIA platform team, and accredited external vendors. (BRD Prerequisites)	Owner / NHCI Steering Committee		
D-14	Vision 2030 Baseline KPI Dataset Loaded	Initial Vision 2030 KPI dataset — current target values, multi-year trendlines, and historical baselines — loaded from the Vision 2030 PMO data feed into the Knowledge Graph for use by the Vision 2030 Alignment Scorer (Section 3.8.2). Data-refresh cadence and access mechanism confirmed with the Vision 2030 Realisation Office; 30-day staleness guard configured per Section 9.2 Table 60. (BRD Prerequisites)	Vision 2030 PMO / MOMAH Data Team / Dr. Ahmed Aldurwiesh	Before Phase 2	Not Started

11. Success Metrics

11.1 Strategic Metrics

Metric Type	Metric	Baseline	Target (6M / Y1 / Y2)	Measurement Method	Frequency	KPI Owner
User Adoption	Number of users actively using the Policy AI Lab platform	0	10 / 20 / 50	Platform login logs / active session tracking	Daily	Business Owner / Product Manager
Platform Adoption	Number of policies generated and simulated through the platform	0	3 / 5 / 10 (80%) (per BRD Strategic and Operational Metrics)	Simulator Review Screen / policy registry count	Quarterly	Dr. Ahmed Aldurwiesh
User Satisfaction	User satisfaction score (platform usability and output quality)	N/A	TBD / >3.5 / >4.0	Quarterly in-app survey	Quarterly	Business Owner / Product Manager

11.2 Operational Metrics

Metric Type	Metric	Baseline	Target (6M / Y1 / Y2)	Measurement Method	Frequency	KPI Owner
Processes	Average time to move a	0	-10% / -30% / -50%	Timestamps at each stage of policy	Quarterly	Dr. Ahmed Aldurwiesh

Efficiency	policy from initial concept to approved draft using the AI platform			drafting lifecycle		
Accuracy	Policy simulation accuracy (proportion of simulation outputs accepted without edit in HITL review)	0	TBD / TBD / 80% (per BRD Strategic and Operational Metrics)	HITL approvals without edits / simulation review screen logs; formal metric definition and threshold to be confirmed with MOMAH data owners	Quarterly	Dr. Ahmed Aldurwiesh
Quality	Policy conflict detection rate — % of actual conflicts flagged by system prior to ministerial review	0	TBD / TBD / ≥85% (per BRD Strategic and Operational Metrics)	HITL approvals cross-referenced with conflict detection logs; ground truth established through: (1) legal expert validation by MOMAH Legal Officers; (2) cross-referencing Ma'rifa and Sanad platforms; (3) known historical policy contradiction	Quarterly	Dr. Ahmed Aldurwiesh

				s; (4) labelled benchmark dataset to be created during system go-live as a Phase 2 deliverable owned by MOMAH Legal Experts		
Quality	Assumption documentation completeness — % of policies with structured assumption logs in the registry	0	TBD / TBD / 90% (per BRD Strategic and Operational Metrics)	Policy registry audit — % of approved policies with complete assumption records; threshold to be confirmed after metadata schema is finalized	Quarterly	Dr. Ahmed Aldurwiesh
Accuracy	Risk threshold alert coverage — % of policies exceeding fiscal/legal risk thresholds for which automated alerts were generated	0	TBD / TBD / >90% (per BRD Strategic and Operational Metrics)	Number of alerts per policy / risk scoring dashboard; coverage target to be confirmed once fiscal/legal thresholds and ground truth are defined	Quarterly	Dr. Ahmed Aldurwiesh

11.3 Technical Metrics

Metric Type	Metric	Baseline	Target (6M / Y1 / Y2)	Measurement Method	Frequency	KPI Owner
AI Model Simulation	ABM simulation accuracy vs. historical outcomes	N/A	Target TBD in SDD after data assessment	Back-testing against available GASTAT/SAM A/MoF historical data where sufficient; statistical accuracy measured using an agreed metric such as MAPE, RMSE, or R ² ; formal threshold depends on historical depth and data quality	Monthly	Dr. Ahmed Aldurwiesh
AI Model Conflict Detection	Precision/recall of regulatory conflict identification	N/A	Target TBD in SDD after labelled dataset creation	Expert review on labelled conflict sample set; recall is the primary metric per BO QA Reply, with precision tracked to manage false positives	Monthly	Dr. Ahmed Aldurwiesh
AI Model Assumption	Accuracy of policy assumption	N/A	Target TBD in SDD after	Human annotation comparison; NLP evaluation	Monthly	Dr. Ahmed Aldurwiesh

Extraction	extraction from text		metadata schema confirmation	pipeline; threshold depends on field definitions, controlled vocabulary, and document quality		
AI Model Risk Alert Coverage	Coverage of genuine risks flagged by alert system	N/A	Target TBD in SDD after risk taxonomy confirmation	Retrospective comparison against expert-identified risks; threshold depends on agreed risk taxonomy, fiscal/legal trigger rules, and labelled examples	Monthly	Dr. Ahmed Aldurwiesh
System Availability	Platform uptime during KSA business hours (07:00–22:00 AST)	N/A	≥99.5% / ≥99.5% / ≥99.5%	Infrastructure APM monitoring dashboard	Weekly	MOMAH IT / DevOps
System Response Time	API response latency (95th percentile)	N/A	≤3s / ≤3s / ≤2s	APM tool; load test reports	Weekly	MOMAH IT / DevOps
System Error Rate	Application-level error rate (5xx HTTP responses)	N/A	<1% / <1% / <0.5%	APM tool error log analysis	Weekly	MOMAH IT / DevOps
System Draft Generation	Time to generate initial policy draft (95th percentile)	N/A	≤60s / ≤60s / ≤45s	Application performance logs; UAT timing records	Weekly	Policy Officers / MOMAH IT

11.4 Methodology Alignment and Governance Metrics

Metric Type	Metric	Baseline	Target	Measurement Method	Frequency	KPI Owner
Methodology Coverage	% of policies with completed case identification and concept builder records	N/A	TBD in SDD / KPI workshop	Policy Registry lifecycle metadata; concept record completion	Quarterly	Policy Owner / GRC
Checklist Quality	% of draft legislation packages with completed 39-item checklist before final review	N/A	TBD in SDD / GRC workshop	Checklist Compliance Engine report and reviewer status	Per policy / Monthly	GRC / Legal Affairs
Gap Analysis	% of identified critical gaps traceable to at least one Vision 2030 KPI and one benchmark precedent	N/A	Aligned to BRD UC-16; threshold to be finalized in SDD	Gap report, opportunity score, benchmark map, audit entry	Quarterly	Policy Strategist / KPI Owner
Consultation Traceability	% of amendment recommendations traceable to consultation themes and reviewer decisions	N/A	TBD in SDD	Consultation Synthesis logs; amendment approval/rejection records	Per consultation cycle	Policy Analyst / Legal Affairs
Vision 2030 Scoring	Composite alignment score, pillar breakdown, and clause-level	N/A	0-100 scoring framework; detailed formulas and	Vision 2030 Alignment Scorer; KPI data freshness checks	Per draft version	KPI Owner

	contribution traceability		weights in SDD			
Public ation Readin ess	% of approved policies with complete publication package, archive package, metadata, and timestamps	N/A	TBD with Legal/GRC	Publication workflow package status; DMS archive confirmation	Per policy	Legal Affairs / GRC
Quality Measu rement	% of enacted policies with actual-vs-simulated outcome monitoring and implementation quality report	N/A	TBD in SDD	Post-Implementation Quality Dashboard; KPI tracking reports	Quarterly / Annually	GRC / KPI Owner
Indepe ndent Audit	Annual independent audit completed for policy quality, benchmarking accuracy, model performance, and governance compliance	N/A	1 audit per year after production go-live	Independent audit report and remediation tracker	Annual	GRC / AI Solution Owner

12. Roadmap & Phasing

12.1 High-Level Delivery Roadmap

Phase	Key Milestone	Est. Start	Est. End
Phase 1 — Assessment	HLSD Approved	04 Apr 2026	28 May 2026
Phase 2 — Design	SDD & Model Card v1 Approved	29 May 2026	29 Jun 2026
Phase 3 — Build	Build Quality Gate Passed	30 Jun 2026	09 Aug 2026
Phase 4 — Test	Proceed to Production Approved	10 Aug 2026	24 Aug 2026
Phase 5 — Deploy	Go-Live Confirmed	25 Aug 2026	31 Aug 2026
Phase 6 — Operate	Handover Completed	01 Sep 2026	30 Sep 2026

12.2 MVP Scope

Integrated Delivery

The Policy AI Lab is planned as one integrated delivery rather than a reduced-scope release and later-phase split. All agreed HLSD capabilities are delivered through the roadmap phases in Section 12.1; those phases are execution sequencing only and do not define a reduced delivery scope.

In MVP

- N/A

Post-MVP (Future Phases)

- N/A

12.3 Execution risks

Risk	Likelihood	Impact	Mitigation Strategy
Government data source access delays (GASTAT, SAMA, MOF) —	High	High	Initiate MOU / data-sharing agreements in Phase 1; develop simulation with synthetic or proxy data fallbacks until live feeds are available

required for ABM simulation agents			
Simulation methodology validation — ABM model assumptions and KPI definitions may be contested by stakeholders	Medium	High	Conduct KPI & simulation methodology workshop in Phase 1 with Dr. Ahmed Aldurwiesh and KPI Owners; lock assumptions in SDD before Phase 3 build commences
KPI definition ambiguity — unclear ownership and inconsistent metric definitions across MOMAH departments	High	Medium	Assign KPI Owner per metric; hold stakeholder alignment workshop; document agreed definitions in SDD with formal sign-off
Integration complexity — connecting Policy AI Lab with MOMAH HCM, ERP, and identity management systems	Medium	High	Conduct integration discovery spike in Phase 2; use API gateway / ESB patterns; engage MOMAH IT from Phase 1 to confirm API availability and access
Legal corpus availability — Sanad and Umm Al-Qura APIs may have limited programmatic access for conflict detection	Medium	Medium	Identify legal corpus sources and API access terms in Phase 1; build offline corpus ETL pipeline as fallback if live APIs are unavailable

Appendix

A. References

- SDAIA National AI Ethics Principles (2025)

- NIST AI Risk Management Framework (AI RMF 1.0)
- Saudi Personal Data Protection Law (PDPL)
- NHCI AI Programme Governance Framework
- Recommendations on AI Evaluation
- Policy_AI_Lab_BRD_v2.0.docx

B. Reference - Agentic evaluation Framework

Document type	Definition	Key Question to Ask	Score (1-5)	Weight
Role in the workflow	How independent is this task within the overall process?	Does the task involve progressing through multi-step workflows?	5	30%
Type of input data	What is the nature of information the task works with?	Does the task rely on unstructured data & content like text, PDFs, or emails?	5	20%
Type of rules	Can the task be executed via fixed business rules, or does it need judgment or context?	Does the task require contextual judgment, or case-by-case handling, to be completed?	4	20%
Need for reasoning before acting (ReAct)	Does the task require thinking before taking the next action?	Does the task require the agent to decide what to do next based on the situation?	5	30%

The overall score reflects the average of all criteria. A score of 1-2 indicates alignment with a digital solution, while 3.5-5 suggests strong suitability for agentic intervention. A score of 1 indicates that the criterion or question is not met or relevant, while a score of 5 signals a strong alignment.

Weighted Assessment Result

Based on the weighted evaluation, the Policy AI Lab achieves a score of 4.8 out of 5. This indicates strong suitability for agentic intervention. The solution requires multi-step workflow orchestration, interpretation of unstructured policy and legal content, contextual judgment, and reasoning before action across drafting, conflict detection, simulation, consultation synthesis, and approval workflows. Therefore, Agentic AI is justified as the appropriate design pattern, subject to the human-in-the-loop controls and governance safeguards defined in Sections 6.4, 6.7, and 8.

C. Approval Sign-Off

Role	Name	Signature	Date
Solution Vendor (Prepared By)			
Solution Owner (Reviewed By)			
NHCI Oversight (Reviewed By)			
Central AI (Reviewed By)			
AI Champion (Reviewed By)			
Sector Head (Reviewed By)			
AI Solutions Steering Group (Approved By)			
Central AI Committee (Approved By)			